A PROJECT REPORT

On

"ONLINE EXAMINATION SYSTEM"

Submitted by

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In partial fulfilment for the award of the degree

Of

BACHELOR OF SCIENCE

In

COMPUTER SCIENCE

Under the guidance of

Mrs. Varsha S. More

Department of Computer Science

B.N.N. College, Bhiwandi.

Sem V

A.Y. 2024-2025

Padmashri Annasaheb Jadhav Bhartiya Samaj Unnati Mandal's

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Department of Computer Science

A.Y. 2024-2025

CERTIFICATE

This is to certify that **Mr. Sumit Niranjan Shahu**, of **T.Y. B.Sc. (Sem.-V)** class has satisfactorily completed the Project "**Online Examination System**" to be submitted in the partial fulfilment for the award of **Bachelor of Science** in **Computer Science** during the academic year **2024– 2025**.

Date of Submission:

Project Guide
(Mrs. Varsha S. More)

Head / In-Charge,
Department Computer Science

College Seal

Signature of Examiner

DECLARATION BY THE STUDENT

I, **Mr. Sumit Niranjan Shahu**, hereby declare that the project entitled "**Online Examination System**" submitted in the partial fulfilment for the award of **Bachelor of Science** in **Computer Science** during the academic year **2024 – 2025** is my original work and the project has not formed the basis for the award of any degree, associateship, fellowship or any other similar titles.

Signature of the Student:

Place:
Date:

ACKNOWLEDGEMENT

I would like to extend our heartiest thanks with a deep sense of gratitude and respect to all those who provides me immense help and guidance during my project work. I would like to thank my project guide of **Mrs. Varsha S. More** for providing a vision about the system.

I have been greatly benefited from their regular critical reviews and inspiration throughout my work. I am grateful to them for their guidance, encouragement, understanding and insightful support in the development process. I would also like to thank my college for giving required resources whenever I wanted and for giving the opportunity to develop the project.

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Technology for their constant Encouragement, suggestions and moral support throughout the duration of my project. Last but not the least I would like to mention here that I am greatly indebted to each and everybody my friends and who has been associated with my project at any stage but whose name does not find a place in this acknowledgement.

With Sincere Regards,
Mr. Shahu Sumit Niranjan



“ON-EX”
ONLINE EXAMINATION SYSTEM

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INTRODUCTION

1 Introduction:-

In the digital age, educational institutions face increasing challenges with traditional examination systems. These challenges include the logistical burden of managing paper-based exams, securing exam content, and timely grading, all of which make the conventional methods inefficient and prone to errors. Furthermore, environmental concerns, such as the extensive use of paper, add pressure on institutions to adopt more sustainable practices.

The "ON-EX" platform aims to address these issues by transitioning the examination process into an online format. This project proposes the development of a LAN-based online examination system using Java, enabling schools and colleges to administer exams digitally. The platform's core functionality includes scheduling exams, setting time limits, randomizing question orders, and offering instant results for multiple-choice questions (MCQs). The system streamlines exam management, reduces administrative overhead, and enhances the overall efficiency of the examination process.

The proposed platform will provide a secure and user-friendly environment for both students and faculty. Students will be able to complete exams within a specified time frame, while teachers can create and administer exams with minimal effort. By automating many of the exam processes, ON-EX not only makes examinations more accessible but also significantly reduces the chances of cheating through its security features.

2 Objectives:-

The primary objectives of the "ON-EX" online examination platform are outlined as follows:

- **Create a Secure and Robust Examination System:** The platform is designed to handle large volumes of users and exams simultaneously while ensuring data integrity. Security measures such as multi-factor authentication and question randomization are implemented to safeguard exam content and results.
- **Improve Accessibility and Usability:** The system aims to make examinations accessible to a broad range of users by providing a user-friendly interface. Students can take exams from designated areas and complete them at their own pace, enhancing focus and reducing stress.
- **Streamline and Automate Examination Processes:** The platform simplifies exam creation, scheduling, and delivery. Teachers can easily configure exams by setting parameters, creating question banks, and scheduling the tests, which will be automatically managed by the system.
- **Ensure Examination Integrity:** The system includes features such as restricted internet access, randomized question orders, and secure login protocols to maintain the integrity of the examination process. These measures ensure that students cannot engage in unethical practices such as cheating.

3 Scope:-

The scope of "ON-EX" is focused on delivering a complete examination management platform with the following core features:

- **User Management:** The system assigns different roles to users, such as administrators, instructors, and students. Administrators have full control over

the system, including the ability to manage users and configure settings. Instructors can create exams, manage question banks, and assess student performance. Students can access their personalized dashboard to view upcoming exams and submit completed assessments.

- **Exam Scheduling and Configuration:** Instructors can schedule exams, set specific time limits, and randomize question and answer sequences to prevent cheating. These configurations ensure a fair and equitable testing environment for all students.
- **Secure Exam Delivery:** The platform ensures secure exam conditions by restricting access to unauthorized applications and websites during the test. The system also shuffles the order of questions for each student, reducing the risk of collaboration or cheating during the exam.
- **Result Generation and Reporting:** Upon completion of the exam, the platform generates detailed reports, including individual scores, performance analysis, and the time taken to complete the test. Teachers can download or export these reports for further analysis.

4 Methodology:-

The development of the "ON-EX" platform follows an **Agile** development methodology, particularly an **Incremental** approach. This ensures continuous delivery of new features, while integrating feedback from stakeholders throughout the project lifecycle. The process of development can be divided into the following stages:

- **Requirement Gathering and Analysis:** In-depth interviews and workshops with students, teachers, and administrators are conducted to identify their specific needs. This data is used to define the platform's core functionalities, roles, and technical requirements, ensuring that the platform is user-centric.
- **System Design and Architecture:** A system architecture is created that includes the user interface (UI), backend logic, and database structure. The design focuses on user accessibility and scalability. Data modeling ensures efficient storage and retrieval of examination-related information.
- **Iterative Development and Testing:** The project is divided into smaller, manageable components known as user stories. These stories are developed in short sprints (1-2 weeks) and undergo thorough testing at each stage. Unit testing and integration testing ensure that the system functions as intended. User Acceptance Testing (UAT) is also conducted to collect feedback and make necessary improvements.
- **Deployment and Maintenance:** Once development is complete, the platform is deployed on a secure server. A Continuous Integration/Continuous Deployment (CI/CD) pipeline is set up to automate the release of new features and updates.

5 Hardware Requirements:-

- **Computers/Laptops:**
  Devices with an Intel i5 processor or higher, at least 4GB of RAM, and 512GB of storage are required to run the platform without performance issues. These specifications ensure smooth operation during exams with multiple users.
- **Internet Connection:** A stable internet connection is essential for accessing the platform, administering exams, and submitting responses. The recommended bandwidth is at least 10 Mbps for both upload and download speeds.

- **Input Devices:** Standard input devices, such as keyboards, mice, or touchpads, are necessary for interacting with the platform during exam creation and completion.

6 Software Requirements :-

- **Programming Languages:**
  The platform will be developed using Java for its robustness and cross-platform.
- **Frontend Technologies:**
  The user interface will be built using **HTML**, **CSS**, and **JavaScript** to ensure an intuitive and responsive experience for users across different devices.
- **Backend Frameworks:**
  Servlet and **JSP** (Java Server Pages) will be used to handle server-side operations and business logic.
- **Database Management:**
  MySQL will be used to manage exam data, including question banks, user information, and test results. MySQL's efficiency in handling relational databases makes it suitable for this project.
- **DevOps Tools:**
  Tomcat Server will be used for hosting the platform, while **JUnit** will handle automated testing.

FUNCTIONAL REQUIREMENTS

The development of an online examination system necessitates clear functional requirements to ensure that the platform meets the needs of all stakeholders—administrators, instructors, and students. These functional requirements outline the expected behavior of the system and define how it should operate in various scenarios.

1 User Management :-

- **Admin Functions:**
  - **User Role Assignment:** The system must allow administrators to manage user accounts.
  - **User Monitoring:** Administrators should have access to logs detailing user activities, such as login times, exam participation, and results, to maintain oversight and identify potential issues.
  - **Account Management:** Admins can update or delete user accounts as needed. They should also have the ability to reset passwords and modify user details.
  - **System Configuration:** The admin should be able to configure system settings, including security protocols, exam formats, and scheduling parameters.
  - **Exam Creation and Management:** Admin must be able to create, modify, and delete exams. This includes defining exam parameters, such as duration, question types (MCQs, descriptive), etc.
  - **Result Analysis:** Instructors should be able to generate and analyze reports on student performance, identifying trends and areas for improvement. They should have the ability to export these reports in various formats for further analysis.
- **Student Functions:**
  - **Account Access and Management:** Students must be able to create their accounts, log in securely, and manage their profile information, including password changes and personal details.
  - **Exam Participation:** The system should enable students to view upcoming exams, take exams within the designated time frame, and submit their answers electronically. They should also receive notifications about exam schedules and deadlines.
  - **Performance Feedback:** After completing an exam, students should be able to view their results, including scores, feedback on individual questions, and overall performance analysis. This information will help them understand their strengths and weaknesses.

2 Exam Management :-

- **Exam Creation:**
  - The system must provide an intuitive interface for instructors to create exams. This should include options for adding various types of questions (MCQs, fill-in-the-blank, essay questions) and setting time limits for each exam.
  - Instructors should be able to set specific instructions for each exam, such as permitted materials and conduct rules during the exam.
- **Exam Scheduling:**
  - The system should allow teachers to schedule exams at specific times and dates, with the ability to define the duration for each exam.
  - Students must receive notifications of upcoming exams, including date, time, and any changes made by instructors.
- **Secure Exam Administration:**
  - To ensure the integrity of the exam process, the system must implement security measures such as browser lockdown to prevent students from accessing unauthorized resources during the exam.
  - Proctoring features, whether live or automated, should be available to monitor students during the exam. This can include webcam monitoring, screen recording, and AI analysis to detect suspicious behavior.
- **Grading and Result Generation:**
  - The platform must support automated grading for objective-type questions, generating results immediately after submission.
  - For subjective questions, instructors should have the ability to review and manually grade submissions, with the system providing a clear interface for feedback and comments.

3 Database Management :-

A robust database management system (DBMS) is essential for storing, retrieving, and managing all data related to users, exams, and results. The database must ensure data integrity, security, and efficient access.

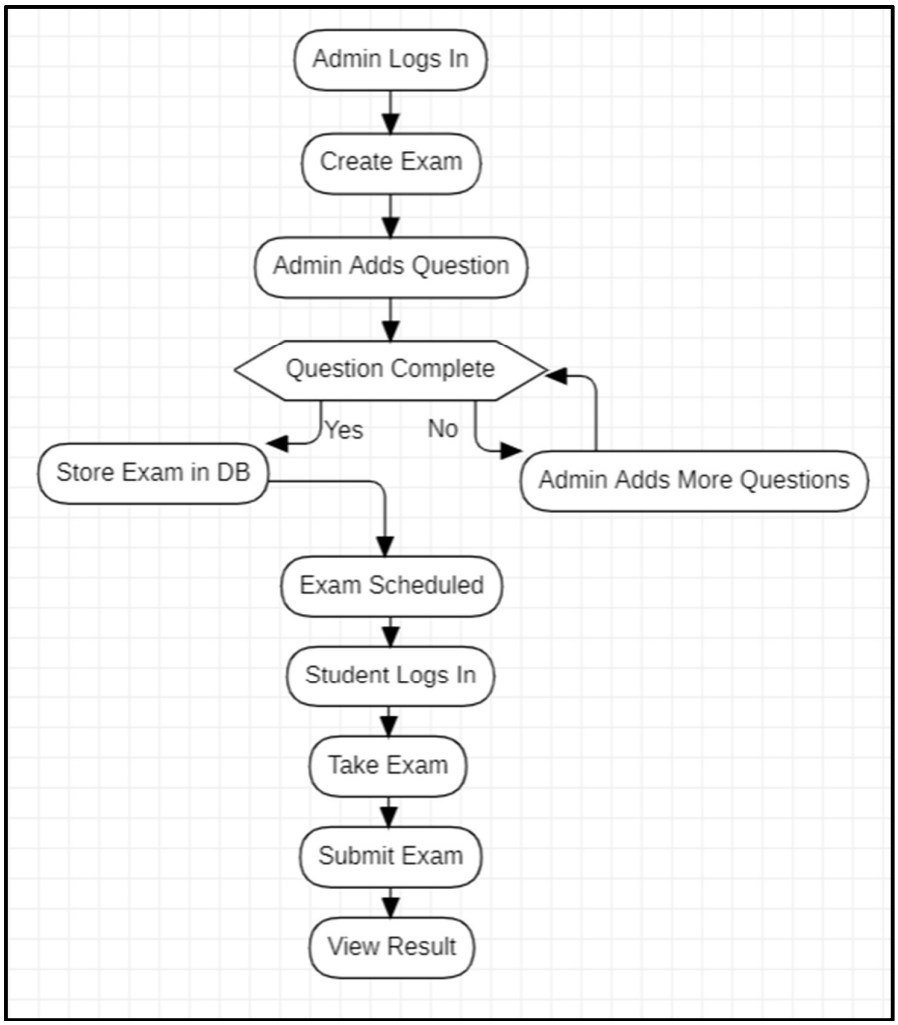
- **User Data Management:**
  - The DBMS must securely store user information, including personal details, login credentials (hashed and encrypted), and role assignments.
  - The system should support user authentication and authorization, ensuring that users can only access features and data relevant to their roles.
- **Exam Data Management:**
  - The database must maintain a comprehensive repository of exams, including metadata such as exam titles, descriptions, schedules, and associated question banks.
  - It should facilitate easy retrieval of exam data for analysis, allowing instructors and administrators to generate reports on exam performance and participation rates.
- **Results Management:**
  - The DBMS should securely store exam results, linking them to individual student profiles for easy access. This data must be protected and only accessible to authorized users (instructors and admins).

- The system should support the generation of performance reports for both students and instructors, summarizing exam results, overall performance metrics, and historical data for trend analysis.

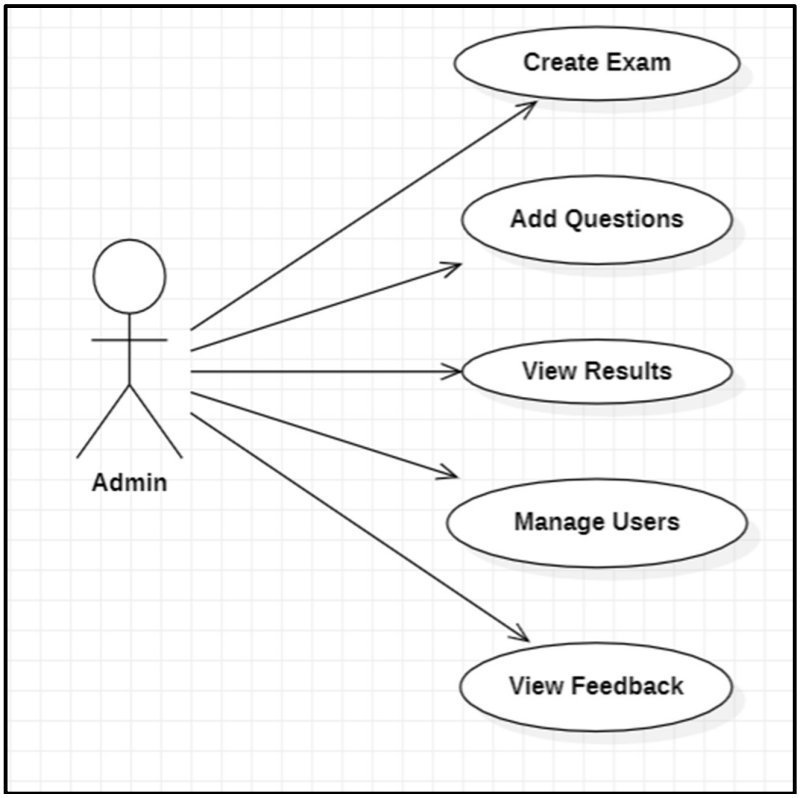
- **Backup and Recovery:**
  - The database management system must implement regular backup procedures to prevent data loss. The recovery protocols should ensure that data can be restored quickly in the event of a system failure or data corruption.

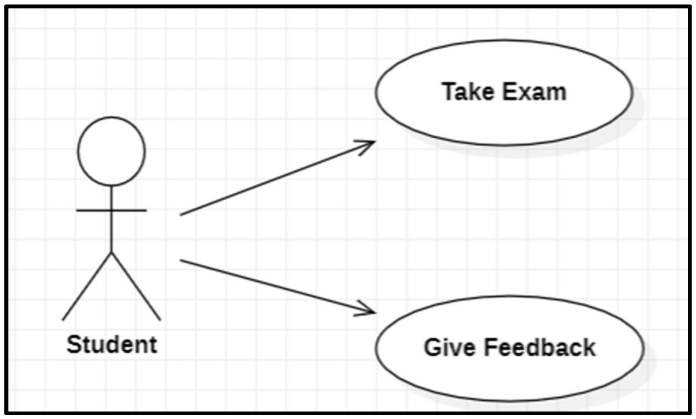
DIAGRAMS

1. **System Flow Chart**

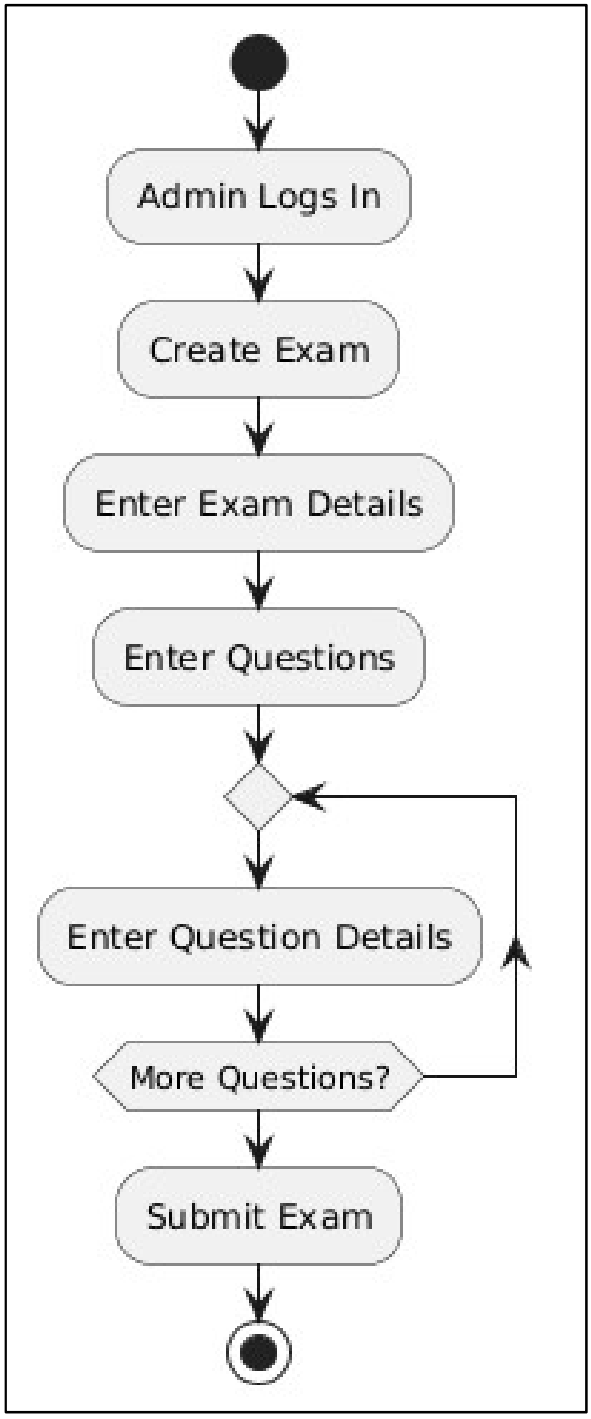


2. Use Case Diagram

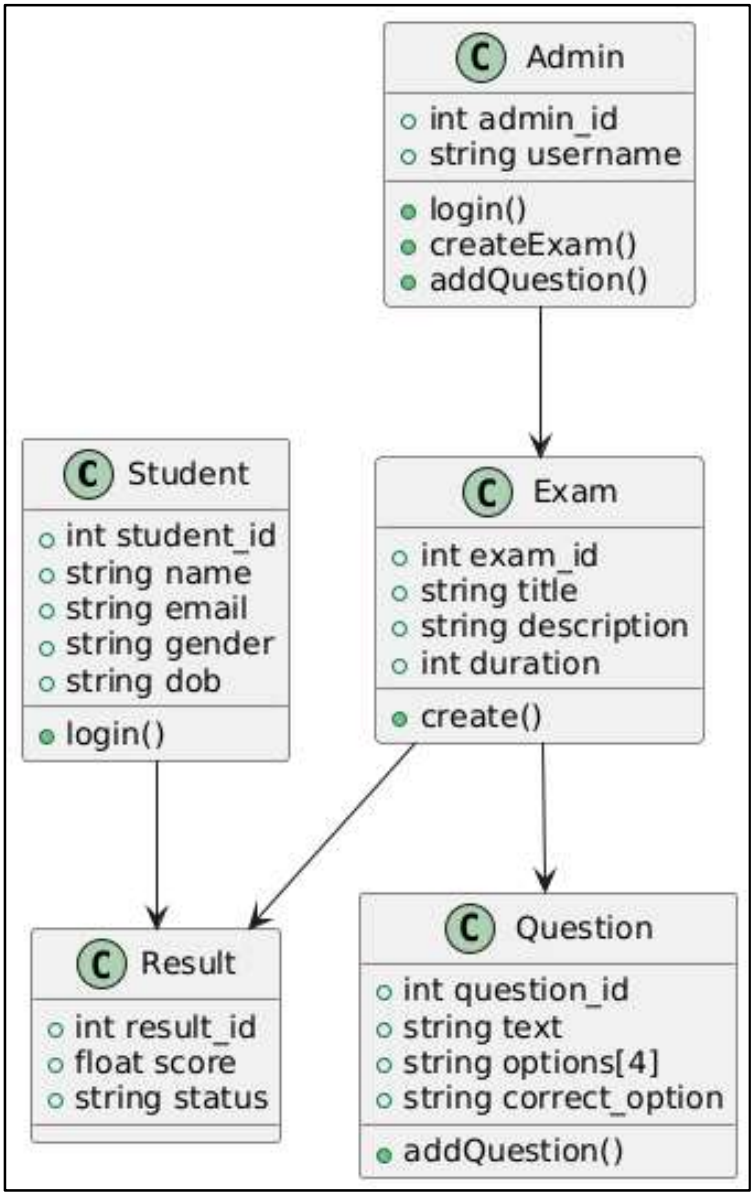




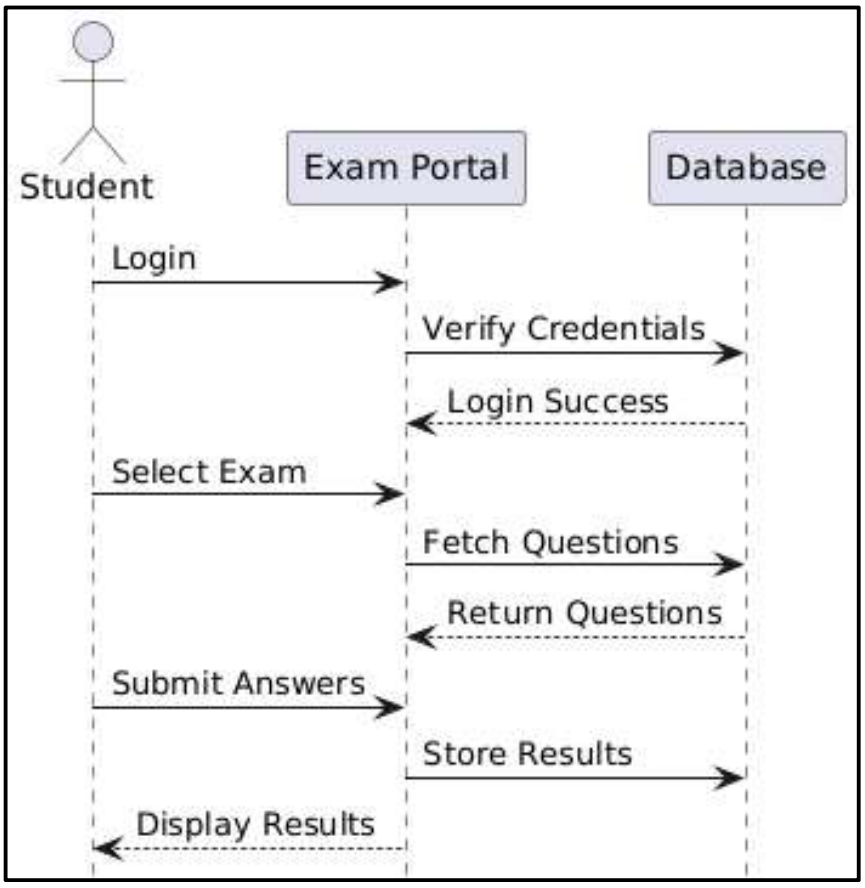
3. Activity Diagram



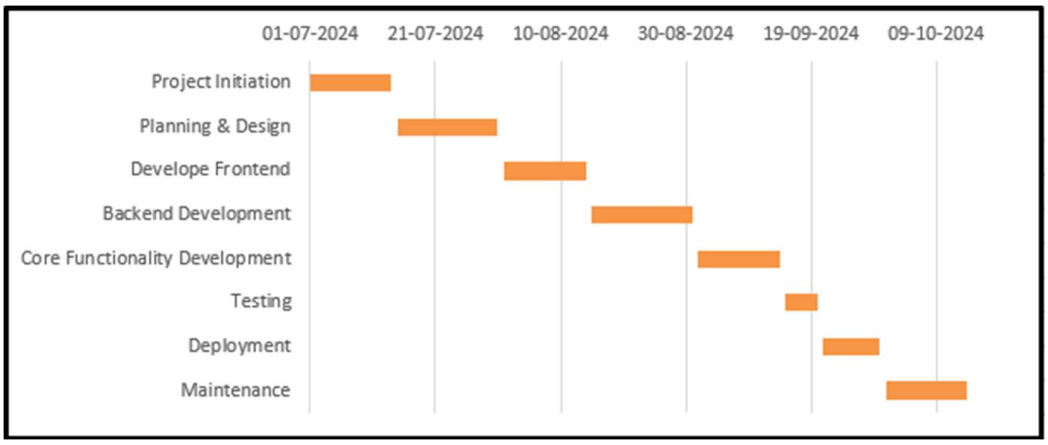
4. Class Diagram



5. **Sequence Diagram**



6. Gantt Chart



PROGRAM CODE

- **index.html (Home Page of Website)**

Code:

```
<!DOCTYPE html>
<html lang='en'>
<head>
   <meta charset='UTF-8'>
   <meta name='viewport' content='width=device-width, initial-scale=1.0'>
   <meta http-equiv='X-UA-Compatible' content='ie=edge'>
   <title>ON-EX - Home</title>
   <link rel='shortcut icon' href='Images/LOGO.png' type='image/x-icon'>
   <link rel='apple-touch-icon' href='Images/LOGO.png'>
   <link rel='preconnect' href='https://fonts.gstatic.com'>
   <link
href='https://fonts.googleapis.com/css?family=Fredericka+the+Great|Josefin+San
s:400,400i,500,500i,600,600i,700,700i&display=swap' rel='stylesheet'>
   <link rel='stylesheet' href='Css/bootstrap.css'>
   <link rel='stylesheet' href='Css/odometer.css'>
   <link rel='stylesheet' href='Css/lightcase.css'>
   <link rel='stylesheet' href='Css/swiper.css'>
   <link rel='stylesheet' href='Css/line.css'>
   <link rel='stylesheet' href='Css/animate.css'>
   <link rel='stylesheet' href='Css/style.css'>
   <link rel='stylesheet' href='Css/basic.css' >
   <script src='https://kit.fontawesome.com/fdf05f1614.js'
crossorigin='anonymous'></script>
</head>
<body>
   <div id='fb-root'></div><script async defer crossorigin='anonymous'
src='https://connect.facebook.net/en_GB/sdk.js#xfbml=1&version=v4.0&appId=7
13047905830100&autoLogAppEvents=1'></script>
   <div class='preloader'>
      <div class='loader book'>
         <figure class='page'></figure>
         <figure class='page'></figure>
         <figure class='page'></figure>
      </div>
   </div>
   <header class='header-section'>
      <div class='header'>
         <div class='header-bottom-area'>
            <div class='container'>
               <div class='header-menu-content'>
                  <nav class='navbar navbar-expand-lg p-0'>
                     <a class='site-logo site-title' href='/OnlineExam'><img
src='Images/LOGO.png' alt='site-logo'></a>
```

```
                    <button class='navbar-toggler ml-auto' type='button' data-
toggle='collapse'
                      data-target='#navbarSupportedContent' aria-
controls='navbarSupportedContent'
                      aria-expanded='false' aria-label='Toggle navigation'>
                      <span class='fas fa-bars'></span>
                    </button>
                    <div class='collapse navbar-collapse'
id='navbarSupportedContent'>
                      <ul class='navbar-nav main-menu ml-auto mr-auto'>
                        <li><a href='#home-section'>Home</a></li>
                        <li><a href='#subject-section'>Courses</a></li>
                        <li><a href='#process-section'>About</a></li>
                        <li><a href='#faq-section'>Faq</a></li>
                        <li><a href='#contact-section'>Contact</a></li>
                      </ul>
                      <div class='header-action'>
                        <a href='Registartion' class='btn--
base'><span>Register</span></a>
                        <a href='Login' class='btn--base
active'><span>Login</span></a>
                      </div>
                    </div>
                  </nav>
                </div>
              </div>
            </div>
          </div>
        </header>

        <section id='home-section' class='banner-section bg-overlay-white bg_img'
data-background='Images/bg.jpg'>
          <div class='container'>
            <div class='row justify-content-center align-items-center mb-30-none'>
              <div class='col-xl-10 text-center mb-30'>
                <div class='banner-content'>
                  <h1 class='title text--base'>The Most Popular Online Exam
Site</h1>
                  <h3 class='sub-title text-white'>We Will Open The World Of
Knowledge For You !</h3>
                  <div class='banner-btn'>
                    <a href='AdminLogin' class='btn--base'>Admin Login</a>
                    <a href='Login' class='btn--base active'>Student Login</a>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </section>
        <section class='feature-section'>
```

```
        <div class='container'>
            <div class='feature-area'>
                <div class='row'>
                    <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 p-0'>
                        <div class='feature-item'>
                            <h2 class='title text-white'>Get Support</h2>
                            <h3 class='sub-title text--base'>Support &amp;amp; Service</h3>
                            <p class='text-white'>Our Support team are at your service. If
need anything, request or support! We are available for You!</p>
                        </div>
                    </div>
                    <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 p-0'>
                        <div class='feature-item'>
                            <h2 class='title text-white'>Attend Your Examination</h2>
                            <h3 class='sub-title text--base'>Subject &amp;amp; Resources</h3>
                            <p class='text-white'>Choose your subject and question bank
with lots of questions. And get ready to give the exam. The random questions you
you will get in your exams.</p>
                        </div>
                    </div>
                    <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 p-0'>
                        <div class='feature-item'>
                            <h2 class='title text-white'>Join Us</h2>
                            <h3 class='sub-title text--base'>Welcome to the ON-EX</h3>
                            <p class='text-white'>Join to our ON-EX community. Get the
latest update and Our support team are always by your side and provide you with
the best possible support!</p>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </section>
    <section id='subject-section' class='subject-section ptb-80'>
        <div class='container'>
            <div class='row justify-content-center'>
                <div class='col-xl-7 text-center'>
                    <div class='section-header'>
                        <h2 class='section-title'>Our Most Popular Subjects</h2>
                        <span class='title-border'><i class='text--base h2 fa-solid fa-book-
open-reader'></i></span>
                    </div>
                </div>
            </div>
            <div class='row justify-content-center mb-30-none'>
                <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                    <div class='subject-item section--bg'>
                        <div class='subject-content'>
                            <h3 class='title'>Operating Systems</h3>
```

```
                    <p>Operating System is a idea of how a device runs by system
software, Operating Sy...</p>
                    <div class='subject-btn'>
                       <a data-toggle='tooltip' title='Operating System is a idea of
how a device runs by system software, Operating System is a idea of how a device
runs by system software,Operating System is a idea of how a device runs by syste'
href='Student?choice=Exam' class='custom-btn'>Exams <i class='fa-solid fa-
angles-right'></i></a>
                    </div>
                 </div>
              </div>
              </div>
                 <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                    <div class='subject-item section--bg'>
                       <div class='subject-content'>
                          <h3 class='title'>Historical Logic</h3>
                          <p>History is the scientific study of the past. Events
occurring before the inventi...</p>
                          <div class='subject-btn'>
                             <a data-toggle='tooltip' title='History is the scientific
study of the past. Events occurring before the invention of writing ... and artistic
dimensions, to theology and liturgy. This subject studies religions from all reg'
href='Student?choice=Exam' class='custom-btn'>Exams <i class='fa-solid fa-
angles-right'></i></a>
                          </div>
                       </div>
                    </div>
                 </div>
                 <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                    <div class='subject-item section--bg'>
                       <div class='subject-content'>
                          <h3 class='title'>Mathematics logic</h3>
                          <p>Mathematical logic, also called formal logic, is a
subfield of mathematics explo...</p>
                          <div class='subject-btn'>
                             <a data-toggle='tooltip' title='Mathematical logic, also
called formal logic, is a subfield of mathematics exploring the applications of
formal logic to mathematics. ... Mathematical logic is often divided into the fields
o' href='Student?choice=Exam' class='custom-btn'>Exams <i class='fa-solid fa-
angles-right'></i></a>
                          </div>
                       </div>
                    </div>
                 </div>
                 <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                    <div class='subject-item section--bg'>
                       <div class='subject-content'>
                          <h3 class='title'>Quiz</h3>
                          <p>This the systematic study of valid rules of inference, i.e.
the relations that l...</p>
```

```
                        <div class='subject-btn'>
                            <a data-toggle='tooltip' title='This the systematic study of valid rules of inference, i.e. the relations that lead to the acceptance of one proposition (the conclusion) on the basis of a set of other propositions (premises' href='Student?choice=Exam' class='custom-btn'>Exams <i class='fa-solid fa-angles-right'></i></a>
                        </div>
                    </div>
                </div>
            </div>
            <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                <div class='subject-item section--bg'>
                    <div class='subject-content'>
                        <h3 class='title'>Basic General Knowledge</h3>
                        <p>General knowledge is information that has been accumulated over time through var...</p>
                        <div class='subject-btn'>
                            <a data-toggle='tooltip' title='General knowledge is information that has been accumulated over time through various mediums. It excludes specialized learning that can only be obtained with extensive training and informatio' href='Student?choice=Exam' class='custom-btn'>Exams <i class='fa-solid fa-angles-right'></i></a>
                        </div>
                    </div>
                </div>
            </div>
            <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                <div class='subject-item section--bg'>
                    <div class='subject-content'>
                        <h3 class='title'>General Chemistry</h3>
                        <p>Chemistry is a branch of science that involves the study of the composition, str...</p>
                        <div class='subject-btn'>
                            <a data-toggle='tooltip' title='Chemistry is a branch of science that involves the study of the composition, structure and properties of matter. Often known as the central science, it is a creative discipline chiefly concer' href='Student?choice=Exam' class='custom-btn'>Exams <i class='fa-solid fa-angles-right'></i></a>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>
</section>
<section class='statistics-section ptb-80 bg-overlay-white bg_img' data-background='Images/bg1.jpg'>
    <div class='container'>
        <div class='row justify-content-center'>
            <div class='col-xl-7 text-center'>
```

```
                <div class='section-header white'>
                    <h2 class='section-title'>Our Achievements</h2>
                    <span class='title-border'><i class='fa-solid fa-book-open-reader
text--base h2'></i></span>
                </div>
            </div>
        </div>
        <div class='statistics-area'>
            <div class='row justify-content-center mb-30-none'>
                <div class='col-xl-3 col-lg-6 col-md-6 col-sm-6 mb-30'>
                    <div class='statistics-item text-center'>
                        <div class='statistics-icon'>
                            <i class='fas fa-users'></i>
                        </div>
                        <div class='statistics-content'>
                            <div class='odo-area'>
                                <h3 class='odo-title odometer' data-odometer-
final='5000+'>0</h3>
                                <h3 class='title'>+</h3>
                            </div>
                        </div>
                        <p>Total Student</p>
                    </div>
                </div>
                    <div class='col-xl-3 col-lg-6 col-md-6 col-sm-6 mb-30'>
                        <div class='statistics-item text-center'>
                            <div class='statistics-icon'>
                                <i class='fas fa-graduation-cap'></i>
                            </div>
                            <div class='statistics-content'>
                                <div class='odo-area'>
                                    <h3 class='odo-title odometer' data-odometer-
final='2555+'>0</h3>
                                    <h3 class='title'>+</h3>
                                </div>
                            </div>
                            <p>Graduation Completed</p>
                        </div>
                    </div>
                    <div class='col-xl-3 col-lg-6 col-md-6 col-sm-6 mb-30'>
                        <div class='statistics-item text-center'>
                            <div class='statistics-icon'>
                                <i class='fa-solid fa-earth-americas'></i>
                            </div>
                            <div class='statistics-content'>
                                <div class='odo-area'>
                                    <h3 class='odo-title odometer' data-odometer-
final='5,342+'>0</h3>
                                    <h3 class='title'>+</h3>
                                </div>
```

```
                        </div>
                        <p>Global Position</p>
                    </div>
                </div>
                <div class='col-xl-3 col-lg-6 col-md-6 col-sm-6 mb-30'>
                    <div class='statistics-item text-center'>
                        <div class='statistics-icon'>
                            <i class='fas fa-book-open'></i>
                        </div>
                        <div class='statistics-content'>
                            <div class='odo-area'>
                                <h3 class='odo-title odometer' data-odometer-
final='255+'>0</h3>
                                <h3 class='title'>+</h3>
                            </div>
                        </div>
                        <p>Total Courses</p>
                    </div>
                </div>
            </div>
        </div>
    </section>
    <section id='process-section' class='process-section ptb-80'>
        <div class='container'>
            <div class='row justify-content-center'>
                <div class='col-xl-7 text-center'>
                    <div class='section-header'>
                        <h2 class='section-title'>Why ON-EX Is Best</h2>
                        <span class='title-border'><i class='fa-solid fa-book-open-reader
text--base h2'></i></span>
                    </div>
                </div>
            </div>
            <div class='process-area'>
                <div class='process-item left'>
                    <div class='process-item-inner'>
                        <div class='process-content'>
                            <h3 class='title'>Choose Your Own Category</h3>
                            <p>Choose the exam category based on your subject. This helps
you typically differentiate between subjects that are essential for studying a
particular course and subjects.</p>
                        </div>
                    </div>
                </div>
                <div class='process-item right'>
                    <div class='process-item-inner'>
                        <div class='process-content'>
                            <h3 class='title'>Select The Preferable Subject</h3>
```

```
                    <p>The aim of this to help you see things more clearly and get a
good impression of the possible options, whether you have your heart set on a
particular career path or not.</p>
                </div>
              </div>
            </div>
            <div class='process-item left'>
              <div class='process-item-inner'>
                <div class='process-content'>
                  <h3 class='title'>Attend The Examination</h3>
                  <p>On a good thing, Here you can give an online exam that is
required based on your preferable subject. This is too easy,  you just need to
register and get ready for the exam.</p>
                </div>
              </div>
            </div>
            <div class='process-item right'>
              <div class='process-item-inner'>
                <div class='process-content'>
                  <h3 class='title'>Get Your Result Fast</h3>
                  <p>After finished your examination, you can get your result
very easily. Go to your dashboard and see the result of the examination you
attend. Isn&amp;#039;t so easy!</p>
                </div>
              </div>
            </div>
          </div>
          <div class='process-area-inner'></div>
        </div>
      </section>
      <section id="faq-section" class='faq-section ptb-80'>
        <div class='container'>
          <div class='row justify-content-center'>
            <div class='col-xl-7 text-center'>
              <div class='section-header'>
                <h2 class='section-title'>Frequently Asked Question</h2>
                <span class='title-border'><i class='fa-solid fa-book-open-reader
text--base h2'></i></span>
              </div>
            </div>
          </div>
          <div class='row justify-content-center mb-30-none'>
            <div class='col-xl-6 mb-30'>
              <div class='faq-thumb bg-overlay-white'>
                <img src='Images/faq.jpg' alt='faq'>
                <div class='faq-video'>
                  <a class='video-icon' data-rel='lightcase:myCollection'
href='https://www.youtube.com/embed/shfeN07ZBJg'>
                    <i class='fa-solid fa-play'></i>
                  </a>
```

```
                </div>
              </div>
            </div>
            <div class='col-xl-6 mb-30'>
              <div class='faq-wrapper'>
                <div class='faq-item'>
                  <h3 class='faq-title'><span class='title'>How to
register?</span><span class='right-icon'></span></h3>
                  <div class='faq-content'>
                    <p>Far far away, behind the word mountains, far from the
countries Vokalia and Consonantia, there live the blind texts. </p>
                  </div>
                </div>
                <div class='faq-item'>
                  <h3 class='faq-title'><span class='title'>How to attend the
exam?</span><span class='right-icon'></span></h3>
                  <div class='faq-content'>
                    <p>Far far away, behind the word mountains, far from the
countries Vokalia and Consonantia, there live the blind texts. </p>
                  </div>
                </div>
                <div class='faq-item'>
                  <h3 class='faq-title'><span class='title'>How will I get my
result?</span><span class='right-icon'></span></h3>
                  <div class='faq-content'>
                    <p>Far far away, behind the word mountains, far from the
countries Vokalia and Consonantia, there live the blind texts. </p>
                  </div>
                </div>
                <div class='faq-item'>
                  <h3 class='faq-title'><span class='title'>How may subjects are
there?</span><span class='right-icon'></span></h3>
                  <div class='faq-content'>
                    <p>Far far away, behind the word mountains, far from the
countries Vokalia and Consonantia, there live the blind texts. </p>
                  </div>
                </div>
                <div class='faq-item'>
                  <h3 class='faq-title'><span class='title'>How I will get the
support?</span><span class='right-icon'></span></h3>
                  <div class='faq-content'>
                    <p>Far far away, behind the word mountains, far from the
countries Vokalia and Consonantia, there live the blind texts. </p>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </section>
```

```
   <section id='contact-section' class='call-to-action-section ptb-80 bg-overlay-
white bg_img' data-background='Images/bg2.jpg'>
      <div class='container'>
         <div class='row justify-content-center'>
            <div class='col-xl-7 text-center'>
               <div class='section-header white'>
                  <h2 class='section-title'>Subscribe for newsletter</h2>
                  <span class='title-border'><i class='fa-solid fa-book-open-reader
text--base h2'></i></span>
               </div>
            </div>
         </div>
         <div class='call-to-action-area'>
            <div class='row justify-content-center align-items-center'>
               <div class='col-lg-10'>
                  <div class='call-to-action-content'>
                     <form class='call-to-action-form'>
                        <div class='row justify-content-center align-items-center'>
                           <div class='col-lg-12'>
                              <form class='subscribe-form'>
                                 <div class='form-group'>
                                    <input type='email' name='email' class='form-
control email'  placeholder='Enter Your Valid Email Adress' required>
                                    <button type='button' class='submit-btn mt-0
subscribe'>subscribe</button>
                                 </div>
                              </form>
                           </div>
                        </div>
                     </form>
                  </div>
               </div>
            </div>
         </div>
      </div>
   </section>
   <section class='blog-section ptb-80'>
      <div class='container'>
         <div class='row justify-content-center'>
            <div class='col-lg-7 text-center'>
               <div class='section-header'>
                  <h2 class='section-title'>Get Every Latest News Feed</h2>
                  <span class='title-border'><i class='fa-solid fa-book-open-reader
text--base h2'></i></span>
               </div>
            </div>
         </div>
         <div class='row justify-content-center mb-30-none'>
               <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                  <div class='blog-item'>
```

```
                    <div class='blog-thumb'>
                        <img src='Images/n.jpg' alt='blog'>
                    </div>
                    <div class='blog-content d-flex flex-wrap align-items-center'>
                        <span class='blog-date'>
                            <div class='date-icon'>
                                <i class='fa-solid fa-calendar-days'></i>
                            </div>
                            <div class='date-content'>
                                20 Mar
                            </div>
                        </span>
                        <div class='blog-content-details'>
                            <h3 class='title'><a href='Login'>Top Exam Preparation
Tips</a>
                            </h3>
                            <div class='blog-btn'>
                                <a href='Login' class='custom-btn'>Read More <i
class='fa-solid fa-angles-right'></i></a>
                            </div>
                        </div>
                    </div>
                </div>
                <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                    <div class='blog-item'>
                        <div class='blog-thumb'>
                            <img src='Images/n1.jpg' alt='blog'>
                        </div>
                        <div class='blog-content d-flex flex-wrap align-items-center'>
                            <span class='blog-date'>
                                <div class='date-icon'>
                                    <i class='fa-solid fa-calendar-days'></i>
                                </div>
                                <div class='date-content'>
                                    20 Mar
                                </div>
                            </span>
                            <div class='blog-content-details'>
                                <h3 class='title'><a href='Login'>15 Exam Preparation
Tips: Key To Success</a>
                                </h3>
                                <div class='blog-btn'>
                                    <a href='Login' class='custom-btn'>Read More <i
class='fa-solid fa-angles-right'></i></a>
                                </div>
                            </div>
                        </div>
                    </div>
                </div>
```

```
                <div class='col-xl-4 col-lg-4 col-md-6 col-sm-6 mb-30'>
                  <div class='blog-item'>
                    <div class='blog-thumb'>
                      <img src='Images/n2.jpg' alt='blog'>
                    </div>
                    <div class='blog-content d-flex flex-wrap align-items-center'>
                      <span class='blog-date'>
                        <div class='date-icon'>
                          <i class='fa-solid fa-calendar-days'></i>
                        </div>
                        <div class='date-content'>
                          20 Mar
                        </div>
                      </span>
                      <div class='blog-content-details'>
                        <h3 class='title'><a href='Login'>Exam Preparation: Ten
Study Tips</a>
                        </h3>
                        <div class='blog-btn'>
                          <a href='Login' class='custom-btn'>Read More <i
class='fa-solid fa-angles-right'></i></a>
                        </div>
                      </div>
                    </div>
                  </div>
                </div>
              </div>
            </section>
            <a href='#0' class='scrollToTop'><i class='fa-solid fa-angles-up'></i></a>
            <footer class='footer-section pt-80'>
              <div class='container'>
                <div class='row justify-content-center'>
                  <div class='col-xl-10 text-center'>
                    <div class='footer-widget widget-menu'>
                      <div class='footer-logo mb-20'>
                        <a href='/OnlineExam' class='site-logo'><img
src='Images/LOGO.png' alt='logo'></a>
                      </div>
                      <p>We Will Open The World Of Knowledge For You! This is the
latest online examination system you will ever need! With our easy online exam
site, you will set up your own engaging exams that fit any kind of difficulty level
and be a learning expert.</p>
                      <div class='social-area'>
                        <ul class='footer-social'>
                          <li><a target='_blank' href='https://www.facebook.com/'><i
class='fab fa-facebook-f'></i></a></li>
                          <li><a target='_blank' href='https://www.linkedin.com/'><i
class='fab fa-linkedin-in'></i></a></li>
```

```
                        <li><a target='_blank' href='https://www.instagram.com/'><i
class='fab fa-instagram'></i></a></li>
                        <li><a target='_blank' href='https://www.twitter.com/'><i
class='fab fa-twitter'></i></a></li>
                    </ul>
                </div>
            </div>
        </div>
    </div>
    <div class='footer-bottom-area'>
        <div class='container'>
            <div class='row'>
                <div class='col-xl-12'>
                    <div class='copyright-area d-flex flex-wrap align-items-center
justify-content-between mb-10-none'>
                        <div class='copyright mb-10'>
                            <p>Copyright © 2024 All Rights reserved by ON-EX</p>
                        </div>
                        <ul class='copyright-list mb-10'>
                            <li><a href='#'>Terms and Condition</a></li>
                            <li><a href='#'>Privacy Policy</a></li>
                        </ul>
                    </div>
                </div>
            </div>
        </div>
    </div>
  </footer>
</body>
</html>
```

Output:



- **db_connection.java (Database Connection for Website)**

Code:

```
package DB;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.Statement;

public class Db_Connection {

   public Connection Con;
   public ResultSet Rst;
   public PreparedStatement Pstmt;
   public Statement Stmt;

   public Db_Connection() {
      try {
         Class.forName("com.mysql.jdbc.Driver");
         String ConDb = "jdbc:mysql://localhost:3306/onlineexams";
         String ConUsername = "root";
         String ConPassword = "895666";
         Con = DriverManager.getConnection(ConDb, ConUsername,
ConPassword);
      } catch (Exception e) {
      }
   }
}
```

- **registration.java (Student Registration Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class Registartion extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      PrintWriter out = response.getWriter();
```

```
        String email = request.getParameter("email"),
                sname = request.getParameter("name"),
                squa = request.getParameter("qua"),
                sdob = request.getParameter("dob"),
                sadd = request.getParameter("addr"),
                sgender = request.getParameter("gender"),
                scity = request.getParameter("city"),
                state = request.getParameter("state"),
                spin = request.getParameter("pin"),
                scont = request.getParameter("cont"),
                password = request.getParameter("pswd"),
                cpassword = request.getParameter("cpswd");
        HttpSession session = request.getSession();
        if (session.getAttribute("Student") != null) {
           response.sendRedirect("Student");
        }
        try {
           out.println("<!DOCTYPE html>\n"
                   + "<html lang='en'>\n"
                   + "  <head>\n"
                   + "    <meta charset='UTF-8' />\n"
                   + "    <meta name='viewport' content='width=device-width, initial-
scale=1.0' />\n"
                   + "    <meta http-equiv='X-UA-Compatible' content='ie=edge' />\n"
                   + "    <title>ExamHub - Registration</title>\n"
                   + "    <link rel='shortcut icon' href='Images/Plogo.png' type='image/x-
icon' />\n"
                   + "    <link rel='apple-touch-icon' href='Images/Plogo.png' />\n"
                   + "    <link rel='preconnect' href='https://fonts.gstatic.com' />\n"
                   + "    <link\n"
+"href='https://fonts.googleapis.com/css?family=Fredericka+the+Great|Josefin+S
ans:400,400i,500,500i,600,600i,700,700i&display=swap'\n"
                   + "      rel='stylesheet'\n"
                   + "    />\n"
                   + "    <link rel='stylesheet' href='Css/bootstrap.css' />\n"
                   + "    <link rel='stylesheet' href='Css/odometer.css' />\n"
                   + "    <link rel='stylesheet' href='Css/lightcase.css' />\n"
                   + "    <link rel='stylesheet' href='Css/swiper.css' />\n"
                   + "    <link rel='stylesheet' href='Css/line.css' />\n"
                   + "    <link rel='stylesheet' href='Css/animate.css' />\n"
                   + "    <link rel='stylesheet' href='Css/style.css' />\n"
                   + "    <link rel='stylesheet' href='Css/basic.css' />\n"
                   + "    <script\n"
                   + "      src='https://kit.fontawesome.com/fdf05f1614.js'\n"
                   + "      crossorigin='anonymous'\n"
                   + "    ></script>\n"
                   + "    <style type='text/css'>\n"
                   + "      .country-code .input-group-prepend .input-group-text {\n"
                   + "        background-color: rgba(255, 255, 255, 0.1) !important;\n"
```

```
                + "       border: 1px solid rgba(255, 255, 255, 0.2);\n"
                + "     }\n"
                + "     .country-code select {\n"
                + "       border: none;\n"
                + "       background-color: transparent;\n"
                + "       color: #fff;\n"
                + "     }\n"
                + "     .country-code select:focus {\n"
                + "       border: none;\n"
                + "       outline: none;\n"
                + "     }\n"
                + "   </style>\n"
                + "  </head>\n"
                + "  <body>\n"
                + "    <div class='preloader'>\n"
                + "      <div class='loader book'>\n"
                + "        <figure class='page'></figure>\n"
                + "        <figure class='page'></figure>\n"
                + "        <figure class='page'></figure>\n"
                + "      </div>\n"
                + "    </div>\n"
                + "\n"
                + "    <section\n"
                + "      class='account-section section--bg bg-overlay-white bg_img
pt-50 pb-30'\n"
                + "      data-background='Images/bg2.jpg'\n"
                + "    >\n"
                + "      <div class='container'>\n"
                + "        <div class='row account-area align-items-center justify-
content-center'>\n"
                + "          <div class='col-lg-8'>\n"
                + "            <div class='account-form-area'>\n"
                + "              <div class='account-logo-area text-center'>\n"
                + "                <div class='account-logo'>\n"
                + "                  <a href='/OnlineExam'\n"
                + "                    ><img src='Images/LOGO.png' alt='logo'\n"
                + "                  /></a>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "              <div class='account-header text-center'>\n"
                + "                <h2 class='title'>Register Your Account Now</h2>\n"
                + "                <h3 class='sub-title'>\n"
                + "                  Already Have An Account ?\n"
                + "                  <a href='Login'>Login Now</a>\n"
                + "                </h3>\n"
                + "              </div>\n"
                + "              <form class='account-form' method='post'
action='Registartion'>\n"
                + "                <div class='row ml-b-20'>\n"
                + "                  <div class='col-lg-6 form-group'>\n"
```

```
                + "              <label>Email Id</label>\n"
                + "              <input\n"
                + "                class='form-control form--control'\n"
                + "                type='email'\n"
                + "                name='email'\n"
                + "                required\n"
                + "              />\n"
                + "            </div>\n"
                + "            <div class='col-lg-6 form-group'>\n"
                + "              <label>Student Name</label>\n"
                + "              <input\n"
                + "                class='form-control form--control'\n"
                + "                type='text'\n"
                + "                name='name'\n"
                + "                required\n"
                + "              />\n"
                + "            </div>\n"
                + "\n"
                + "            <div class='col-lg-6 form-group'>\n"
                + "              <label>Qualification</label>\n"
                + "              <select name='qua' class='form-control form--
control'>\n"
                + "                <option value='Bsc'>Bsc</option>\n"
                + "                <option value='B.Tech'>B.Tech</option>\n"
                + "                <option value='B.COM'>B.Com</option>\n"
                + "                <option value='BBA'>BBA</option>\n"
                + "                <option value='BMS'>BMS</option>\n"
                + "                <option value='Diploma'>Diploma</option>\n"
                + "                <option value='MBA'>MBA</option>\n"
                + "                <option value='B.Voc'>B.Voc</option>\n"
                + "              </select>\n"
                + "            </div>\n"
                + "            <div class='col-lg-6 form-group'>\n"
                + "              <label>Date Of Birth</label>\n"
                + "              <input\n"
                + "                class='form-control form--control'\n"
                + "                type='date'\n"
                + "                name='dob'\n"
                + "                required\n"
                + "              />\n"
                + "            </div>\n"
                + "            <div class='col-lg-6 form-group'>\n"
                + "              <label>Address</label>\n"
                + "              <textarea\n"
                + "                rows='1'\n"
                + "                class='form-control form--control'\n"
                + "                name='addr'\n"
                + "                required\n"
                + "              ></textarea>\n"
                + "            </div>\n"
```

```
                + "              <div class='col-lg-6 form-group'>\n"
                + "                <label>Gender</label>\n"
                + "                <div class='form-check'>\n"
                + "                  <input\n"
                + "                    class='form-check-input'\n"
                + "                    type='radio'\n"
                + "                    name='gender'\n"
                + "                    value='male'\n"
                + "                    checked\n"
                + "                  />\n"
                + "                  <label class='form-check-label' for='male'> Male
</label>\n"
                + "                </div>\n"
                + "                <div class='form-check'>\n"
                + "                  <input\n"
                + "                    class='form-check-input'\n"
                + "                    type='radio'\n"
                + "                    name='gender'\n"
                + "                    value='female'\n"
                + "                  />\n"
                + "                  <label class='form-check-label' for='female'>\n"
                + "                    Female\n"
                + "                  </label>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "\n"
                + "              <div class='col-lg-6 form-group'>\n"
                + "                <label>State</label>\n"
                + "                <select name='state' class='form-control form--
control'>\n"
                + "                  <option value='Andhra Pradesh'>Andhra
Pradesh</option>\n"
                + "                  <option value='Andaman and Nicobar Islands'>\n"
                + "                    Andaman and Nicobar Islands\n"
                + "                  </option>\n"
                + "                  <option value='Arunachal Pradesh'>\n"
                + "                    Arunachal Pradesh\n"
                + "                  </option>\n"
                + "                  <option value='Assam'>Assam</option>\n"
                + "                  <option value='Bihar'>Bihar</option>\n"
                + "                  <option
value='Chandigarh'>Chandigarh</option>\n"
                + "                  <option
value='Chhattisgarh'>Chhattisgarh</option>\n"
                + "                  <option value='Dadar and Nagar Haveli'>\n"
                + "                    Dadar and Nagar Haveli\n"
                + "                  </option>\n"
                + "                  <option value='Daman and Diu'>Daman and
Diu</option>\n"
                + "                  <option value='Delhi'>Delhi</option>\n"
```

```
            + "                <option
value='Lakshadweep'>Lakshadweep</option>\n"
            + "                <option
value='Puducherry'>Puducherry</option>\n"
            + "                <option value='Goa'>Goa</option>\n"
            + "                <option value='Gujarat'>Gujarat</option>\n"
            + "                <option value='Haryana'>Haryana</option>\n"
            + "                <option value='Himachal Pradesh'>Himachal
Pradesh</option>\n"
            + "                <option value='Jammu and Kashmir'>\n"
            + "                  Jammu and Kashmir\n"
            + "                </option>\n"
            + "                <option value='Jharkhand'>Jharkhand</option>\n"
            + "                <option value='Karnataka'>Karnataka</option>\n"
            + "                <option value='Kerala'>Kerala</option>\n"
            + "                <option value='Madhya Pradesh'>Madhya
Pradesh</option>\n"
            + "                <option
value='Maharashtra'>Maharashtra</option>\n"
            + "                <option value='Manipur'>Manipur</option>\n"
            + "                <option value='Meghalaya'>Meghalaya</option>\n"
            + "                <option value='Mizoram'>Mizoram</option>\n"
            + "                <option value='Nagaland'>Nagaland</option>\n"
            + "                <option value='Odisha'>Odisha</option>\n"
            + "                <option value='Punjab'>Punjab</option>\n"
            + "                <option value='Rajasthan'>Rajasthan</option>\n"
            + "                <option value='Sikkim'>Sikkim</option>\n"
            + "                <option value='Tamil Nadu'>Tamil
Nadu</option>\n"
            + "                <option value='Telangana'>Telangana</option>\n"
            + "                <option value='Tripura'>Tripura</option>\n"
            + "                <option value='Uttar Pradesh'>Uttar
Pradesh</option>\n"
            + "                <option
value='Uttarakhand'>Uttarakhand</option>\n"
            + "                <option value='West Bengal'>West
Bengal</option>\n"
            + "              </select>\n"
            + "            </div>\n"
            + "            <div class='col-lg-6 form-group'>\n"
            + "              <label>City</label>\n"
            + "              <select name='city' class='form-control form--
control'>\n"
            + "                <option value='Agra'>Agra</option>\n" +
                               "<option
value='Ahmedabad'>Ahmedabad</option>\n" +
                               "<option
value='Ahmednagar'>Ahmednagar</option>\n" +
                               "<option value='Akola'>Akola</option>\n" +
                               "<option value='Alibag'>Alibag</option>\n" +
```

```
                                "<option value='Allahabad'>Allahabad</option>\n"
+                                "<option value='Amravati'>Amravati</option>\n"
+                                "<option
value='Aurangabad'>Aurangabad</option>\n" +
                                "<option value='Bahraich'>Bahraich</option>\n" +
                                "<option value='Bangalore'>Bangalore</option>\n"
+                                "<option value='Badlapur'>Badlapur</option>\n"
+                                "<option value='Bhiwandi'>Bhiwandi</option>\n"
+                                "<option value='Bhopal'>Bhopal</option>\n" +
                                "<option
value='Chandigarh'>Chandigarh</option>\n" +
                                "<option
value='Chandrapur'>Chandrapur</option>\n" +
                                "<option value='Chennai'>Chennai</option>\n" +
                                "<option
value='Coimbatore'>Coimbatore</option>\n" +
                                "<option value='Dehradun'>Dehradun</option>\n"
+                                "<option value='Delhi'>Delhi</option>\n" +
                                "<option value='Dombivli'>Dombivli</option>\n" +
                                "<option value='Faridabad'>Faridabad</option>\n"
+                                "<optionvalue='Ghaziabad'>Ghaziabad</option>\n"
+                             "<optionvalue='Gandhinagar'>Gandhinagar</option>\n"
+                               "<option value='Gonda'>Gonda</option>\n" +
                                "<option
value='Gorakhpur'>Gorakhpur</option>\n" +
                                "<option value='Goa'>Goa</option>\n" +
                                "<option value='Gurgaon'>Gurgaon</option>\n" +
                                "<option value='Haridwar'>Haridwar</option>\n" +
                                "<option
value='Hyderabad'>Hyderabad</option>\n" +
                                "<option value='Indore'>Indore</option>\n" +
                                "<option value='Jabalpur'>Jabalpur</option>\n" +
                                "<option value='Jaipur'>Jaipur</option>\n" +
                                "<option value='Jalgaon'>Jalgaon</option>\n" +
                                "<option value='Jodhpur'>Jodhpur</option>\n" +
                                "<option value='Kanpur'>Kanpur</option>\n" +
                                "<option value='Kalyan'>Kalyan</option>\n" +
                                "<option value='Kochi'>Kochi</option>\n" +
                                "<option value='Kolhapur'>Kolhapur</option>\n" +
                                "<option value='Kolkata'>Kolkata</option>\n" +
                                "<option value='Latur'>Latur</option>\n" +
                                "<option value='Leh'>Leh</option>\n" +
                                "<option value='Lucknow'>Lucknow</option>\n" +
                                "<option value='Madurai'>Madurai</option>\n" +
                                "<option value='Manali'>Manali</option>\n" +
                                "<option
value='Mangalore'>Mangalore</option>\n" +
                                "<option value='Meerut'>Meerut</option>\n" +
                                "<option value='Mumbai'>Mumbai</option>\n" +
                                "<option value='Mysore'>Mysore</option>\n" +
```

```
                                "<option value='Nagpur'>Nagpur</option>\n" +
                                "<option value='Nashik'>Nashik</option>\n" +
                                "<option value='Navi Mumbai'>Navi
Mumbai</option>\n" +
                                "<option value='Noida'>Noida</option>\n" +
                                "<option value='Palghar'>Palghar</option>\n" +
                                "<option value='Panvel'>Panvel</option>\n" +
                                "<option value='Patna'>Patna</option>\n" +
                                "<option
value='Pondicherry'>Pondicherry</option>\n" +
                                "<option value='Pune'>Pune</option>\n" +
                                "<option value='Rajkot'>Rajkot</option>\n" +
                                "<option value='Ratnagiri'>Ratnagiri</option>\n" +
                                "<option value='Satara'>Satara</option>\n" +
                                "<option value='Shimla'>Shimla</option>\n" +
                                "<option value='Sitapur'>Sitapur</option>\n" +
                                "<option value='Solapur'>Solapur</option>\n" +
                                "<option value='Srinagar'>Srinagar</option>\n" +
                                "<option value='Surat'>Surat</option>\n" +
                                "<option value='Thane'>Thane</option>\n" +
                                "<option
value='Thiruvananthapuram'>Thiruvananthapuram</option>\n" +
                                "<option
value='Tiruchirappalli'>Tiruchirappalli</option>\n" +
                                "<option value='Udaipur'>Udaipur</option>\n" +
                                "<option
value='Ulhasnagar'>Ulhasnagar</option>\n" +
                                "<option value='Varanasi'>Varanasi</option>\n" +
                                "<option value='Vasai-Virar'>Vasai-
Virar</option>\n" +
                                "<option value='Vadodara'>Vadodara</option>\n"
                    + "              </select>\n"
                    + "            </div>\n"
                    + "            <div class='col-lg-6 form-group'>\n"
                    + "              <label>Pin</label>\n"
                    + "              <input\n"
                    + "                class='form-control form--control'\n"
                    + "                type='text'\n"
                    + "                name='pin'\n"
                    + "                required\n"
                    + "              />\n"
                    + "            </div>\n"
                    + "            <div class='col-lg-6 form-group'>\n"
                    + "              <label>Contact</label>\n"
                    + "              <input\n"
                    + "                type='text'\n"
                    + "                name='cont'\n"
                    + "                class='form-control form--control'\n"
                    + "                required\n"
                    + "              />\n"
```

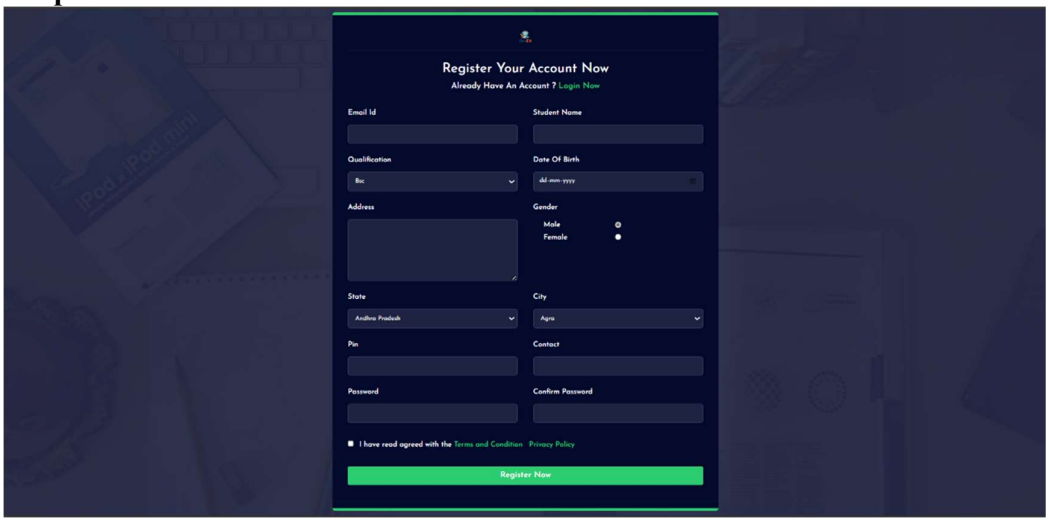
```
                + "            </div>\n"
                + "\n"
                + "            <div class='col-lg-6 form-group'>\n"
                + "              <label>Password</label>\n"
                + "              <input\n"
                + "                class='form-control form--control'\n"
                + "                type='password'\n"
                + "                name='pswd'\n"
                + "                required\n"
                + "              />\n"
                + "            </div>\n"
                + "            <div class='col-lg-6 form-group'>\n"
                + "              <label>Confirm Password</label>\n"
                + "              <input\n"
                + "                class='form-control form--control'\n"
                + "                type='password'\n"
                + "                name='cpswd'\n"
                + "                required\n"
                + "              />\n"
                + "            </div>\n"
                + "            <div class='col-lg-12 form-group'>\n"
                + "              <div\n"
                + "                class='checkbox-wrapper d-flex flex-wrap align-
items-center'\n"
                + "              >\n"
                + "                <div class='checkbox-item'>\n"
                + "                  <input type='checkbox' id='c1' name='terms'
required/>\n"
                + "                  <label for='c1'\n"
                + "                    >I have read agreed with the\n"
                + "                    <a\n"
                + "                      href='#'\n"
                + "                      class='mr-2'\n"
                + "                      >Terms and Condition</a\n"
                + "                    >\n"
                + "                    <a\n"
                + "                      href='#'\n"
                + "                      class='mr-2'\n"
                + "                      >Privacy Policy</a\n"
                + "                    >\n"
                + "                  </label>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "            </div>\n"
                + "            <div class='col-lg-12 form-group text-center'>\n"
                + "              <button type='submit' class='submit-btn'>\n"
                + "                Register Now\n"
                + "              </button>\n"
                + "            </div>\n"
                + "          </div>\n"
```

```
                + "            </form>\n"
                + "          </div>\n"
                + "        </div>\n"
                + "      </div>\n"
                + "    </div>\n"
                + "  </section>\n"
                + "  <a href='#' class='scrollToTop'><i class='fa-solid fa-angles-
up'></i></a>\n"
                + "  <script src='Js/jquery.js'></script>\n"
                + "  <script src='Js/bootstrap.js'></script>\n"
                + "  <script src='Js/swiper.js'></script>\n"
                + "  <script src='Js/viewport.js'></script>\n"
                + "  <script src='Js/odometer.js'></script>\n"
                + "  <script src='Js/lightcase.js'></script>\n"
                + "  <script src='Js/wow.js'></script>\n"
                + "  <script src='Js/main.js'></script>\n"
                + " </body>\n"
                + "</html>");

        if (email != null
                || sname != null
                || squa != null
                || sadd != null
                || sgender != null
                || scity != null
                || state != null
                || spin != null
                || password != null
                || cpassword != null) {
            if (password.equals(cpassword)) {
                Db_Connection db = new Db_Connection();
                String Query = "insert into stu_info values(?, ?, ?, ?, ?, ?, ?, ?, ?, ?,
?)";
                String Query1 = "insert into login values(?, ?, 'Student')";
                db.Pstmt = db.Con.prepareStatement(Query);
                db.Pstmt.setString(1, email);
                db.Pstmt.setString(2, sname);
                db.Pstmt.setString(3, squa);
                db.Pstmt.setString(4, sdob);
                db.Pstmt.setString(5, sgender);
                db.Pstmt.setString(6, sadd);
                db.Pstmt.setString(7, scity);
                db.Pstmt.setString(8, state);
                db.Pstmt.setString(9, spin);
                db.Pstmt.setString(10, scont);
                db.Pstmt.setString(11, password);
                int i1 = db.Pstmt.executeUpdate();
                db.Pstmt = db.Con.prepareStatement(Query1);
                db.Pstmt.setString(1, email);
                db.Pstmt.setString(2, password);
```

```
                int i2 = db.Pstmt.executeUpdate();
                if (i1 > 0 && i2 > 0) {
                    session.setAttribute("Student", sname);
                    session.setMaxInactiveInterval(5 * 60);
                    response.sendRedirect("Student?msg=You Are Successfully
Registered");
                }
            } else {
                response.sendRedirect("Registartion?msg=Password Mismatch");
            }
        }
    } catch (IOException | SQLException e) {
        System.out.println(e);
        response.sendRedirect("Registartion?msg=User Already Exist");
    }
  }
  @Override
  protected void doGet(HttpServletRequest request, HttpServletResponse
response)
          throws ServletException, IOException {
      processRequest(request, response);
  }
  @Override
  protected void doPost(HttpServletRequest request, HttpServletResponse
response)
          throws ServletException, IOException {
      processRequest(request, response);
  }
  @Override
  public String getServletInfo() {
      return "Short description";
  }
}
```

Output:



- **AdminLogin.java (Admin Login Page)**

Code:

```
package Admin;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class AdminLogin extends HttpServlet {

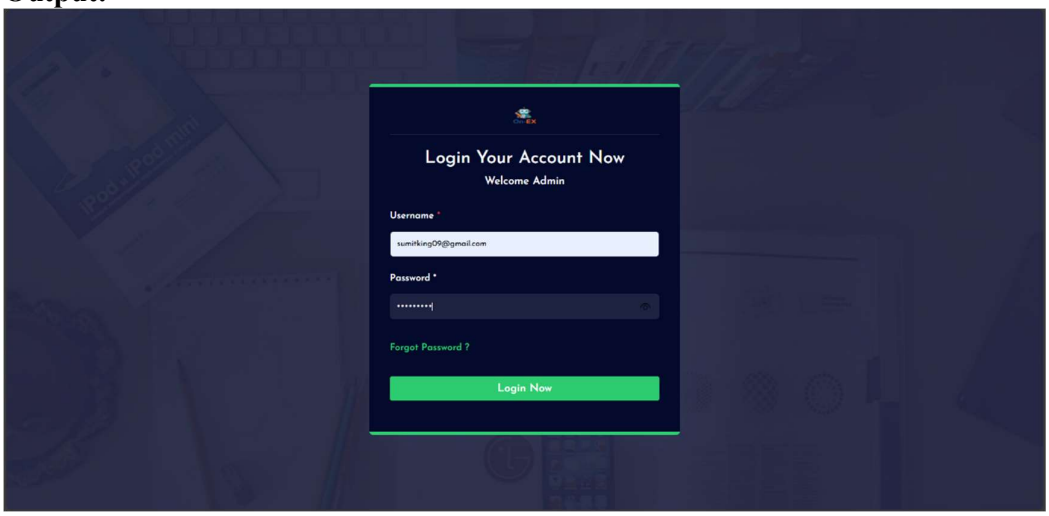
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
          throws ServletException, IOException {
       response.setContentType("text/html;charset=UTF-8");
       HttpSession session = request.getSession();
       String sname = request.getParameter("email"),
             password = request.getParameter("pswd");
       try {
          PrintWriter out = response.getWriter();
          out.println("<!DOCTYPE html>\n"
                + "<html lang='en'>\n"
                + "<head>\n"
                + "    <meta charset='UTF-8'>\n"
                + "    <meta name='viewport' content='width=device-width, initial-
scale=1.0'>\n"
                + "    <meta http-equiv='X-UA-Compatible' content='ie=edge'>\n"
                + "    <title>ExamHub - Home</title>\n"
                + "    <link rel='shortcut icon' href='Images/LOGO.png'
type='image/x-icon'>\n"
                + "    <link rel='apple-touch-icon' href='Images/LOGO.png'>    \n"
                + "    <link rel='preconnect' href='https://fonts.gstatic.com'>\n"
                + "    <link
href='https://fonts.googleapis.com/css?family=Fredericka+the+Great|Josefin+San
s:400,400i,500,500i,600,600i,700,700i&display=swap' rel='stylesheet'>\n"
                + "    <link rel='stylesheet' href='Css/bootstrap.css'>\n"
                + "    <link rel='stylesheet' href='Css/odometer.css'>\n"
                + "    <link rel='stylesheet' href='Css/lightcase.css'>\n"
                + "    <link rel='stylesheet' href='Css/swiper.css'>\n"
                + "    <link rel='stylesheet' href='Css/line.css'>\n"
                + "    <link rel='stylesheet' href='Css/animate.css'>\n"
                + "    <link rel='stylesheet' href='Css/style.css'>\n"
                + "    <link rel='stylesheet' href='Css/basic.css' >\n"
                + "    <script src='https://kit.fontawesome.com/fdf05f1614.js'
crossorigin='anonymous'></script>\n"
```

```
                + "</head>\n"
                + "<body>\n"
                + "    <div class='preloader'>\n"
                + "      <div class='loader book'>\n"
                + "        <figure class='page'></figure>\n"
                + "        <figure class='page'></figure>\n"
                + "        <figure class='page'></figure>\n"
                + "      </div>\n"
                + "    </div>\n"
                + "\n"
                + "    <section\n"
                + "      class='account-section section--bg bg-overlay-white
bg_img'\n"
                + "      data-background='Images/bg2.jpg'\n"
                + "    >\n"
                + "      <div class='container'>\n"
                + "        <div class='row account-area align-items-center justify-
content-center'>\n"
                + "          <div class='col-lg-5'>\n"
                + "            <div class='account-form-area'>\n"
                + "              <div class='account-logo-area text-center'>\n"
                + "                <div class='account-logo'>\n"
                + "                  <a href='/OnlineExam'\n"
                + "                    ><img src='Images/LOGO.png' alt='logo'
width=\"40px\"\n"
                + "                  /></a>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "              <div class='account-header text-center'>\n"
                + "                <h2 class='title'>Login Your Account Now</h2>\n"
                + "                <h3 class='sub-title'>\n"
                + "                  Welcome Admin\n"
                + "                </h3>\n"
                + "              </div>\n"
                + "              <form\n"
                + "                class='account-form'\n"
                + "                action='AdminLogin'\n"
                + "                method='POST'\n"
                + "              >\n"
                + "                <div class='row ml-b-20'>\n"
                + "                  <div class='col-lg-12 form-group'>\n"
                + "                    <label>Username <span class='text-
danger'>*</span></label>\n"
                + "                    <input\n"
                + "                      type='text'\n"
                + "                      class='form-control form--control'\n"
                + "                      name='email'\n"
                + "                      required\n"
                + "                    />\n"
                + "                  </div>\n"
```

```
                + "              <div class='col-lg-12 form-group'>\n"
                + "                <label>Password <span>*</span></label>\n"
                + "                <input\n"
                + "                  type='password'\n"
                + "                  class='form-control form--control'\n"
                + "                  name='pswd'\n"
                + "                  required\n"
                + "                />\n"
                + "              </div>\n"
                + "\n"
                + "              <div class='col-lg-12 form-group'>\n"
                + "                <div\n"
                + "                  class='checkbox-wrapper d-flex flex-wrap align-
items-center'\n"
                + "                >\n"
                + "                  <div class='checkbox-item'>\n"
                + "                    <label\n"
                + "                      ><a\n"
                + "                        href='Reset'\n"
                + "                      >\n"
                + "                        Forgot Password ?</a\n"
                + "                      ></label\n"
                + "                    >\n"
                + "                  </div>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "              <div class='col-lg-12 form-group text-center'>\n"
                + "                <button type='submit' class='submit-btn'>Login
Now</button>\n"
                + "              </div>\n"
                + "            </div>\n"
                + "          </form>\n"
                + "        </div>\n"
                + "      </div>\n"
                + "    </div>\n"
                + "  </section>\n"
                + "  <a href='#' class='scrollToTop'><i class='fa-solid fa-angles-
up'></i></a>\n"
                + "<script src='Js/jquery.js'></script>\n"
                + "<script src='Js/bootstrap.js'></script>\n"
                + "<script src='Js/swiper.js'></script>\n"
                + "<script src='Js/viewport.js'></script>\n"
                + "<script src='Js/odometer.js'></script>\n"
                + "<script src='Js/lightcase.js'></script>\n"
                + "<script src='Js/wow.js'></script>\n"
                + "<script src='Js/main.js'></script>\n"
                + "\n"
                + "  </body>\n"
                + "</html>");
```

```
            if (sname != null
                    || password != null) {
                Db_Connection db = new Db_Connection();
                String Query = "select email, password from login where email=? and
password=? and usertype='Admin'";
                db.Pstmt = db.Con.prepareStatement(Query);
                db.Pstmt.setString(1, sname);
                db.Pstmt.setString(2, password);
                db.Rst = db.Pstmt.executeQuery();
                if (db.Rst.isBeforeFirst()) {
                    session.setAttribute("Admin", sname);
                    session.setMaxInactiveInterval(5 * 60);
                    response.sendRedirect("AdminHome");
                } else {
                    response.sendRedirect("index.html?msg=Invalid UserName Or
Password");
                }
            }
        } catch (IOException | SQLException e) {
            System.out.println(e);
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



- **Login.java (Student Login Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class Login extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
          throws ServletException, IOException {
       response.setContentType("text/html;charset=UTF-8");
       String sname = request.getParameter("email"),
             password = request.getParameter("pswd");
       HttpSession session = request.getSession();
       if (session.getAttribute("Student") != null) {
          response.sendRedirect("Student");
       }
       PrintWriter out = response.getWriter();
       try {
          out.println("<!DOCTYPE html>\n"
                + "<html lang='en'>\n"
                + "  <head>\n"
                + "<meta charset='UTF-8'>\n"
                + "    <meta name='viewport' content='width=device-width, initial-
scale=1.0'>\n"
                + "    <meta http-equiv='X-UA-Compatible' content='ie=edge'>\n"
```

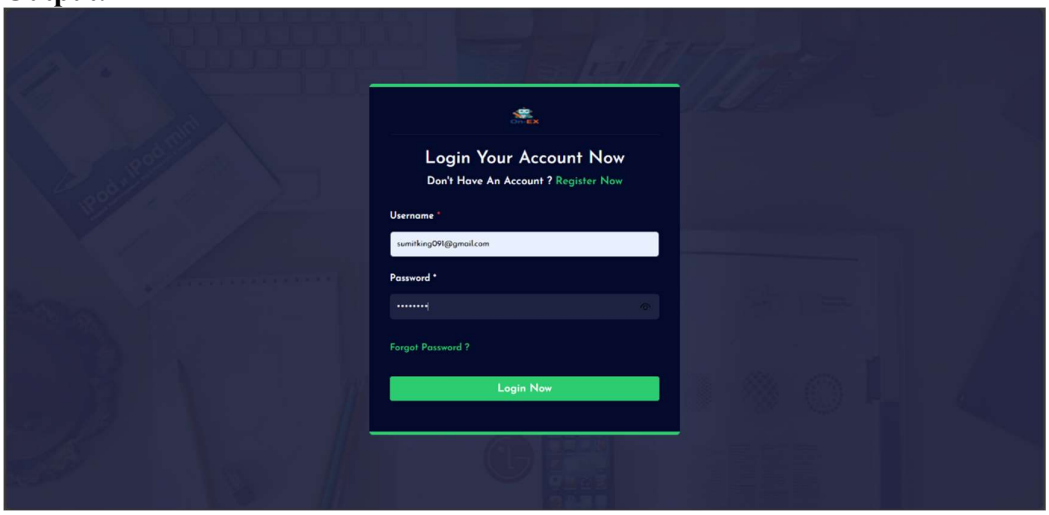
```
                + "    <title>ExamHub - Home</title>\n"
                + "    <link rel='shortcut icon' href='Images/LOGO.png'
type='image/x-icon'>\n"
                + "    <link rel='apple-touch-icon' href='Images/LOGO.png'>    \n"
                + "    <link rel='preconnect' href='https://fonts.gstatic.com'>\n"
                + "    <link
href='https://fonts.googleapis.com/css?family=Fredericka+the+Great|Josefin+San
s:400,400i,500,500i,600,600i,700,700i&display=swap' rel='stylesheet'>\n"
                + "    <link rel='stylesheet' href='Css/bootstrap.css'>\n"
                + "    <link rel='stylesheet' href='Css/odometer.css'>\n"
                + "    <link rel='stylesheet' href='Css/lightcase.css'>\n"
                + "    <link rel='stylesheet' href='Css/swiper.css'>\n"
                + "    <link rel='stylesheet' href='Css/line.css'>\n"
                + "    <link rel='stylesheet' href='Css/animate.css'>\n"
                + "    <link rel='stylesheet' href='Css/style.css'>\n"
                + "    <link rel='stylesheet' href='Css/basic.css' >\n"
                + "    <script src='https://kit.fontawesome.com/fdf05f1614.js'
crossorigin='anonymous'></script>\n"
                + "  </head>\n"
                + "  <body>\n"
                + "    <div class='preloader'>\n"
                + "      <div class='loader book'>\n"
                + "        <figure class='page'></figure>\n"
                + "        <figure class='page'></figure>\n"
                + "        <figure class='page'></figure>\n"
                + "      </div>\n"
                + "    </div>\n"
                + "\n"
                + "    <section\n"
                + "      class='account-section section--bg bg-overlay-white
bg_img'\n"
                + "      data-background='Images/bg2.jpg'\n"
                + "    >\n"
                + "      <div class='container'>\n"
                + "        <div class='row account-area align-items-center justify-
content-center'>\n"
                + "          <div class='col-lg-5'>\n"
                + "            <div class='account-form-area'>\n"
                + "              <div class='account-logo-area text-center'>\n"
                + "                <div class='account-logo'>\n"
                + "                  <a href='/OnlineExam'\n"
                + "                    ><img src='Images/LOGO.png' alt='logo'
width=\"50px\"\n"
                + "                  /></a>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "              <div class='account-header text-center'>\n"
                + "                <h2 class='title'>Login Your Account Now</h2>\n"
                + "                <h3 class='sub-title'>\n"
                + "                  Don't Have An Account ?\n"
```

```
                + "              <a href='Registartion'\n"
                + "                >Register Now</a\n"
                + "              >\n"
                + "            </h3>\n"
                + "          </div>\n"
                + "          <form\n"
                + "            class='account-form'\n"
                + "            action='Login'\n"
                + "            method='POST'\n"
                + "          >\n"
                + "            <div class='row ml-b-20'>\n"
                + "              <div class='col-lg-12 form-group'>\n"
                + "                <label>Username <span class='text-
danger'>*</span></label>\n"
                + "                <input\n"
                + "                  type='text'\n"
                + "                  class='form-control form--control'\n"
                + "                  name='email'\n"
                + "                  required\n"
                + "                />\n"
                + "              </div>\n"
                + "              <div class='col-lg-12 form-group'>\n"
                + "                <label>Password <span>*</span></label>\n"
                + "                <input\n"
                + "                  type='password'\n"
                + "                  class='form-control form--control'\n"
                + "                  name='pswd'\n"
                + "                  required\n"
                + "                />\n"
                + "              </div>\n"
                + "\n"
                + "              <div class='col-lg-12 form-group'>\n"
                + "                <div\n"
                + "                  class='checkbox-wrapper d-flex flex-wrap align-
items-center'\n"
                + "                >\n"
                + "                  <div class='checkbox-item'>\n"
                + "                    <label\n"
                + "                      ><a\n"
                + "                        href='Reset'\n"
                + "                      >\n"
                + "                        Forgot Password ?</a\n"
                + "                      ></label\n"
                + "                    >\n"
                + "                  </div>\n"
                + "                </div>\n"
                + "              </div>\n"
                + "              <div class='col-lg-12 form-group text-center'>\n"
                + "                <button type='submit' class='submit-btn'>Login
Now</button>\n"
```

```
                + "                </div>\n"
                + "              </div>\n"
                + "            </form>\n"
                + "          </div>\n"
                + "        </div>\n"
                + "      </div>\n"
                + "    </section>\n"
                + "    <a href='#' class='scrollToTop'><i class='fa-solid fa-angles-
up'></i></a>\n"
                + "<script src='Js/jquery.js'></script>\n"
                + "<script src='Js/bootstrap.js'></script>\n"
                + "<script src='Js/swiper.js'></script>\n"
                + "<script src='Js/viewport.js'></script>\n"
                + "<script src='Js/odometer.js'></script>\n"
                + "<script src='Js/lightcase.js'></script>\n"
                + "<script src='Js/wow.js'></script>\n"
                + "<script src='Js/main.js'></script>\n"
                + "\n"
                + "  </body>\n"
                + "</html>");
            if (sname != null
                || password != null) {
              Db_Connection db = new Db_Connection();
              String Query = "select email, password from login where email=? and
password=? and usertype='Student'";
              db.Pstmt = db.Con.prepareStatement(Query);
              db.Pstmt.setString(1, sname);
              db.Pstmt.setString(2, password);
              db.Rst = db.Pstmt.executeQuery();
              if (db.Rst.isBeforeFirst()) {
                session.setAttribute("Student", sname);
                session.setMaxInactiveInterval(5 * 60);
                response.sendRedirect("Student");
              } else {
                response.sendRedirect("index.html?msg=Invalid UserName Or
Password");
              }
            }
        } catch (IOException | SQLException e) {
          System.out.println(e);
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

```
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



- **AdminHome.java (Home Page For Admin)**

Code:

```
package Admin;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class AdminHome extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        HttpSession session = request.getSession();
        Object name = session.getAttribute("Admin");
        String msg = request.getParameter("msg"),
                msgtype = request.getParameter("msgtype"),
                choice = request.getParameter("choice");
```

```
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<!doctype html>\n"
                    + "<html>\n"
                    + "<head>\n"
                    + "  <meta charset='UTF-8'>\n"
                    + "  <meta name='viewport' content='width=device-width, initial-
scale=1.0'>\n"
                    + "  <script
src=\"https://unpkg.com/flowbite@1.5.3/dist/flowbite.js\"></script>\n"
                    + "<script src=\"https://cdn.tailwindcss.com\"></script>"
                    + "<script src=\"https://kit.fontawesome.com/fdf05f1614.js\"
crossorigin=\"anonymous\"></script>"
                    + "</head>\n"
                    + "<body class='bg-gray-400'>\n");
            out.println("<nav class='bg-purple-600 px-2 sm:px-20 py-2.5 rounded
dark:bg-gray-900'>\n"
                    + "  <div class='flex flex-wrap justify-between items-center mx-
auto'>\n"
                    + "  <a href='AdminHome' class='flex items-center'>\n"
                    + "      <img src='Images/LOGO.png' class='mr-3 h-6 sm:h-9'
alt='Flowbite Logo'>\n"
                    + "  </a>\n"
                    + "  <div class='flex items-center md:order-2'>\n"
                    + "      <button type='button' class='flex mr-3 text-sm bg-gray-800
rounded-full md:mr-0 focus:ring-4 focus:ring-gray-300 dark:focus:ring-gray-600'
id='user-menu-button' aria-expanded='false' data-dropdown-toggle='user-
dropdown' data-dropdown-placement='bottom'>\n"
                    + "        <span class='sr-only'>Open user menu</span>\n"
                    + "        <img class='w-8 h-8 rounded-full'
src='Images/AdminImg.png' alt='user photo'>\n"
                    + "      </button>\n"
                    + "      <!-- Dropdown menu -->\n"
                    + "      <div class='hidden z-50 my-4 text-base list-none bg-white
rounded divide-y divide-gray-100 shadow dark:bg-gray-700 dark:divide-gray-600'
id='user-dropdown' data-popper-reference-hidden='' data-popper-escaped='' data-
popper-placement='bottom' style='position: absolute; inset: 0px auto auto 0px;
margin: 0px; transform: translate3d(0px, 10.4px, 0px);'>\n"
                    + "        <div class='py-3 px-4'>\n"
                    + "          <span class='block text-sm text-gray-900 dark:text-
white'>Sumit Shahu</span>\n"
                    + "          <span class='block text-sm font-medium text-gray-500
truncate dark:text-gray-400'>"
                    + name
                    + "</span>\n"
                    + "        </div>\n"
                    + "        <ul class='py-1' aria-labelledby='user-menu-button'>\n"
                    + "          <li>\n"
```

```
                + "          <a href='AdminHome?choice=AdminProfile' class='block
py-2 px-4 text-sm text-gray-700 hover:bg-gray-100 dark:hover:bg-gray-600
dark:text-gray-200 dark:hover:text-white'>Profile</a>\n"
                + "        </li>\n"
                + "        <li>\n"
                + "          <a href='AdminHome?choice=AdminChangePassword'
class='block py-2 px-4 text-sm text-gray-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:text-gray-200 dark:hover:text-white'>Change Password</a>\n"
                + "        </li>\n"
                + "        <li>\n"
                + "          <a href='Logout' class='block py-2 px-4 text-sm text-gray-
700 hover:bg-gray-100 dark:hover:bg-gray-600 dark:text-gray-200
dark:hover:text-white'>Logout</a>\n"
                + "        </li>\n"
                + "       </ul>\n"
                + "     </div>\n"
                + "     <button data-collapse-toggle='mobile-menu-2' type='button'
class='inline-flex items-center p-2 ml-1 text-sm text-gray-500 rounded-lg
md:hidden hover:bg-gray-100 focus:outline-none focus:ring-2 focus:ring-gray-
200 dark:text-gray-400 dark:hover:bg-gray-700 dark:focus:ring-gray-600' aria-
controls='mobile-menu-2' aria-expanded='false'>\n"
                + "       <span class='sr-only'>Open main menu</span>\n"
                + "       <svg class='w-6 h-6' aria-hidden='true' fill='currentColor'
viewBox='0 0 20 20' xmlns='http://www.w3.org/2000/svg'><path fill-
rule='evenodd' d='M3 5a1 1 0 011-1h12a1 1 0 110 2H4a1 1 0 01-1-1zM3 10a1 1 0
011-1h12a1 1 0 110 2H4a1 1 0 01-1-1zM3 15a1 1 0 011-1h12a1 1 0 110 2H4a1 1
0 01-1-1z' clip-rule='evenodd'></path></svg>\n"
                + "    </button>\n"
                + "  </div>\n"
                + "  <div class='hidden justify-between items-center w-full md:flex
md:w-auto md:order-1' id='mobile-menu-2'>\n"
                + "    <ul class='flex flex-col p-4 mt-4 bg-orange-400 rounded-lg
md:flex-row md:space-x-8 md:mt-0 md:text-sm md:font-medium dark:bg-gray-
800 md:dark:bg-gray-900 '>\n"
                + "     <li>\n"
                + "       <a href='AdminHome?choice=DashBoard' class='block py-2
pr-4 pl-3 text-gray-700 rounded hover:bg-gray-100 md:hover:bg-transparent
hover:text-blue-700 md:p-0 dark:text-gray-400 md:dark:hover:text-white
dark:hover:bg-gray-700 dark:hover:text-white md:dark:hover:bg-transparent
'>Home</a>\n"
                + "     </li>\n"
                + "<li>\n"
                + "          <button id='dropdownNavbarLink' data-dropdown-
toggle='DropdownExams' class='flex justify-between items-center py-2 pr-4 pl-3
w-full font-medium text-gray-700  hover:bg-gray-50 md:hover:bg-transparent
hover:text-blue-700 md:p-0 md:w-auto dark:text-gray-400 dark:hover:text-white
dark:focus:text-white  dark:hover:bg-gray-700 md:dark:hover:bg-
transparent'>Manage Exams <svg class='ml-1 w-4 h-4' aria-hidden='true'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M5.293
```

```
7.293a1 1 0 011.414 0L10 10.586l3.293-3.293a1 1 0 111.414 1.414l-4 4a1 1 0 01-
1.414 0l-4-4a1 1 0 010-1.414z' clip-rule='evenodd'></path></svg></button>\n"
            + "          <div id='DropdownExams' class='hidden z-10 w-44 font-
normal bg-white rounded divide-y divide-gray-100 shadow dark:bg-gray-700
dark:divide-gray-600'>\n"
            + "              <ul class='py-1 text-sm text-gray-700 dark:text-gray-
400' aria-labelledby='dropdownLargeButton'>\n"
            + "                <li>\n"
            + "                  <a href='AdminHome?choice=AddExams'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Add Exams</a>\n"
            + "                </li>\n"
            + "                <li>\n"
            + "                  <a href='AdminHome?choice=UpdateExams'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Update Exams</a>\n"
            + "                </li>\n"
            + "                <li>\n"
            + "                  <a href='AdminHome?choice=DeleteExams'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Delete Exams</a>\n"
            + "                </li>\n"
            + "                <li>\n"
            + "                  <a href='AdminHome?choice=ViewExamsList'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Exams List</a>\n"
            + "                </li>\n"
            + "                  <a href='AdminHome?choice=ViewFeedback'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>View Feedback</a>\n"
            + "                </li>\n"
            + "              </ul>\n"
            + "          </div>\n"
            + "       </li>"
            + "<li>\n"
            + "          <button id='dropdownNavbarLink' data-dropdown-
toggle='DropdownStudents' class='flex justify-between items-center py-2 pr-4 pl-
3 w-full font-medium text-gray-700 hover:bg-gray-50 md:hover:bg-transparent
hover:text-blue-700 md:p-0 md:w-auto dark:text-gray-400 dark:hover:text-white
dark:focus:text-white  dark:hover:bg-gray-700 md:dark:hover:bg-
transparent'>Manage Students <svg class='ml-1 w-4 h-4' aria-hidden='true'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M5.293
7.293a1 1 0 011.414 0L10 10.586l3.293-3.293a1 1 0 111.414 1.414l-4 4a1 1 0 01-
1.414 0l-4-4a1 1 0 010-1.414z' clip-rule='evenodd'></path></svg></button>\n"
            + "          <div id='DropdownStudents' class='hidden z-10 w-44 font-
normal bg-white rounded divide-y divide-gray-100 shadow dark:bg-gray-700
dark:divide-gray-600'>\n"
            + "              <ul class='py-1 text-sm text-gray-700 dark:text-gray-
400' aria-labelledby='dropdownLargeButton'>\n"
```

```
                + "                <li>\n"
                + "                  <a href='AdminHome?choice=StudentList'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Students List</a>\n"
                + "                </li>\n"
                + "                <li>\n"
                + "                  <a href='AdminHome?choice=UpdateStudent'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Update Student</a>\n"
                + "                </li>\n"
                + "                <li>\n"
                + "                  <a href='AdminHome?choice=DeleteStudent'
class='block py-2 px-4 hover:text-blue-700 hover:bg-gray-100 dark:hover:bg-
gray-600 dark:hover:text-white'>Delete Student</a>\n"
                + "                </li>\n"
                + "              </ul>\n"
                + "            </div>\n"
                + "        </li>"
                + "      <li>\n"
                + "        <a href='AdminHome?choice=Result' class='block py-2 pr-4
pl-3 text-gray-700 rounded hover:bg-gray-100 md:hover:bg-transparent
hover:text-blue-700 md:p-0 dark:text-gray-400 md:dark:hover:text-white
dark:hover:bg-gray-700 dark:hover:text-white md:dark:hover:bg-transparent
'>Result</a>\n"
                + "      </li>\n"
                + "    </ul>\n"
                + "  </div>\n"
                + "  </div>\n"
                + "</nav>");
            if (msg != null) {
                if (msgtype != null) {
                    if (msgtype.equals("success")) {
                        out.println("<div class='relative w-full'>"
                                + "<div id='toast-success' class='absolute top-5 right-5 flex
items-center p-4 mb-4 w-full max-w-xs text-white text-bold bg-green-400
rounded-lg shadow dark:text-gray-400 dark:bg-gray-800' role='alert'>\n"
                                + "    <div class='inline-flex flex-shrink-0 justify-center items-
center w-8 h-8 text-green-500 bg-green-100 rounded-lg dark:bg-green-800
dark:text-green-200'>\n"
                                + "        <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M16.707
5.293a1 1 0 010 1.414l-8 8a1 1 0 01-1.414 0l-4-4a1 1 0 011.414-1.414L8
12.586l7.293-7.293a1 1 0 011.414 0z' clip-rule='evenodd'></path></svg>\n"
                                + "        <span class='sr-only'>Check icon</span>\n"
                                + "    </div>\n"
                                + "    <div class='ml-3 text-sm font-normal'>"
                                + msg
                                + "</div>\n"
```

```
                        + "    <button type='button' class='ml-auto -mx-1.5 -my-1.5
bg-white text-gray-400 hover:text-gray-900 rounded-lg focus:ring-2 focus:ring-
gray-300 p-1.5 hover:bg-gray-100 inline-flex h-8 w-8 dark:text-gray-500
dark:hover:text-white dark:bg-gray-800 dark:hover:bg-gray-700' data-dismiss-
target='#toast-success' aria-label='Close'>\n"
                        + "        <span class='sr-only'>Close</span>\n"
                        + "        <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z' clip-
rule='evenodd'></path></svg>\n"
                        + "    </button>\n"
                        + "</div>"
                        + "</div>");
                } else if (msgtype.equals("error")) {
                    out.println("<div class='relative w-full'><div id='toast-warning'
class='absolute top-5 right-5 flex items-center p-4 w-full max-w-xs text-white bg-
red-600 rounded-lg shadow dark:text-gray-400 dark:bg-gray-800' role='alert'>\n"
                        + "    <div class='inline-flex flex-shrink-0 justify-center items-
center w-8 h-8 text-orange-500 bg-red-100 rounded-lg dark:bg-orange-700
dark:text-orange-200'>\n"
                        + "        <svg xmlns='http://www.w3.org/2000/svg' class='w-8
h-8 text-red-600' viewBox='0 0 20 20' fill='currentColor'>\n"
                        + "      <path fill-rule='evenodd'\n"
                        + "        d='M18 10a8 8 0 11-16 0 8 8 0 0116 0zm-7 4a1 1 0
11-2 0 1 1 0 012 0zm-1-9a1 1 0 00-1 1v4a1 1 0 102 0V6a1 1 0 00-1-1z'\n"
                        + "        clip-rule='evenodd' />\n"
                        + "    </svg>\n"
                        + "<span class='sr-only'>Warning icon</span>\n"
                        + "    </div>\n"
                        + "    <div class='ml-3 text-sm font-normal'>"
                        + msg
                        + "</div>\n"
                        + "    <button type='button' class='ml-auto -mx-1.5 -my-1.5
bg-white text-gray-400 hover:text-gray-900 rounded-lg focus:ring-2 focus:ring-
gray-300 p-1.5 hover:bg-gray-100 inline-flex h-8 w-8 dark:text-gray-500
dark:hover:text-white dark:bg-gray-800 dark:hover:bg-gray-700' data-dismiss-
target='#toast-warning' aria-label='Close'>\n"
                        + "        <span class='sr-only'>Close</span>\n"
                        + "        <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z' clip-
rule='evenodd'></path></svg>\n"
                        + "    </button>\n"
                        + "</div></div>");
```

```
            } else if (msgtype.equals("delete")) {
                out.println("<div class='relative w-full'><div id='toast-danger'
class='absolute top-5 right-5 flex items-center p-4 mb-4 w-full max-w-xs text-
white bg-pink-600 rounded-lg shadow dark:text-gray-400 dark:bg-gray-800'
role='alert'>\n"
                        + "    <div class='inline-flex flex-shrink-0 justify-center items-
center w-8 h-8 text-red-500 bg-pink-100 rounded-lg dark:bg-red-800 dark:text-
red-200'>\n"
                        + "      <svg aria-hidden=\"true\" class=\"w-5 h-5\"
fill=\"currentColor\" viewBox=\"0 0 20 20\"
xmlns=\"http://www.w3.org/2000/svg\"><path fill-rule=\"evenodd\" d=\"M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z\" clip-
rule=\"evenodd\"></path></svg>\n"
                        + "       \n"
                        + "     <path fill-rule='evenodd'\n"
                        + "       d='M18 10a8 8 0 11-16 0 8 8 0 0116 0zm-7 4a1 1 0
11-2 0 1 1 0 012 0zm-1-9a1 1 0 00-1 1v4a1 1 0 102 0V6a1 1 0 00-1-1z'\n"
                        + "       clip-rule='evenodd' />\n"
                        + "    </svg>\n"
                        + "<span class='sr-only'>Error icon</span>\n"
                        + "    </div>\n"
                        + "    <div class='ml-3 text-sm font-normal'>"
                        + msg
                        + "</div>\n"
                        + "    <button type='button' class='ml-auto -mx-1.5 -my-1.5
bg-white text-gray-400 hover:text-gray-900 rounded-lg focus:ring-2 focus:ring-
gray-300 p-1.5 hover:bg-gray-100 inline-flex h-8 w-8 dark:text-gray-500
dark:hover:text-white dark:bg-gray-800 dark:hover:bg-gray-700' data-dismiss-
target='#toast-danger' aria-label='Close'>\n"
                        + "       <span class='sr-only'>Close</span>\n"
                        + "       <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z' clip-
rule='evenodd'></path></svg>\n"
                        + "    </button>\n"
                        + "</div></div>");
            }
        }
    }
    out.println("<div class='flex justify-center'>");
    if (name != null) {

        if (choice != null) {
            switch (choice) {
                case "UpdateExams":
```

```
                    RequestDispatcher rd =
request.getRequestDispatcher("UpdateExams");
                    rd.include(request, response);
                    break;
                case "ViewExamsList":
                    RequestDispatcher rd1 =
request.getRequestDispatcher("ViewExamsList");
                    rd1.include(request, response);
                    break;
                case "AddExams":
                    RequestDispatcher rd2 =
request.getRequestDispatcher("AddExams");
                    rd2.include(request, response);
                    break;
                case "DeleteExams":
                    RequestDispatcher rd4 =
request.getRequestDispatcher("DeleteExams");
                    rd4.include(request, response);
                    break;
                case "ViewFeedback":
                    RequestDispatcher rd5 =
request.getRequestDispatcher("ViewFeedback");
                    rd5.include(request, response);
                    break;
                case "Result":
                    RequestDispatcher rd6 =
request.getRequestDispatcher("Result");
                    rd6.include(request, response);
                    break;
                case "StudentList":
                    RequestDispatcher rd7 =
request.getRequestDispatcher("StudentList");
                    rd7.include(request, response);
                    break;
                case "UpdateStudent":
                    RequestDispatcher rd8 =
request.getRequestDispatcher("UpdateStudent");
                    rd8.include(request, response);
                    break;
                case "DeleteStudent":
                    RequestDispatcher rd9 =
request.getRequestDispatcher("DeleteStudent");
                    rd9.include(request, response);
                    break;
                case "AdminProfile":
                    RequestDispatcher rd13 =
request.getRequestDispatcher("AdminProfile");
                    rd13.include(request, response);
                    break;
                case "AdminChangePassword":
```

```
                    RequestDispatcher rd10 =
request.getRequestDispatcher("AdminChangePassword");
                    rd10.include(request, response);
                    break;
                default:
                    RequestDispatcher rd3 =
request.getRequestDispatcher("AdminDashBoard");
                    rd3.include(request, response);
                }
            } else {
                response.sendRedirect("AdminHome?choice=AdminDashBoard");
            }
        } else {
            response.sendRedirect("/OnlineExam");
        }
        out.println("</div></body>\n"
                + "</html>");
    }
  }

  @Override
  protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
    processRequest(request, response);
  }
  @Override
  protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
    processRequest(request, response);
  }
  @Override
  public String getServletInfo() {
    return "Short description";
  }
}
```

- **AdminDashBoard.java (DashBoard Page For Admin)**

Code:

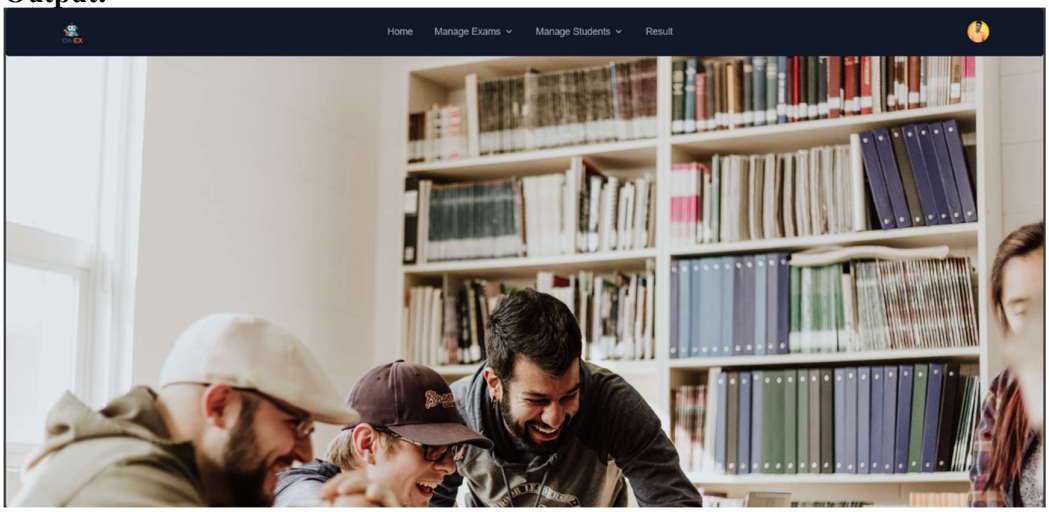
```
package Admin;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class AdminDashBoard extends HttpServlet {
```

```
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<style>\n" +
                    "    body { font-family: Arial, sans-serif; background-color: rgba(0,
0, 0, 0.329); background-image: url(Images/Student7.jpg); background-size:
cover; }\n");
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



➢ **AdminChangePassword.java (Change Password Page For Admin)**

Code:

```
package Admin;

import DB.Db_Connection;
```

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

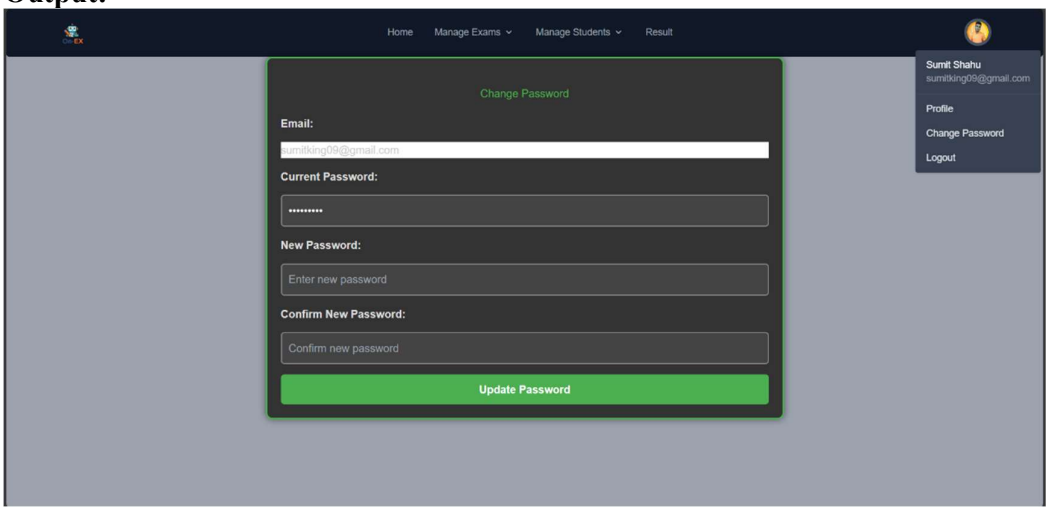
public class AdminChangePassword extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
          throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      Db_Connection db = new Db_Connection();
      try (PrintWriter out = response.getWriter()) {
         HttpSession session = request.getSession();
         String email = (String) session.getAttribute("Admin");
         String query = "SELECT password, usertype FROM login WHERE email
= ?";
         db.Pstmt = db.Con.prepareStatement(query);
         db.Pstmt.setString(1, email);
         db.Rst = db.Pstmt.executeQuery();
         out.println("<!DOCTYPE html>");
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Admin Change Password</title>");
         out.println("<style>");
         out.println("body { font-family: Arial, sans-serif; background-color:
#2e2e2e; color: #ccc; margin: 0; padding: 0; }");
         out.println("h1 { color: #4CAF50; text-align: center; padding: 20px 0; }");
         out.println(".container { width: 50%; margin: 0 auto; padding: 20px;
background-color: #333; border: 3px solid #4CAF50; border-radius: 10px; box-
shadow: 0 0 10px rgba(0, 0, 0, 0.5); }");
         out.println("form { display: flex; flex-direction: column; gap: 15px; }");
         out.println("input[type='password'], input[type='submit'] { padding: 10px;
font-size: 1em; border: 2px solid #888; border-radius: 5px; background-color:
#444; color: white; }");
         out.println("input[type='password']:focus, input[type='submit']:focus {
border-color: #4CAF50; outline: none; }");
         out.println("input[type='submit'] { background-color: #4CAF50; color:
white; border: none; cursor: pointer; font-weight: bold; }");
         out.println("input[type='submit']:hover { background-color: #45a049; }");
         out.println("label { font-weight: bold; color: #ddd; }");
         out.println("</style>");
         out.println("</head>");
         out.println("<body>");

         out.println("<div class='container'>");
         out.println("<h1>Change Password</h1>");
         if (db.Rst.next()) {
```

```
            String currentPassword = db.Rst.getString("password");
            String userType = db.Rst.getString("usertype");
            out.println("<form method='POST' action='AdminChangePassword'>");
            out.println("<label for='email'>Email:</label>");
            out.println("<input type='text' name='email' value='" + email + "'
readonly />");
            out.println("<label for='currentPassword'>Current Password:</label>");
            out.println("<input type='password' name='currentPassword' value='" +
currentPassword + "' readonly />");
            out.println("<label for='newPassword'>New Password:</label>");
            out.println("<input type='password' name='newPassword'
placeholder='Enter new password' required />");
            out.println("<label for='confirmPassword'>Confirm New
Password:</label>");
            out.println("<input type='password' name='confirmPassword'
placeholder='Confirm new password' required />");
            out.println("<input type='submit' value='Update Password' />");
            out.println("</form>");
          } else {
            out.println("<p>No user found with the given email.</p>");
          }
          out.println("</div>");
          out.println("</body>");
          out.println("</html>");
        } catch (Exception e) {
          e.printStackTrace();
        } finally {
          if (db.Pstmt != null) try { db.Pstmt.close(); } catch (Exception e) { }
          if (db.Rst != null) try { db.Rst.close(); } catch (Exception e) { }
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        String email = request.getParameter("email");
        String currentPassword = request.getParameter("currentPassword");
        String newPassword = request.getParameter("newPassword");
        String confirmPassword = request.getParameter("confirmPassword");
        if (newPassword.equals(confirmPassword)) {
          Db_Connection db = new Db_Connection();
          try {
            String updateQuery = "UPDATE login SET password = ? WHERE
email = ?";
```

```
                db.Pstmt = db.Con.prepareStatement(updateQuery);
                db.Pstmt.setString(1, newPassword);
                db.Pstmt.setString(2, email);
                int rowsUpdated = db.Pstmt.executeUpdate();
                if (rowsUpdated > 0) {
               response.sendRedirect("AdminHome?choice=AdminChangePassword");
                } else {
                   response.getWriter().println("<p>Failed to update the password.
Please try again.</p>");
                }
            } catch (Exception e) {
                e.printStackTrace();
            } finally {
                if (db.Pstmt != null) try { db.Pstmt.close(); } catch (Exception e) {  }
            }
        } else {
            response.getWriter().println("<p>Password confirmation does not match.
Please try again.</p>");
        }
    }
    @Override
    public String getServletInfo() {
        return "Servlet to allow admin to change user password";
    }
}
```

Output:



➢ **AddExams.java (Exam Creating Page)**

Code:

```
package Question;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
```

```
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class AddExams extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
          throws ServletException, IOException, SQLException {
       response.setContentType("text/html;charset=UTF-8");
       PrintWriter out = response.getWriter();
       String examname = request.getParameter("examname");
       String examdescription = request.getParameter("examdescription");
       String total_questions = request.getParameter("total_questions");
       String exam_date = request.getParameter("exam_date");
       String exam_time = request.getParameter("exam_time");
      try {
          out.println("<style>\n" +
                  "    body { font-family: Arial, sans-serif; background-color: rgba(0, 0,
0, 0.329); background-image: url(Images/Student7.jpg); background-size: cover;
}\n" +
                  "    .container { max-width: 800px; margin: auto; padding: 20px;
background: rgba(255, 255, 255, 0.8); }\n" +
                  "    .section { border: 2px solid blue; border-radius: 8px; padding:
20px; margin-bottom: 20px; }\n" +
                  "    h1, h2 {\n" +
                  "    text-align: center; \n" +
                  "    font-weight: bold; \n" +
                  "    text-decoration: underline; \n" +
                  "    font-size: 2.5em;\n" +
                  "    margin: 20px 0; \n" +
                  "    color: #333; \n" +
                  "     }\n" +
                  "    h2 {\n" +
                  "    font-size: 2em; /* Increase font size for h2 */\n" +
                  "     }\n" +
                  "    .btn { padding: 10px 15px; border: none; border-radius: 5px;
cursor: pointer; }\n" +
                  "    .btn-submit { background-color: #007BFF; color: white; }\n" +
                  "    .btn-add { background-color: #28A745; color: white; }\n" +
                  "    .space-y-4 > div { margin-bottom: 10px; }\n" +
                  "    input[type='text'], input[type='number'], input[type='date'],
input[type='time'], textarea { width: 100%; padding: 10px; border: 1px solid #ccc;
border-radius: 4px; }\n" +
                  "    .question-item { margin-bottom: 20px; border-bottom: 1px solid
#ccc; padding-bottom: 20px; }\n" +
                  "</style>");
          out.println("<div class='container'>\n" +
```

```
                "    <h1 class='text-xl font-bold'>Add Exam & Questions</h1>\n" +
                "    <form method='POST' action='AddExams' class='space-y-4'>\n"
+               "        div class='section'>\n" +
                "            <h2>Exam Details</h2>\n" +
                "            <div>\n" +
                "                <label for='examname'>Exam Name</label>\n" +
                "                <input type='text' name='examname' id='examname'
placeholder='Enter Exam Name' required=''>\n" +
                "            </div>\n" +
                "            <div>\n" +
                "                <label for='examdescription'>Exam
Description</label>\n" +
                "                <textarea name='examdescription' id='examdescription'
placeholder='Enter Exam Description' required=''></textarea>\n" +
                "            </div>\n" +
                "            <div>\n" +
                "                <label for='total_questions'>Total Questions</label>\n" +
                "                <input type='number' name='total_questions'
id='total_questions' placeholder='Enter Number of Questions' required=''>\n" +
                "            </div>\n" +
                "            <div>\n" +
                "                <label for='exam_date'>Exam Date</label>\n" +
                "                <input type='date' name='exam_date' id='exam_date'
required=''>\n" +
                "            </div>\n" +
                "            <div>\n" +
                "                <label for='exam_time'>Exam Time</label>\n" +
                "                <input type='time' name='exam_time' id='exam_time'
required=''>\n" +
                "            </div>\n" +
                "        </div>\n" +
                "        <div class='section'>\n" +
                "            <h2>Questions</h2>\n" +
                "            <div id='questions'>\n" +
                "                <div class='question-item space-y-2'>\n" +
                "                    <div>\n" +
                "                        <label>Question</label>\n" +
                "                        <input type='text' name='que[]' placeholder='Enter
Question' required=''>\n" +
                "                    </div>\n" +
                "                    <div>\n" +
                "                        <label>Option 1</label>\n" +
                "                        <input type='text' name='opt1[]' placeholder='Enter
Option 1' required=''>\n" +
                "                    </div>\n" +
                "                    <div>\n" +
                "                        <label>Option 2</label>\n" +
                "                        <input type='text' name='opt2[]' placeholder='Enter
Option 2' required=''>\n" +
                "                    </div>\n" +
```

```
                "                <div>\n" +
                "                    <label>Option 3</label>\n" +
                "                    <input type='text' name='opt3[]' placeholder='Enter
Option 3' required=">\n" +
                "                </div>\n" +
                "                <div>\n" +
                "                    <label>Option 4</label>\n" +
                "                    <input type='text' name='opt4[]' placeholder='Enter
Option 4' required=">\n" +
                "                </div>\n" +
                "                <div>\n" +
                "                    <label>Correct Option</label>\n" +
                "                    <input type='text' name='ans[]' placeholder='Enter
Correct Option' required=">\n" +
                "                </div>\n" +
                "            </div>\n" +
                "        </div>\n" +
                "        <button type='button' id='addQuestion' class='btn btn-
add'>+</button>\n" +
                "    </div>\n" +
                "    <button type='submit' class='btn btn-submit'>Submit
Exam</button>\n" +
                "  </form>\n" +
                "</div>\n" +
                "<script>\n" +
                "  document.getElementById('addQuestion').onclick = function()
{\n" +
                "    const questionSection =
document.getElementById('questions');\n" +
                "    const newQuestion = document.createElement('div');\n" +
                "    newQuestion.className = 'question-item space-y-2';\n" +
                "    newQuestion.innerHTML = `\n" +
                "        <div>\n" +
                "            <label>Question</label>\n" +
                "            <input type='text' name='que[]' placeholder='Enter
Question' required=">\n" +
                "        </div>\n" +
                "        <div>\n" +
                "            <label>Option 1</label>\n" +
                "            <input type='text' name='opt1[]' placeholder='Enter Option
1' required=">\n" +
                "        </div>\n" +
                "        <div>\n" +
                "            <label>Option 2</label>\n" +
                "            <input type='text' name='opt2[]' placeholder='Enter Option
2' required=">\n" +
                "        </div>\n" +
                "        <div>\n" +
                "            <label>Option 3</label>\n" +
```

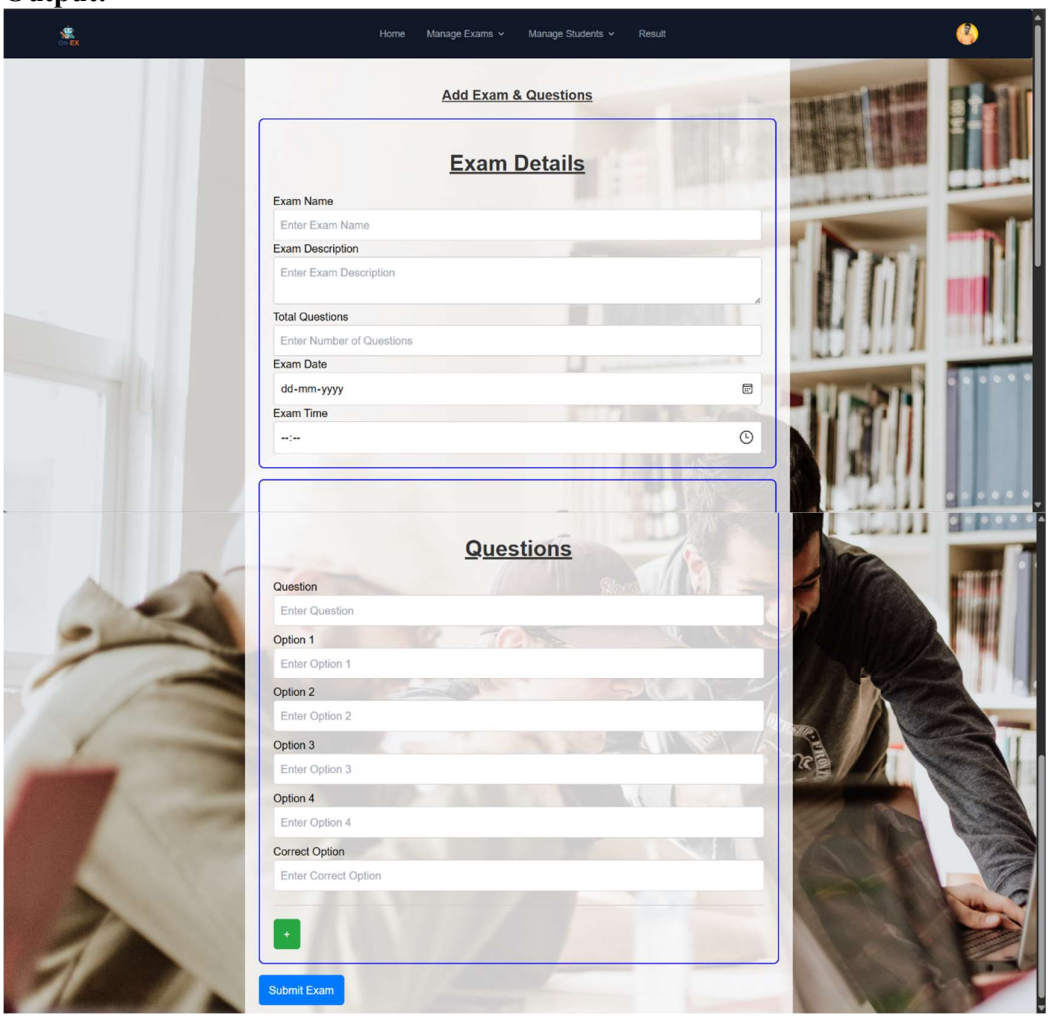
```
                "            <input type='text' name='opt3[]' placeholder='Enter Option
3' required=">\n" +
                "        </div>\n" +
                "        <div>\n" +
                "            <label>Option 4</label>\n" +
                "            <input type='text' name='opt4[]' placeholder='Enter Option
4' required=">\n" +
                "        </div>\n" +
                "        <div>\n" +
                "            <label>Correct Option</label>\n" +
                "            <input type='text' name='ans[]' placeholder='Enter Correct
Option' required=">\n" +
                "        </div>\n" +
                "    `;\n" +
                "    questionSection.appendChild(newQuestion);\n" +
                "};\n" +
                "</script>");
        if (examname == null || examdescription == null || total_questions == null
|| exam_date == null || exam_time == null) {
            out.println("<div class='alert'></div>");
            return;
        }
        Db_Connection dbconn = new Db_Connection();
        String query = "INSERT INTO exam (exam_name, exam_description,
total_questions, exam_date, exam_duration) VALUES (?,?,?,?,?)";
        dbconn.Pstmt = dbconn.Con.prepareStatement(query);
        dbconn.Pstmt.setString(1, examname);
        dbconn.Pstmt.setString(2, examdescription);
        dbconn.Pstmt.setString(3, total_questions);
        dbconn.Pstmt.setString(4, exam_date);
        dbconn.Pstmt.setString(5, exam_time);
        dbconn.Pstmt.executeUpdate();
        String examTable = examname.replace(" ", "_");
        String createTableSQL = "CREATE TABLE IF NOT EXISTS " +
examTable + " (" +
            "question_number INT AUTO_INCREMENT PRIMARY KEY, "+
            "question TEXT, " +
            "opt1 VARCHAR(255), " +
            "opt2 VARCHAR(255), " +
            "opt3 VARCHAR(255), " +
            "opt4 VARCHAR(255), " +
            "ans VARCHAR(255)" +
            ")";
        dbconn.Stmt = dbconn.Con.createStatement();
        dbconn.Stmt.executeUpdate(createTableSQL);

        String insertQuestionSQL = "INSERT INTO " + examTable + " (question,
opt1, opt2, opt3, opt4, ans) " +
                    "VALUES (?, ?, ?, ?, ?, ?)";
        dbconn.Pstmt = dbconn.Con.prepareStatement(insertQuestionSQL);
```

```
            String[] questions = request.getParameterValues("que[]");
            String[] options1 = request.getParameterValues("opt1[]");
            String[] options2 = request.getParameterValues("opt2[]");
            String[] options3 = request.getParameterValues("opt3[]");
            String[] options4 = request.getParameterValues("opt4[]");
            String[] answers = request.getParameterValues("ans[]");
            if (questions != null && options1 != null && options2 != null &&
    options3 != null && options4 != null && answers != null) {
                for (int i = 0; i < questions.length; i++) {
                    // Ensure you are setting valid values before executing the statement
                    dbconn.Pstmt.setString(1, questions[i]);
                    dbconn.Pstmt.setString(2, options1[i]);
                    dbconn.Pstmt.setString(3, options2[i]);
                    dbconn.Pstmt.setString(4, options3[i]);
                    dbconn.Pstmt.setString(5, options4[i]);
                    dbconn.Pstmt.setString(6, answers[i]);
                    dbconn.Pstmt.executeUpdate();
                }
            }
            response.sendRedirect("AdminHome?choice=ViewExamsList");
        } catch (SQLException | IOException e) {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<style>");
            out.println(".popup {");
            out.println("    display: block;");
            out.println("    position: fixed;");
            out.println("    top: 0;");
            out.println("    left: 0;");
            out.println("    width: 100%;");
            out.println("    height: 100%;");
            out.println("    background-color: rgba(0,0,0,0.5);");
            out.println("    z-index: 9999;");
            out.println("    text-align: center;");
            out.println("}");
            out.println(".popup-content {");
            out.println("    background-color: white;");
            out.println("    padding: 20px;");
            out.println("    border-radius: 8px;");
            out.println("    display: inline-block;");
            out.println("    margin-top: 200px;");
            out.println("    box-shadow: 0px 4px 6px rgba(0, 0, 0, 0.2);");
            out.println("}");
            out.println(".popup-button {");
            out.println("    background-color: #007BFF;");
            out.println("    color: white;");
            out.println("    padding: 10px 20px;");
            out.println("    border: none;");
            out.println("    border-radius: 4px;");
```

```
            out.println("    cursor: pointer;");
            out.println("}");
            out.println("</style>");
            out.println("</head>");
            out.println("<body>");
            out.println("<div id='myPopup' class='popup'>");
            out.println("  <div class='popup-content'>");
            out.println("    <h2>Error</h2>");
            out.println("    <p>" + e.getMessage() + "</p>");
            out.println("    <button class='popup-button'
onclick='redirectToAddExams()'>OK</button>");
            out.println("  </div>");
            out.println("</div>");
            out.println("<script>");
            out.println("function redirectToAddExams() {");
            out.println("document.getElementById('myPopup').style.display='none';");
            out.println("    window.location.href='AdminHome?choice=AddExams';");
            out.println("}");
            out.println("</script>");
            out.println("</body>");
            out.println("</html>");
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        try {
            processRequest(request, response);
        } catch (SQLException ex) {
            Logger.getLogger(AddExams.class.getName()).log(Level.SEVERE, null,
ex);
        }
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        try {
            processRequest(request, response);
        } catch (SQLException ex) {
            Logger.getLogger(AddExams.class.getName()).log(Level.SEVERE, null,
ex);
        }
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



➢ **UpdateExams.java (Exam Updating Page)**

Code:

```
package Question;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class UpdateExams extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
```

```
        throws ServletException, IOException, SQLException {
    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();
    String examname = request.getParameter("examname");
    try {
        if (examname == null || examname.isEmpty()) {
            out.println("<div class='md:mx-20 lg:my-36 md:my-28 sm:my-16
sm:mx-6 my-10 mx-5 w-full'>\n"
            + "    <div class='bg-white rounded-lg shadow dark:border md:mt-0
xl:p-0 dark:bg-gray-800 dark:border-gray-700'>\n"
            + "        <div class='px-6 sm:px-8 py-4'>\n"
            + "            <h1 class='text-xl text-center font-bold leading-tight tracking-
tight text-gray-900 md:text-2xl dark:text-white'>\n"
            + "                Provide Exam Name to Update\n"
            + "            </h1>\n"
            + "            <form class='space-y-2' action='UpdateExams'
method='post'>\n"
            + "                <div>\n"
            + "                    <label for='examname' class='block mb-2 text-sm font-
medium text-gray-900 dark:text-white'>Exam Name</label>\n"
            + "                    <input type='text' name='examname' id='examname'
class='bg-gray-50 border border-gray-300 text-gray-900 sm:text-sm rounded-lg
focus:ring-primary-600 focus:border-primary-600 block w-full p-2.5 dark:bg-
gray-700 dark:border-gray-600 dark:placeholder-gray-400 dark:text-white
dark:focus:ring-blue-500 dark:focus:border-blue-500' placeholder='Enter Exam
Name' required>\n"
            + "                </div>\n"
            + "                <button type='submit' class='w-full text-white bg-blue-600
hover:bg-blue-700 focus:ring-4 focus:outline-none focus:ring-blue-300 font-
medium rounded-lg text-sm px-5 py-2.5 text-center dark:bg-blue-600
dark:hover:bg-blue-700 dark:focus:ring-blue-800'>Submit</button>\n"
            + "            </form>\n"
            + "        </div>\n"
            + "    </div>\n"
            + "</div>");
            response.sendRedirect("UpdateExams?examname=" + examname);
            return;
        }
        Db_Connection dbconn = new Db_Connection();
        String examQuery = "SELECT * FROM exam WHERE exam_name = ?";
        dbconn.Pstmt = dbconn.Con.prepareStatement(examQuery);
        dbconn.Pstmt.setString(1, examname);
        ResultSet rs = dbconn.Pstmt.executeQuery();
        if (!rs.next()) {
            out.println("<div class='alert'>Exam not found.</div>");
            return;
        }
        String examTable = examname.replace(" ", "_");
        String questionQuery = "SELECT * FROM " + examTable;
        dbconn.Stmt = dbconn.Con.createStatement();
```

```
        ResultSet questionRs = dbconn.Stmt.executeQuery(questionQuery);
        out.println("<style>\n" +
                "    body { font-family: Arial, sans-serif; background-color: rgba(0, 0,
0, 0.329); background-image: url(Images/Student7.jpg); background-size: cover;
}\n" +
                "    .container { max-width: 800px; margin: auto; padding: 20px;
background: rgba(255, 255, 255, 0.8); }\n" +
                "    .section { border: 2px solid blue; border-radius: 8px; padding:
20px; margin-bottom: 20px; }\n" +
                "    h1, h2 {\n" +
                "    text-align: center;\n" +
                "    font-weight: bold;\n" +
                "    text-decoration: underline;\n" +
                "    font-size: 2.5em;\n" +
                "    margin: 20px 0;\n" +
                "    color: #333;\n" +
                "    }\n" +
                "    h2 {\n" +
                "    font-size: 2em;\n" +
                "    }\n" +
                "    .btn { padding: 10px 15px; border: none; border-radius: 5px;
cursor: pointer; }\n" +
                "    .btn-submit { background-color: #007BFF; color: white; }\n" +
                "    .btn-add { background-color: #28A745; color: white; }\n" +
                "    .space-y-4 > div { margin-bottom: 10px; }\n" +
                "    input[type='text'], input[type='number'], input[type='date'],
input[type='time'], textarea { width: 100%; padding: 10px; border: 1px solid #ccc;
border-radius: 4px; }\n" +
                "    .question-item { margin-bottom: 20px; border-bottom: 1px solid
#ccc; padding-bottom: 20px; }\n" +
                "</style>");
        out.println("<div class='container'>\n" +
                "    <h1 class='text-xl font-bold'>Update Exam & Questions</h1>\n"
+                 "    <form method='POST' action='UpdateExams' class='space-y-
4'>\n" +
                "        <div class='section'>\n" +
                "            <h2>Exam Details</h2>\n" +
                "            <div>\n" +
                "                <label for='examname'>Exam Name</label>\n" +
                "                <input type='text' name='examname' id='examname'
value='" + examname + "' readonly>\n" +
                "            </div>\n" +
                "            <div>\n" +
                "                <label for='examdescription'>Exam
Description</label>\n" +
                "                <textarea name='examdescription' id='examdescription'>"
+ rs.getString("exam_description") + "</textarea>\n" +
                "            </div>\n" +
                "            <div>\n" +
                "                <label for='total_questions'>Total Questions</label>\n" +
```

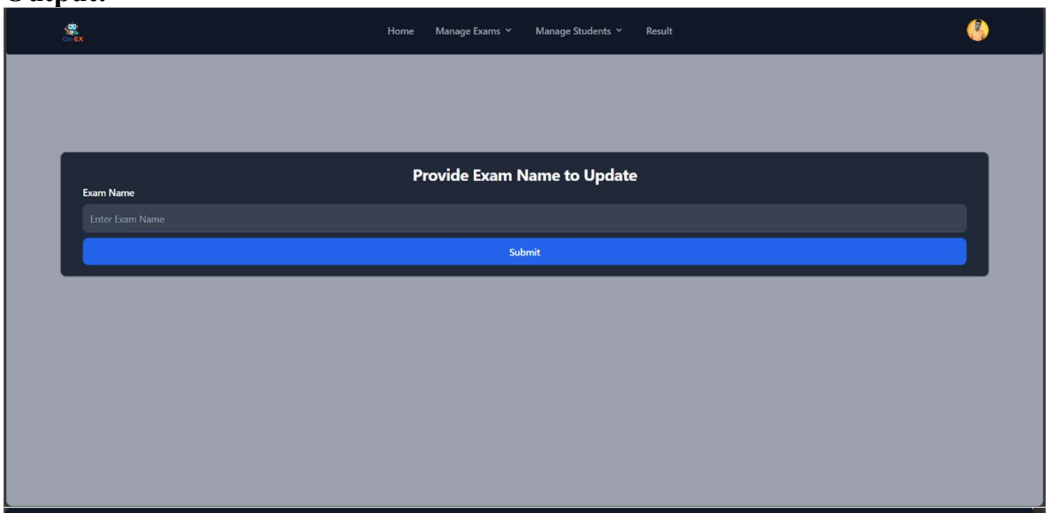
```
                "              <input type='number' name='total_questions'
id='total_questions' value='" + rs.getString("total_questions") + "' required=">\n"
+                 "          </div>\n" +
                "          <div>\n" +
                "             <label for='exam_date'>Exam Date</label>\n" +
                "             <input type='date' name='exam_date' id='exam_date'
value='" + rs.getString("exam_date") + "' required=">\n" +
                "          </div>\n" +
                "          <div>\n" +
                "             <label for='exam_time'>Exam Time</label>\n" +
                "             <input type='time' name='exam_time' id='exam_time'
value='" + rs.getString("exam_duration") + "' required=">\n" +
                "          </div>\n" +
                "       </div>\n" +
                "       <div class='section'>\n" +
                "          <h2>Questions</h2>\n");
        int questionCounter = 1;
        while (questionRs.next()) {
           out.println("<div class='question-item space-y-2'>\n" +
                "    <div>\n" +
                "       <label>Question " + questionCounter + "</label>\n" +
                "       <input type='text' name='que[]' value='" +
questionRs.getString("question") + "' required=">\n" +
                "    </div>\n" +
                "    <div>\n" +
                "       <label>Option 1</label>\n" +
                "       <input type='text' name='opt1[]' value='" +
questionRs.getString("opt1") + "' required=">\n" +
                "    </div>\n" +
                "    <div>\n" +
                "       <label>Option 2</label>\n" +
                "       <input type='text' name='opt2[]' value='" +
questionRs.getString("opt2") + "' required=">\n" +
                "    </div>\n" +
                "    <div>\n" +
                "       <label>Option 3</label>\n" +
                "       <input type='text' name='opt3[]' value='" +
questionRs.getString("opt3") + "' required=">\n" +
                "    </div>\n" +
                "    <div>\n" +
                "       <label>Option 4</label>\n" +
                "       <input type='text' name='opt4[]' value='" +
questionRs.getString("opt4") + "' required=">\n" +
                "    </div>\n" +
                "    <div>\n" +
                "       <label>Correct Option</label>\n" +
                "       <input type='text' name='ans[]' value='" +
questionRs.getString("ans") + "' required=">\n" +
                "    </div>\n" +
                "</div>");
```

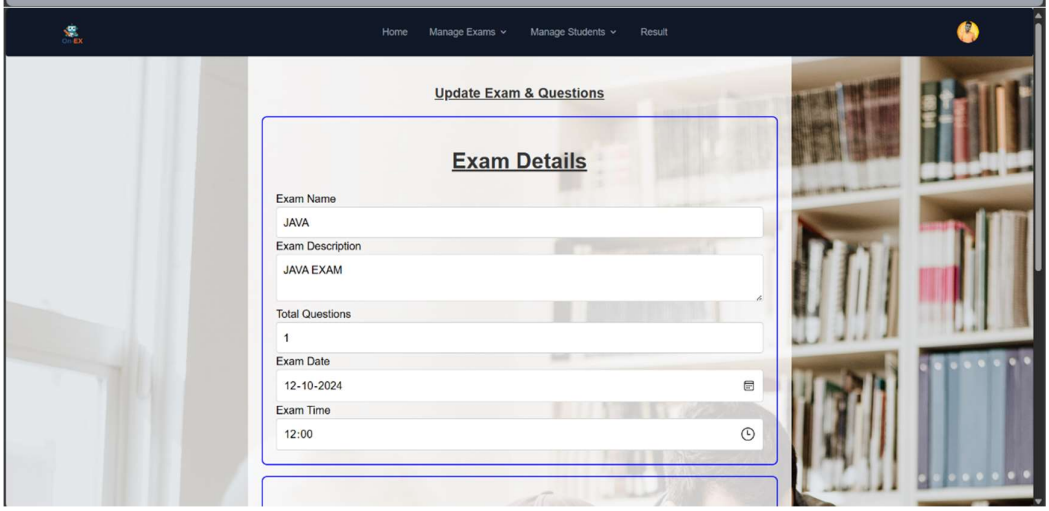
```
            questionCounter++;
        }
        out.println("        </div>\n" +
                "        <button type='button' id='addQuestion' class='btn btn-
add'>+</button>\n" +
                "        <button type='submit' class='btn btn-submit'>Update
Exam</button>\n" +
                "    </form>\n" +
                "</div>\n" +
                "<script>\n" +
                "
document.getElementById('addQuestion').addEventListener('click', function()
{\n" +
                "        const newQuestion = document.createElement('div');\n" +
                "        newQuestion.innerHTML = `<div class='question-item space-
y-2'>\n" +
                "            <div><label>Question</label><input type='text'
name='que[]' required></div>\n" +
                "            <div><label>Option 1</label><input type='text'
name='opt1[]' required></div>\n" +
                "            <div><label>Option 2</label><input type='text'
name='opt2[]' required></div>\n" +
                "            <div><label>Option 3</label><input type='text'
name='opt3[]' required></div>\n" +
                "            <div><label>Option 4</label><input type='text'
name='opt4[]' required></div>\n" +
                "            <div><label>Correct Option</label><input type='text'
name='ans[]' required></div>\n" +
                "        </div>`;\n" +
                "        document.querySelector('.section:last-
child').appendChild(newQuestion);\n" +
                "    });\n" +
                "</script>");
      } catch (Exception e) {
          Logger.getLogger(UpdateExams.class.getName()).log(Level.SEVERE,
null, e);
          out.println("<div class='alert'>Error: " + e.getMessage() + "</div>");
      }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
          throws ServletException, IOException {
        try {
          processRequest(request, response);
      } catch (SQLException ex) {
          Logger.getLogger(UpdateExams.class.getName()).log(Level.SEVERE,
null, ex);
      }
    }
```

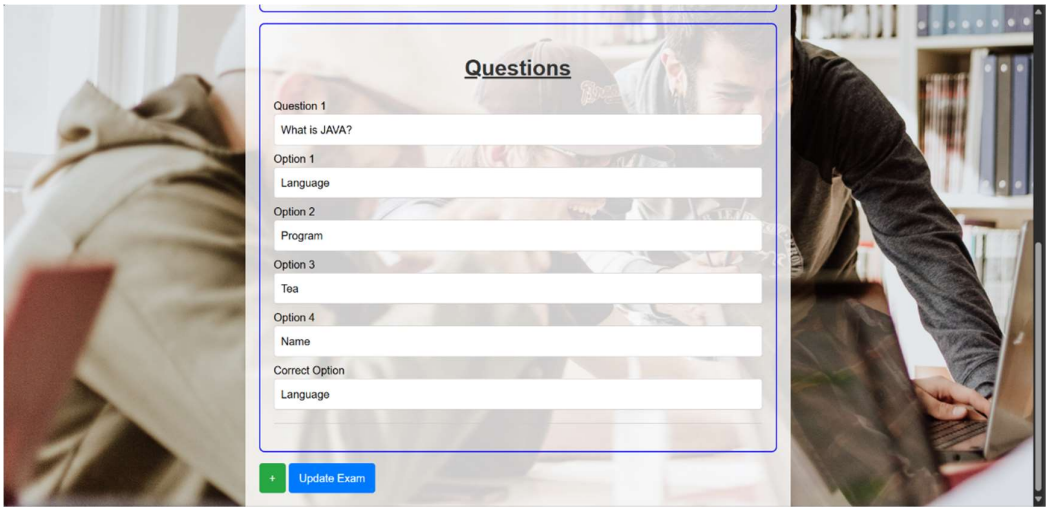
```
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        PrintWriter out = response.getWriter();
        String examname = request.getParameter("examname");
        String examdescription = request.getParameter("examdescription");
        String totalQuestions = request.getParameter("total_questions");
        String examDate = request.getParameter("exam_date");
        String examTime = request.getParameter("exam_time");
        String[] questions = request.getParameterValues("que[]");
        String[] option1 = request.getParameterValues("opt1[]");
        String[] option2 = request.getParameterValues("opt2[]");
        String[] option3 = request.getParameterValues("opt3[]");
        String[] option4 = request.getParameterValues("opt4[]");
        String[] answers = request.getParameterValues("ans[]");
        try {
            Db_Connection dbconn = new Db_Connection();
            String updateExamQuery = "UPDATE exam SET exam_description = ?,
total_questions = ?, exam_date = ?, exam_duration = ? WHERE exam_name = ?";
            dbconn.Pstmt = dbconn.Con.prepareStatement(updateExamQuery);
            dbconn.Pstmt.setString(1, examdescription);
            dbconn.Pstmt.setInt(2, Integer.parseInt(totalQuestions));
            dbconn.Pstmt.setString(3, examDate);
            dbconn.Pstmt.setString(4, examTime);
            dbconn.Pstmt.setString(5, examname);
            int examUpdateResult = dbconn.Pstmt.executeUpdate();
            if (examUpdateResult > 0) {
                out.println("<div class='alert'>Exam updated successfully.</div>");
            } else {
                out.println("<div class='alert'>Failed to update exam.</div>");
            }
            String examTable = examname.replace(" ", "_");
            dbconn.Con.setAutoCommit(false);
            String clearQuestionsQuery = "DELETE FROM " + examTable;
            dbconn.Stmt = dbconn.Con.createStatement();
            dbconn.Stmt.executeUpdate(clearQuestionsQuery);
            String insertQuestionQuery = "INSERT INTO " + examTable + "
(question, opt1, opt2, opt3, opt4, ans) VALUES (?, ?, ?, ?, ?, ?)";
            dbconn.Pstmt = dbconn.Con.prepareStatement(insertQuestionQuery);
            for (int i = 0; i < questions.length; i++) {
                dbconn.Pstmt.setString(1, questions[i]);
                dbconn.Pstmt.setString(2, option1[i]);
                dbconn.Pstmt.setString(3, option2[i]);
                dbconn.Pstmt.setString(4, option3[i]);
                dbconn.Pstmt.setString(5, option4[i]);
                dbconn.Pstmt.setString(6, answers[i]);
                dbconn.Pstmt.addBatch(); // Add the question update to the batch
            }
            int[] updateResults = dbconn.Pstmt.executeBatch();
```

```
            dbconn.Con.commit();
            out.println("<div class='alert'>Questions updated successfully.</div>");
        } catch (Exception e) {
            try {
                // Rollback the transaction in case of an error
                Db_Connection dbconn = new Db_Connection();
                dbconn.Con.rollback();
            } catch (SQLException rollbackEx) {
                Logger.getLogger(UpdateExams.class.getName()).log(Level.SEVERE,
null, rollbackEx);
            }
            Logger.getLogger(UpdateExams.class.getName()).log(Level.SEVERE,
null, e);
            out.println("<div class='alert'>Error: " + e.getMessage() + "</div>");
        }
    }
}
```

Output:







➢ **DeleteExams.java (Exam Deletion Page)**

Code:

```
package Question;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

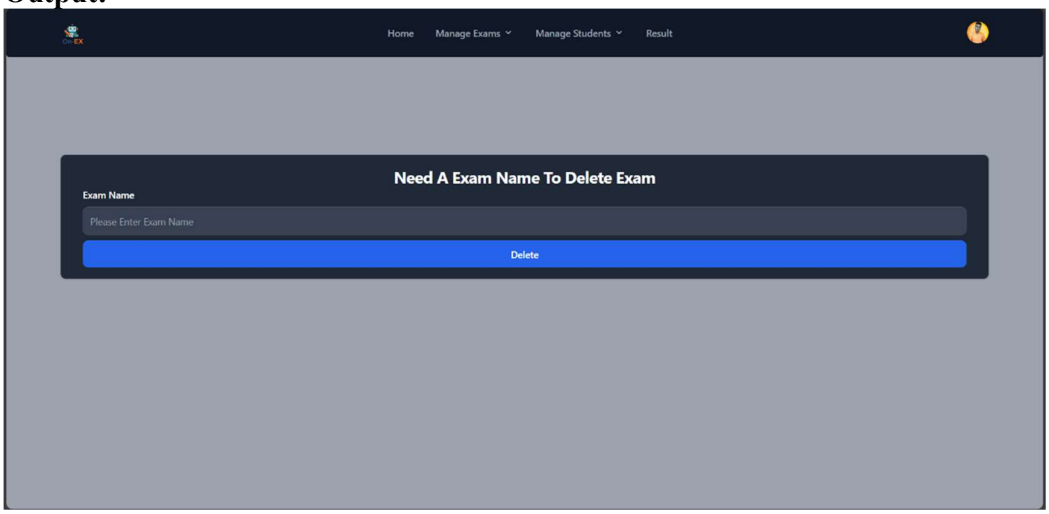
public class DeleteExams extends HttpServlet {

   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      String exam = request.getParameter("delexam");
      PrintWriter out = response.getWriter();
      try {
         if (exam != null) {
            Db_Connection db = new Db_Connection();
            String Query1 = "select * from "+exam;
            String DelQuery1 = "delete from "+exam;
            db.Pstmt = db.Con.prepareStatement(Query1);
            db.Rst = db.Pstmt.executeQuery();
            if (db.Rst.next()) {
               db.Pstmt = db.Con.prepareStatement(DelQuery1);
               int i = db.Pstmt.executeUpdate();
            }else {
response.sendRedirect("AdminHome?choice=DeleteExams&msg=Wrong Exam
Name&msgtype=error");
            }
```

```
            String Query = "select * from exam where exam_name=?";
            String DelQuery = "delete from exam where exam_name=?";
            db.Pstmt = db.Con.prepareStatement(Query);
            db.Pstmt.setString(1, exam);
            db.Rst = db.Pstmt.executeQuery();
            if (db.Rst.next()) {
               db.Pstmt = db.Con.prepareStatement(DelQuery);
               db.Pstmt.setString(1, exam);
               int i = db.Pstmt.executeUpdate();
               if (i > 0) {
response.sendRedirect("AdminHome?choice=ViewExamsList&msg=Exam
Successfully Deleted&msgtype=delete");
               } else {
response.sendRedirect("AdminHome?choice=ViewExamsList&msg=Error In
Exam Deletion&msgtype=error");
               }
            } else {
response.sendRedirect("AdminHome?choice=DeleteExams&msg=Wrong Exam
Name&msgtype=error");
            }
         } else {
            out.println("\n"
                  + "  <div class='md:mx-20 lg:my-36 md:my-28 sm:my-16 sm:mx-
6 my-10 mx-5 w-full'>\n"
                  + "     <div class='bg-white rounded-lg shadow dark:border
md:mt-0  xl:p-0 dark:bg-gray-800 dark:border-gray-700'>\n"
                  + "        <div class='px-6 sm:px-8 py-4'>\n"
                  + "           <h1 class='text-xl text-center font-bold leading-tight
tracking-tight text-gray-900 md:text-2xl dark:text-white'>\n"
                  + "              Need A Exam Name To Delete Exam\n"
                  + "           </h1>\n"
                  + "           <form class='space-y-2' action='DeleteExams'
method='post'>\n"
                  + "              <div>\n"
                  + "                 <label for='delexam' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Exam Name</label>\n"
                  + "                 <input type='text' name='delexam' id='delexam'
class='bg-gray-50 border border-gray-300 text-gray-900 sm:text-sm rounded-lg
focus:ring-primary-600 focus:border-primary-600 block w-full p-2.5 dark:bg-
gray-700 dark:border-gray-600 dark:placeholder-gray-400 dark:text-white
dark:focus:ring-blue-500 dark:focus:border-blue-500' placeholder='Please Enter
Exam Name' required=''>\n"
                  + "              </div>\n"
                  + "              <button type='submit' class='w-full text-white bg-
blue-600 hover:bg-blue-700 focus:ring-4 focus:outline-none focus:ring-blue-300
font-medium rounded-lg text-sm px-5 py-2.5 text-center dark:bg-blue-600
dark:hover:bg-blue-700 dark:focus:ring-blue-800'>Delete</button>\n"
                  + "           </form>\n"
                  + "        </div>\n"
                  + "     </div>\n"
```

```
                    + "  </div>\n"
                    + "");
            }
        } catch (SQLException | IOException e) {
            System.out.println(e);
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



➢ **ViewExamsList.java (Exam List Page)**

Code:

```
package Question;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import java.util.logging.Level;
```

```
import java.util.logging.Logger;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

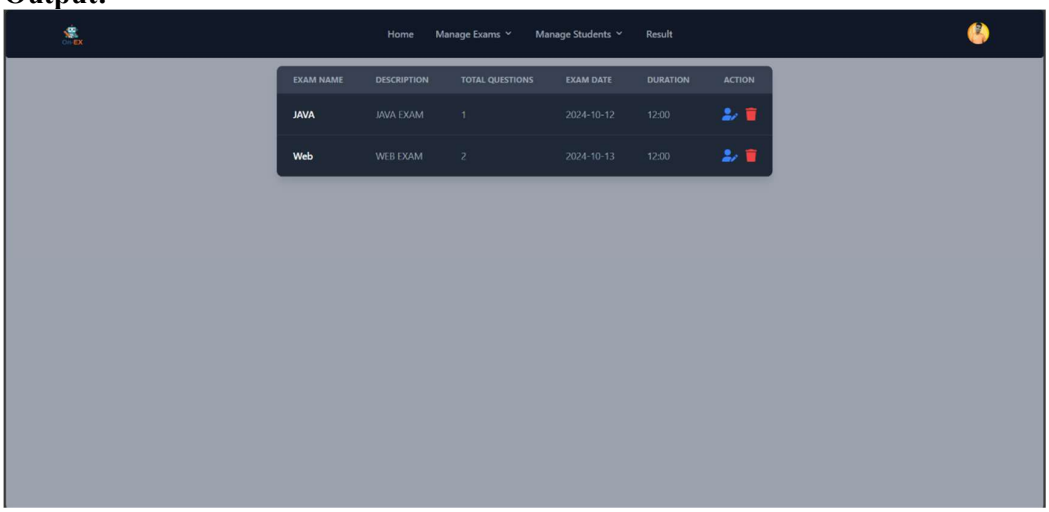
public class ViewExamsList extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         Db_Connection db = new Db_Connection();
         String query = "SELECT * FROM exam";
         db.Pstmt = db.Con.prepareStatement(query);
         db.Rst = db.Pstmt.executeQuery();
         out.println("<div class='overflow-x-auto shadow-lg sm:rounded-lg mx-3
my-3'>\n"
               + "    <table class='w-full rounded-lg text-sm text-left text-gray-600
dark:text-gray-400'>\n"
               + "       <thead class='text-xs text-gray-700 uppercase bg-blue-600
dark:bg-gray-700 dark:text-gray-400'>\n"
               + "          <tr>\n"
               + "             <th scope='col' class='py-3 px-6'>\n"
               + "                Exam Name\n"
               + "             </th>\n"
               + "             <th scope='col' class='py-3 px-6'>\n"
               + "                Description\n"
               + "             </th>\n"
               + "             <th scope='col' class='py-3 px-6'>\n"
               + "                Total Questions\n"
               + "             </th>\n"
               + "             <th scope='col' class='py-3 px-6'>\n"
               + "                Exam Date\n"
               + "             </th>\n"
               + "             <th scope='col' class='py-3 px-6'>\n"
               + "                Duration\n"
               + "             </th>\n"
               + "             <th scope='col' class='py-3 px-6 text-center'>\n"
               + "                Action\n"
               + "             </th>\n"
               + "          </tr>\n"
               + "       </thead>\n"
               + "       <tbody>\n");
         if (db.Rst.isBeforeFirst()) {
            while (db.Rst.next()) {
               out.println("          <tr class='bg-teal-400 border-y border-gray-500
dark:bg-gray-800 dark:border-gray-700 hover:bg-teal-200 dark:hover:bg-gray-
600'>\n"
```

```
                    + "            <th scope='row' class='py-4 px-6 font-medium text-
gray-900 whitespace-nowrap dark:text-white'>\n"
                    + db.Rst.getString(1)  // Exam Name
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(2)  // Exam Description
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(3)  // Total Questions
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(4)  // Exam Date
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(5)  // Exam Duration
                    + "            </td>\n"
                    + "            <td class='flex items-center py-4 px-6 space-x-
3'>\n"
                    + "                <a
href='AdminHome?choice=UpdateExams&examname="
                    + db.Rst.getString(1)  // Exam Name for update link
                    + "' class='font-medium text-pink-600 dark:text-blue-500'><i
class='fa-solid text-lg fa-user-pen'></i></a>\n"
                    + "                <a
href='AdminHome?choice=DeleteExams&delexam="
                    + db.Rst.getString(1)  // Exam Name for delete link
                    + "' class='font-medium text-red-600 dark:text-red-500'><i
class='fa-solid text-lg fa-trash'></i></a>\n"
                    + "            </td>\n"
                    + "        </tr>\n");
            }
        } else {
            out.println("        <tr>\n"
                    + "            <td colspan='6' class='py-4 px-6 text-center'>No
exams found.</td>\n"
                    + "        </tr>\n");
        }

        out.println("    </tbody>\n"
                + "  </table>\n"
                + "</div>");
    } catch (SQLException ex) {
        Logger.getLogger(ViewExamsList.class.getName()).log(Level.SEVERE,
null, ex);
    }
  }
  @Override
  protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
```

```
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Servlet to display exam details";
    }
}
```

Output:



- **Result.java (Result List Page)**

Code:

```
package App;

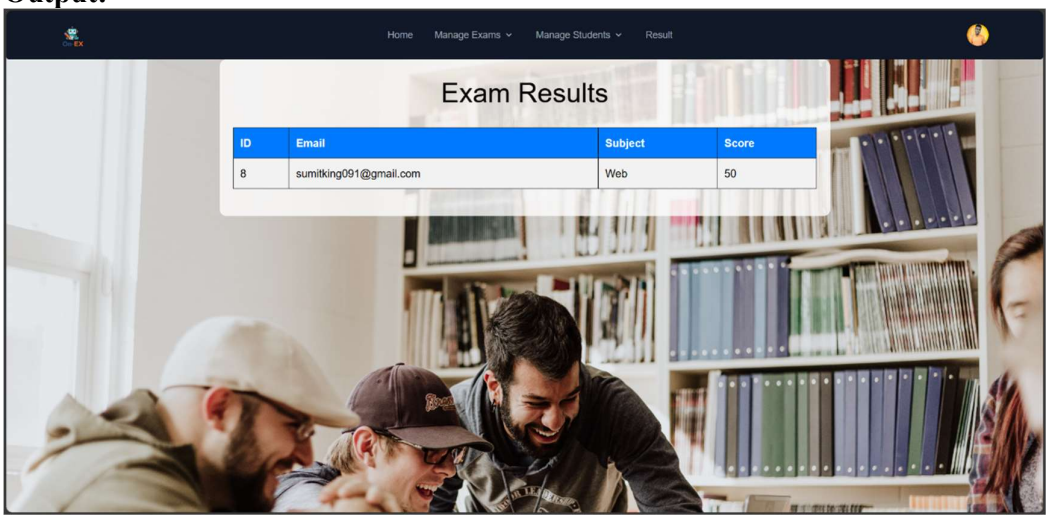
import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.ResultSet;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class Result extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            try {
```

```
            Db_Connection db = new Db_Connection();
            String sql = "SELECT id, email, exam_name, score FROM result";
            db.Pstmt = db.Con.prepareStatement(sql);
            ResultSet rs = db.Pstmt.executeQuery();
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>View Results</title>");
            out.println("<style>");
            out.println("body { font-family: Arial, sans-serif; background-color:
rgba(0, 0, 0, 0.329); background-image: url(Images/Student7.jpg); background-
size: cover; }");
            out.println(".container { max-width: 900px; margin: auto; padding:
20px; background: rgba(255, 255, 255, 0.8); border-radius: 10px; }");
            out.println("h1 { text-align: center; font-size: 2.5em; margin-bottom:
20px; }");
            out.println("table { width: 100%; border-collapse: collapse; margin:
20px 0; }");
            out.println("table, th, td { border: 1px solid #000; }");
            out.println("th, td { padding: 10px; text-align: left; }");
            out.println("th { background-color: #007BFF; color: white; }");
            out.println("tr:nth-child(even) { background-color: #f2f2f2; }");
            out.println(".btn { display: block; margin: 20px auto; padding: 10px
20px; background-color: #007BFF; color: white; border: none; border-radius: 5px;
text-align: center; text-decoration: none; }");
            out.println(".btn:hover { background-color: #0056b3; }");
            out.println("</style>");
            out.println("</head>");
            out.println("<body>");
            out.println("<div class='container'>");
            out.println("<h1>Exam Results</h1>");
            if (rs.next()) {
                out.println("<table>");
out.println("<tr><th>ID</th><th>Email</th><th>Subject</th><th>Score</th></t
r>");
                do {
                    String studentId = rs.getString("id");
                    String email = rs.getString("email");
                    String exam_name = rs.getString("exam_name");
                    int score = rs.getInt("score");
                    out.println("<tr>");
                    out.println("<td>" + studentId + "</td>");
                    out.println("<td>" + email + "</td>");
                    out.println("<td>" + exam_name + "</td>");
                    out.println("<td>" + score + "</td>");
                    out.println("</tr>");
                } while (rs.next());
                out.println("</table>");
            } else {
                out.println("<p>No results available at the moment.</p>");
```

```
                }
                out.println("</div>");
                out.println("</body>");
                out.println("</html>");
                rs.close();
                db.Pstmt.close();
                db.Con.close();
            } catch (Exception e) {
                e.printStackTrace();
                response.getWriter().println("<h1>Error occurred while fetching the
results. Please try again later.</h1>");
            }
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Servlet to display exam results.";
    }
}
```

Output:



- **StudentList.java (Student List Page)**

Code:

```
package Student;

import DB.Db_Connection;
import Question.ViewExamsList;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import java.util.logging.Level;
import java.util.logging.Logger;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

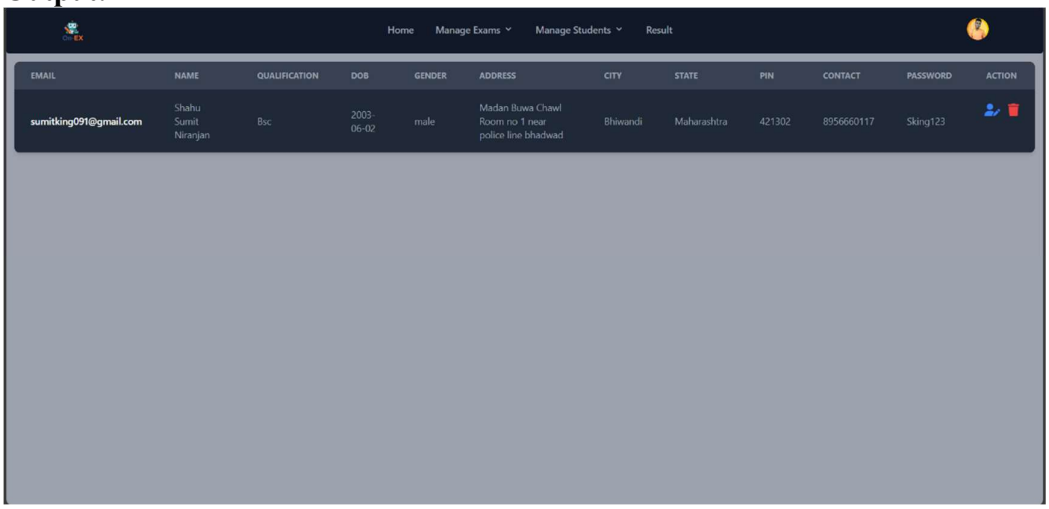
public class StudentList extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
          throws ServletException, IOException {
       response.setContentType("text/html;charset=UTF-8");
       try (PrintWriter out = response.getWriter()) {
          Db_Connection db = new Db_Connection();
          String Query = "select * from stu_info";
          db.Pstmt = db.Con.prepareStatement(Query);
          db.Rst = db.Pstmt.executeQuery();
          out.println("<div class='overflow-x-auto shadow-md  sm:rounded-lg mx-3
my-3'>\n"
                  + "    <table class='w-full text-sm text-left text-gray-500 dark:text-
gray-400 bg-red-700'>\n"
                  + "        <thead class='text-xs text-gray-700 bg-blue-600 uppercase
dark:bg-gray-700 dark:text-gray-400'>\n"
                  + "            <tr>\n"
                  + "                <th scope='col' class='py-3 px-6'>\n"
                  + "                    Email\n"
                  + "                </th>\n"
                  + "                <th scope='col' class='py-3 px-6'>\n"
                  + "                    Name\n"
                  + "                </th>\n"
                  + "                <th scope='col' class='py-3 px-6'>\n"
                  + "                    Qualification\n"
                  + "                </th>\n"
                  + "                <th scope='col' class='py-3 px-6'>\n"
                  + "                    Dob\n"
                  + "                </th>\n"
                  + "                <th scope='col' class='py-3 px-6'>\n"
                  + "                    Gender\n"
                  + "                </th>\n"
                  + "                <th scope='col' class='py-3 px-6'>\n"
                  + "                    Address\n"
```

```
                + "            </th>\n"
                + "            <th scope='col' class='py-3 px-6'>\n"
                + "               City\n"
                + "            <th scope='col' class='py-3 px-6'>\n"
                + "               State\n"
                + "            </th>\n"
                + "            <th scope='col' class='py-3 px-6'>\n"
                + "               Pin\n"
                + "            </th>\n"
                + "            <th scope='col' class='py-3 px-6'>\n"
                + "               Contact\n"
                + "            </th>\n"
                + "            <th scope='col' class='py-3 px-6'>\n"
                + "               Password\n"
                + "            </th>\n"
                + "            <th scope='col' class='py-3 px-6 text-center'>\n"
                + "               Action\n"
                + "            </th>\n"
                + "         </tr>\n"
                + "      </thead>\n"
                + "      <tbody>\n");
        if (db.Rst.isBeforeFirst()) {
           while (db.Rst.next()) {
              out.println("          <tr class='bg-teal-400 border-y border-gray-500
dark:bg-gray-800 dark:border-gray-700 hover:bg-teal-200 dark:hover:bg-gray-
600'>\n"
                    + "            <th scope='row' class='py-4 px-6 font-medium text-
gray-900 whitespace-nowrap dark:text-white'>\n"
                    + db.Rst.getString(1)
                    + "               \n"
                    + "            </th>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(2)
                    + "               \n"
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(3)
                    + "               \n"
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(4)
                    + "               \n"
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(5)
                    + "               \n"
                    + "            </td>\n"
                    + "            <td class='py-4 px-6'>\n"
                    + db.Rst.getString(6)
                    + "               \n"
```

```
                    + "             </td>\n"
                    + "             <td class='py-4 px-6'>\n"
                    + db.Rst.getString(7)
                    + "                 \n"
                    + "             </td>\n"
                    + "             <td class='py-4 px-6'>\n"
                    + db.Rst.getString(8)
                    + "                 \n"
                    + "             </td>\n"
                    + "             <td class='py-4 px-6'>\n"
                    + db.Rst.getString(9)
                    + "                 \n"
                    + "             </td>\n"
                    + "             <td class='py-4 px-6'>\n"
                    + db.Rst.getString(10)
                    + "                 \n"
                    + "             </td>\n"
                    + "             <td class='py-4 px-6'>\n"
                    + db.Rst.getString(11)
                    + "                 \n"
                    + "             </td>\n"
                    + "             <td class='flex items-center py-4 px-6 space-x-
3'>\n"
                    + "                 <a
href='AdminHome?choice=UpdateStudent&upemail="
                    + db.Rst.getString(1)
                    + "' class='font-medium text-pink-600 dark:text-blue-500'><i
class='fa-solid text-lg fa-user-pen'></i></a>\n"
                    + "                 <a href='DeleteStudent?delemail="
                    + db.Rst.getString(1)
                    + "' class='font-medium text-red-600 dark:text-red-500'><i
class='fa-solid  text-lg fa-trash'></i></a>\n"
                    + "             </td>\n"
                    + "         </tr>\n");
            }
        }
        out.println("       </tbody>\n"
              + "    </table>\n"
              + "</div>");
    } catch (SQLException ex) {
        Logger.getLogger(ViewExamsList.class.getName()).log(Level.SEVERE,
null, ex);
    }
  }
  @Override
  protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
      processRequest(request, response);
  }
```

```
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



- **UpdateStudent.java (Update Student Details Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class UpdateStudent extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        String upemail = request.getParameter("upemail"),
                email = request.getParameter("email"),
                name = request.getParameter("name"),
                qua = request.getParameter("qua"),
```

```
            dob = request.getParameter("dob"),
            gender = request.getParameter("gender"),
            add = request.getParameter("add"),
            city = request.getParameter("city"),
            state = request.getParameter("state"),
            pin = request.getParameter("pin"),
            contact = request.getParameter("contact"),
            password = request.getParameter("password"),
            check = request.getParameter("check");
        PrintWriter out = response.getWriter();
        try {
           if (upemail != null) {
            Db_Connection db = new Db_Connection();
            String Query = "select * from stu_info where email=?";
            db.Pstmt = db.Con.prepareStatement(Query);
            db.Pstmt.setString(1, upemail);
            db.Rst = db.Pstmt.executeQuery();
            if (db.Rst.next()) {
               if (check != null && check.equals("true")) {
response.sendRedirect("AdminHome?choice=UpdateStudent&upemail=" +
upemail);
                }
                out.println("\n"
                      + "  <div class='md:mx-20 my-5 sm:mx-6 mx-5 w-full'>\n"
                      + "     <div class='bg-white rounded-lg shadow dark:border
md:mt-0  xl:p-0 dark:bg-gray-800 dark:border-gray-700'>\n"
                      + "        <div class='px-6 sm:px-8 py-4'>\n"
                      + "           <h1 class='text-xl text-center font-bold leading-tight
tracking-tight text-gray-900 md:text-2xl dark:text-white'>\n"
                      + "              Update Student\n"
                      + "           </h1>\n"
                      + "           <form class='space-y-2' action='UpdateStudent'
method='post'>\n"
                      + "              <div>\n"
                      + "                 <label for='email' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Emali Id</label>\n"
                      + "                 <input value='"
                      + db.Rst.getString(1)
                      + "' type='email' name='email' id='email' class='bg-gray-50
border border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-
600 focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-
gray-600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Question Id'
required=''>\n"
                      + "              </div>\n"
                      + "              <div class='flex justify-between space-x-4'><div
class='w-full'>\n"
                      + "                 <label for='name' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Name</label>\n"
                      + "                 <input value='"
```

```
                + db.Rst.getString(2)
                + "' type='text' name='name' id='name' class='bg-gray-50 border
border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-600
focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-gray-
600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Question' required=''>\n"
                + "            </div>\n"
                + "            <div class='w-full'>\n"
                + "               <label for='qua' class='block mb-2 text-sm font-
medium text-gray-900 dark:text-white'>Qualification</label>\n"
                + "               <input value='"
                + db.Rst.getString(3)
                + "' type='text' name='qua' id='qua' class='bg-gray-50 border
border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-600
focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-gray-
600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Option1' required=''>\n"
                + "            </div></div>\n"
                + "            <div class='flex justify-between space-x-4'><div
class='w-full'>\n"
                + "               <label for='dob' class='block mb-2 text-sm font-
medium text-gray-900 dark:text-white'>Date Of Birth</label>\n"
                + "               <input value='"
                + db.Rst.getString(4)
                + "' type='text' name='dob' id='dob' class='bg-gray-50 border
border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-600
focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-gray-
600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Option2' required=''>\n"
                + "            </div>\n"
                + "            <div class='w-full'>\n"
                + "               <label for='gender' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Gender</label>\n"
                + "               <input value='"
                + db.Rst.getString(5)
                + "' type='text' name='gender' id='gender' class='bg-gray-50
border border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-
600 focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-
gray-600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Option3' required=''>\n"
                + "            </div></div>\n"
                + "            <div>\n"
                + "               <label for='add' class='block mb-2 text-sm font-
medium text-gray-900 dark:text-white'>Address</label>\n"
                + "               <textarea  type='text' name='add' id='add'
class='bg-gray-50 border border-gray-300 text-gray-900 sm:text-sm rounded-lg
focus:ring-primary-600 focus:border-primary-600 block w-full p-2.5 dark:bg-
gray-700 dark:border-gray-600 dark:placeholder-gray-400 dark:text-white
dark:focus:ring-blue-500 dark:focus:border-blue-500' placeholder='Please Enter
Option4' required=''>"
```

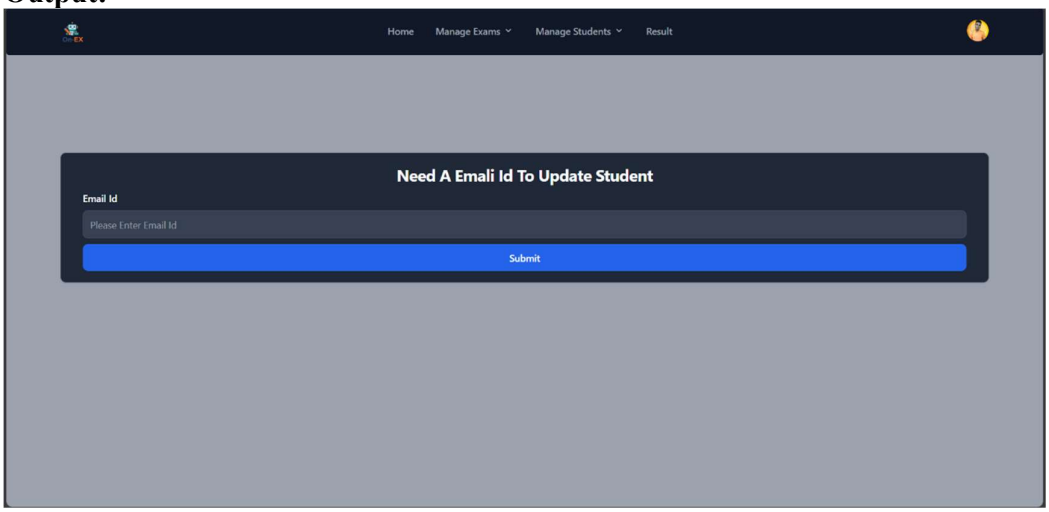
```
                    + db.Rst.getString(6)
                    + "</textarea>\n"
                    + "                </div>\n"
                    + "                <div class='flex justify-between space-x-4'><div
class='w-full'>\n"
                    + "                    <label for='city' class='block mb-2 text-sm font-
medium text-gray-900 dark:text-white'>City</label>\n"
                    + "                    <input value='"
                    + db.Rst.getString(7)
                    + "' type='text' name='city' id='city' class='bg-gray-50 border
border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-600
focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-gray-
600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Answer' required=">\n"
                    + "                </div>\n"
                    + "                <div class='w-full'>\n"
                    + "                    <label for='state' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>State</label>\n"
                    + "                    <input value='"
                    + db.Rst.getString(8)
                    + "' type='text' name='state' id='state' class='bg-gray-50 border
border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-600
focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-gray-
600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Answer' required=">\n"
                    + "                </div>\n"
                    + "                <div class='w-full'>\n"
                    + "                    <label for='pin' class='block mb-2 text-sm font-
medium text-gray-900 dark:text-white'>Pin</label>\n"
                    + "                    <input value='"
                    + db.Rst.getString(9)
                    + "' type='text' name='pin' id='pin' class='bg-gray-50 border
border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-600
focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-gray-
600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Answer' required=">\n"
                    + "                </div></div>\n"
                    + "                <div class='flex justify-between space-x-4'><div
class='w-full'>\n"
                    + "                    <label for='contact' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Contact</label>\n"
                    + "                    <input value='"
                    + db.Rst.getString(10)
                    + "' type='text' name='contact' id='contact' class='bg-gray-50
border border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-
600 focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-
gray-600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Answer' required=">\n"
                    + "                </div>\n"
                    + "                <div class='w-full'>\n"
```

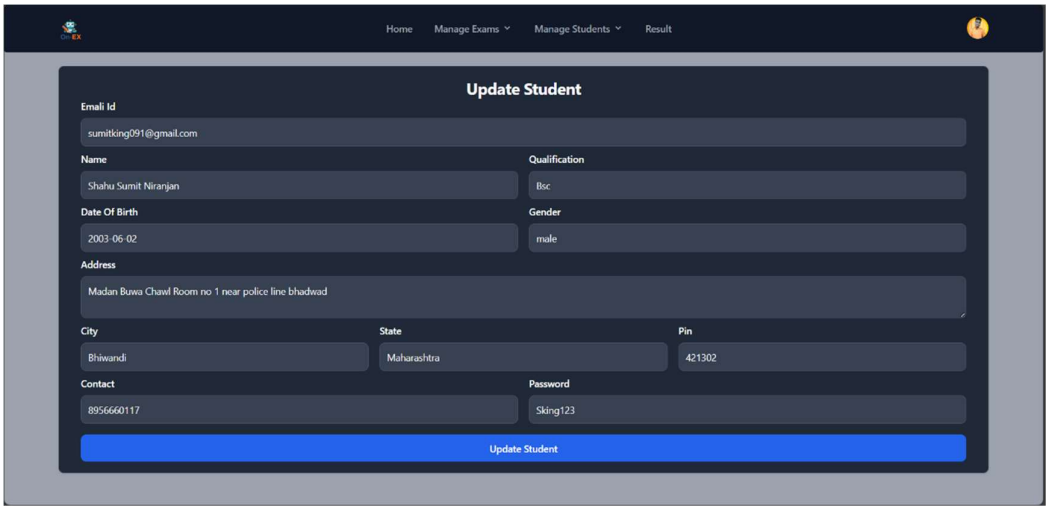
```
                    + "                    <label for='password' class='block mb-2 text-
sm font-medium text-gray-900 dark:text-white'>Password</label>\n"
                    + "                    <input value='"
                    + db.Rst.getString(11)
                    + "' type='text' name='password' id='password' class='bg-gray-50
border border-gray-300 text-gray-900 sm:text-sm rounded-lg focus:ring-primary-
600 focus:border-primary-600 block w-full p-2.5 dark:bg-gray-700 dark:border-
gray-600 dark:placeholder-gray-400 dark:text-white dark:focus:ring-blue-500
dark:focus:border-blue-500' placeholder='Please Enter Answer' required=''>\n"
                    + "                </div></div>\n"
                    + "                <div>\n"
                    + "                    <input type='hidden' name='upemail' value='"
                    + upemail
                    + "'>\n"
                    + "                </div>\n"
                    + "                <button type='submit' class='w-full text-white bg-
blue-600 hover:bg-blue-700 focus:ring-4 focus:outline-none focus:ring-blue-300
font-medium rounded-lg text-sm px-5 py-2.5 text-center dark:bg-blue-600
dark:hover:bg-blue-700 dark:focus:ring-blue-800'>Update Student</button>\n"
                    + "            </form>\n"
                    + "        </div>\n"
                    + "    </div>\n"
                    + "  </div>\n"
                    + "");
            } else {
response.sendRedirect("AdminHome?choice=UpdateStudent&msg=Wrong Email
Id&msgtype=error");
            }
        } else {
            out.println("\n"
                    + "  <div class='md:mx-20 lg:my-36 md:my-28 sm:my-16 sm:mx-
6 my-10 mx-5 w-full'>\n"
                    + "     <div class='bg-white rounded-lg shadow dark:border
md:mt-0  xl:p-0 dark:bg-gray-800 dark:border-gray-700'>\n"
                    + "         <div class='px-6 sm:px-8 py-4'>\n"
                    + "             <h1 class='text-xl text-center font-bold leading-tight
tracking-tight text-gray-900 md:text-2xl dark:text-white'>\n"
                    + "                 Need A Emali Id To Update Student\n"
                    + "             </h1>\n"
                    + "             <form class='space-y-2' action='UpdateStudent'
method='post'>\n"
                    + "                 <div>\n"
                    + "                     <input type='hidden' name='check'
value='true'>\n"
                    + "                 </div>\n"
                    + "                 <div>\n"
                    + "                     <label for='upemail' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Email Id</label>\n"
                    + "                     <input type='email' name='upemail' id='upemail'
class='bg-gray-50 border border-gray-300 text-gray-900 sm:text-sm rounded-lg
```

```
focus:ring-primary-600 focus:border-primary-600 block w-full p-2.5 dark:bg-
gray-700 dark:border-gray-600 dark:placeholder-gray-400 dark:text-white
dark:focus:ring-blue-500 dark:focus:border-blue-500' placeholder='Please Enter
Email Id' required=">\n"
                    + "                </div>\n"
                    + "                <button type='submit' class='w-full text-white bg-
blue-600 hover:bg-blue-700 focus:ring-4 focus:outline-none focus:ring-blue-300
font-medium rounded-lg text-sm px-5 py-2.5 text-center dark:bg-blue-600
dark:hover:bg-blue-700 dark:focus:ring-blue-800'>Submit</button>\n"
                    + "            </form>\n"
                    + "        </div>\n"
                    + "    </div>\n"
                    + "  </div>\n"
                    + "");
            }
            if (email != null
                    || name != null
                    || qua != null
                    || gender != null
                    || add != null
                    || city != null
                    || pin != null
                    || contact != null
                    || password != null
                    || state != null
                    || dob != null) {
                Db_Connection db = new Db_Connection();
                String Query = "update stu_info set email=?, sname=?, squa=?, sdob=?,
sgender=?, sadd=?, scity=?, spin=?, state=?, scont=?, password=? where
email=?";
                db.Pstmt = db.Con.prepareStatement(Query);
                db.Pstmt.setString(1, email);
                db.Pstmt.setString(2, name);
                db.Pstmt.setString(3, qua);
                db.Pstmt.setString(4, dob);
                db.Pstmt.setString(5, gender);
                db.Pstmt.setString(6, add);
                db.Pstmt.setString(7, city);
                db.Pstmt.setString(8, pin);
                db.Pstmt.setString(9, state);
                db.Pstmt.setString(10, contact);
                db.Pstmt.setString(11, password);
                db.Pstmt.setString(12, upemail);
                int i = db.Pstmt.executeUpdate();
                if (i > 0) {
response.sendRedirect("AdminHome?choice=StudentList&msg=Student
Successfully Updated&msgtype=success");
                } else {
response.sendRedirect("AdminHome?choice=StudentList&msg=Error In Student
Updation&msgtype=error");
```

```
            }
          }
        } catch (SQLException | IOException e) {
          response.sendRedirect("AdminHome?choice=UpdateStudent&msg=Email
Should Be Unique&msgtype=error&upemail=" + upemail);
          System.out.println(e);
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
          throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
          throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:





➢ **DeleteStudent.java (Delete Student Details Page)**

Code:

```
package Student;

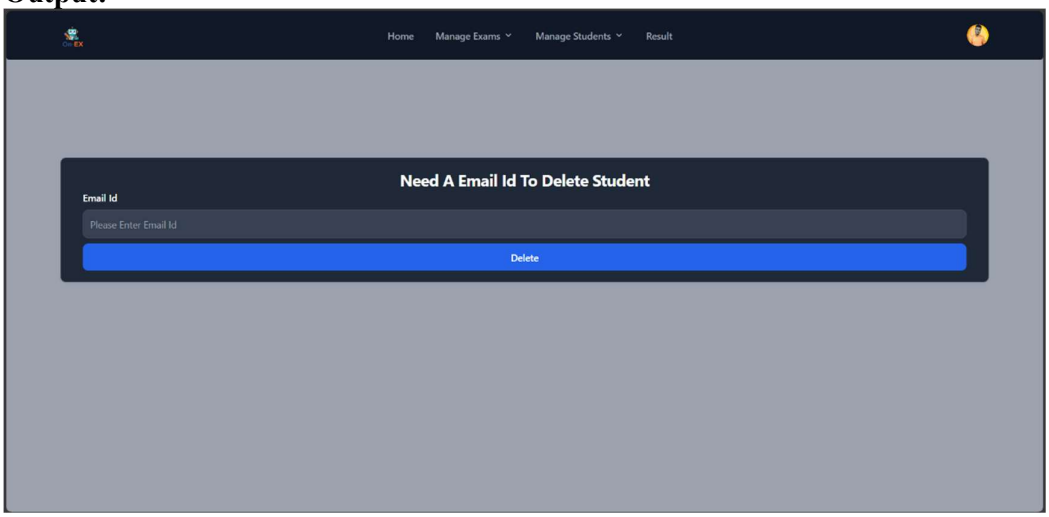
import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class DeleteStudent extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      String email = request.getParameter("delemail");
      PrintWriter out = response.getWriter();
      try {
         if (email != null) {
            Db_Connection db = new Db_Connection();
            String Query1 = "select * from login where email=?";
            String DelQuery1 = "delete from login where email=?";
            db.Pstmt = db.Con.prepareStatement(Query1);
            db.Pstmt.setString(1, email);
            db.Rst = db.Pstmt.executeQuery();
            if (db.Rst.next()) {
               db.Pstmt = db.Con.prepareStatement(DelQuery1);
               db.Pstmt.setString(1, email);
               int i = db.Pstmt.executeUpdate();
            } else {
response.sendRedirect("AdminHome?choice=DeleteStudent&msg=Wrong Email
Id&msgtype=error");
            }
```

```
            String Query = "select * from stu_info where email=?";
            String DelQuery = "delete from stu_info where email=?";
            db.Pstmt = db.Con.prepareStatement(Query);
            db.Pstmt.setString(1, email);
            db.Rst = db.Pstmt.executeQuery();
            if (db.Rst.next()) {
               db.Pstmt = db.Con.prepareStatement(DelQuery);
               db.Pstmt.setString(1, email);
               int i = db.Pstmt.executeUpdate();
               if (i > 0) {
response.sendRedirect("AdminHome?choice=StudentList&msg=Student
Successfully Deleted&msgtype=delete");
               } else {
response.sendRedirect("AdminHome?choice=StudentList&msg=Error In Student
Deletion&msgtype=error");
               }
            } else {
response.sendRedirect("AdminHome?choice=DeleteStudent&msg=Wrong Email
Id&msgtype=error");
            }
         } else {
            out.println("\n"
                  + "  <div class='md:mx-20 lg:my-36 md:my-28 sm:my-16 sm:mx-
6 my-10 mx-5 w-full'>\n"
                  + "     <div class='bg-white rounded-lg shadow dark:border
md:mt-0  xl:p-0 dark:bg-gray-800 dark:border-gray-700'>\n"
                  + "        <div class='px-6 sm:px-8 py-4'>\n"
                  + "           <h1 class='text-xl text-center font-bold leading-tight
tracking-tight text-gray-900 md:text-2xl dark:text-white'>\n"
                  + "              Need A Email Id To Delete Student\n"
                  + "           </h1>\n"
                  + "           <form class='space-y-2' action='DeleteStudent'
method='post'>\n"
                  + "              <div>\n"
                  + "                 <label for='delemail' class='block mb-2 text-sm
font-medium text-gray-900 dark:text-white'>Email Id</label>\n"
                  + "                 <input type='email' name='delemail' id='delemail'
class='bg-gray-50 border border-gray-300 text-gray-900 sm:text-sm rounded-lg
focus:ring-primary-600 focus:border-primary-600 block w-full p-2.5 dark:bg-
gray-700 dark:border-gray-600 dark:placeholder-gray-400 dark:text-white
dark:focus:ring-blue-500 dark:focus:border-blue-500' placeholder='Please Enter
Email Id' required=''>\n"
                  + "              </div>\n"
                  + "              <button type='submit' class='w-full text-white bg-
blue-600 hover:bg-blue-700 focus:ring-4 focus:outline-none focus:ring-blue-300
font-medium rounded-lg text-sm px-5 py-2.5 text-center dark:bg-blue-600
dark:hover:bg-blue-700 dark:focus:ring-blue-800'>Delete</button>\n"
                  + "           </form>\n"
                  + "        </div>\n"
                  + "     </div>\n"
```

```
                    + "  </div>\n"
                    + "");
            }
        } catch (Exception e) {
            System.out.println(e);
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

Output:



- **View.java (View Feedback from Students Page)**

Code:

```
package Feedback;

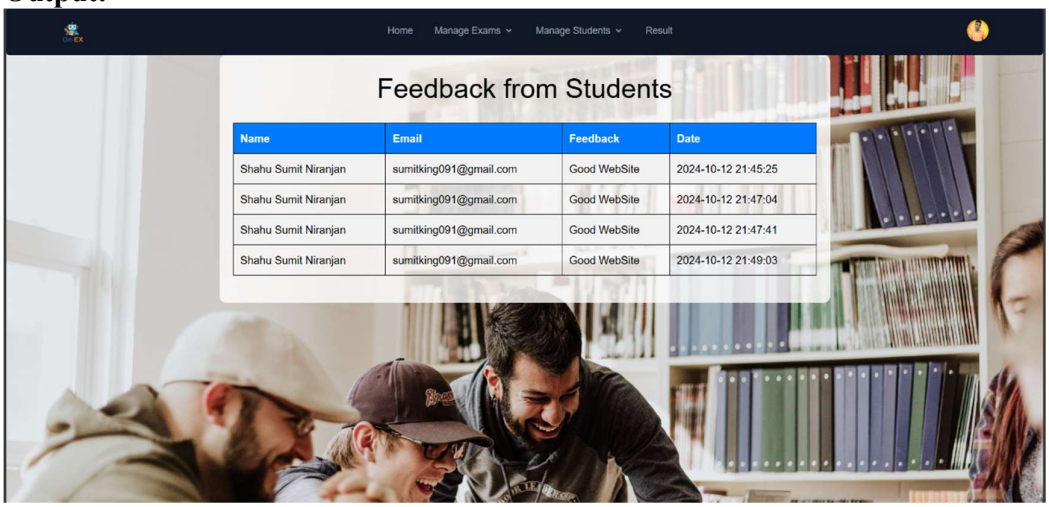
import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.ResultSet;
import javax.servlet.ServletException;
```

```
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class View extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         try {
            Db_Connection db = new Db_Connection();
            String sql = "SELECT name, email, feedback, date FROM feedback";
            db.Pstmt = db.Con.prepareStatement(sql);
            ResultSet rs = db.Pstmt.executeQuery();
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>View Feedback</title>");
            out.println("<style>");
            out.println("body { font-family: Arial, sans-serif; background-color:
rgba(0, 0, 0, 0.329); background-image: url(Images/Student7.jpg); background-
size: cover; }");
            out.println(".container { max-width: 900px; margin: auto; padding:
20px; background: rgba(255, 255, 255, 0.8); border-radius: 10px; }");
            out.println("h1 { text-align: center; font-size: 2.5em; margin-bottom:
20px; }");
            out.println("table { width: 100%; border-collapse: collapse; margin:
20px 0; }");
            out.println("table, th, td { border: 1px solid #000; }");
            out.println("th, td { padding: 10px; text-align: left; }");
            out.println("th { background-color: #007BFF; color: white; }");
            out.println("tr:nth-child(even) { background-color: #f2f2f2; }");
            out.println(".btn { display: block; margin: 20px auto; padding: 10px
20px; background-color: #007BFF; color: white; border: none; border-radius: 5px;
text-align: center; text-decoration: none; }");
            out.println(".btn:hover { background-color: #0056b3; }");
            out.println("</style>");
            out.println("</head>");
            out.println("<body>");
            out.println("<div class='container'>");
            out.println("<h1>Feedback from Students</h1>");
            if (rs.next()) {
               out.println("<table>");
out.println("<tr><th>Name</th><th>Email</th><th>Feedback</th><th>Date</th
></tr>");
               do {
                  String name = rs.getString("name");
                  String email = rs.getString("email");
                  String feedback = rs.getString("feedback");
```

```
                String timestamp = rs.getString("date");
                out.println("<tr>");
                out.println("<td>" + name + "</td>");
                out.println("<td>" + email + "</td>");
                out.println("<td>" + feedback + "</td>");
                out.println("<td>" + timestamp + "</td>");
                out.println("</tr>");
            } while (rs.next());
            out.println("</table>");
        } else {
            out.println("<p>No feedback available at the moment.</p>");
        }
        out.println("</div>");
        out.println("</body>");
        out.println("</html>");
        rs.close();
        db.Pstmt.close();
        db.Con.close();
      } catch (Exception e) {
        e.printStackTrace();
      }
    }
  }
 protected void doGet(HttpServletRequest request, HttpServletResponse response)
      throws ServletException, IOException {
    processRequest(request, response);
  }
  @Override
  protected void doPost(HttpServletRequest request, HttpServletResponse
response)
      throws ServletException, IOException {
    processRequest(request, response);
  }
```

Output:



- **AdminProfile.java (View & Update Admin Details Page)**

Code:

```
package App;

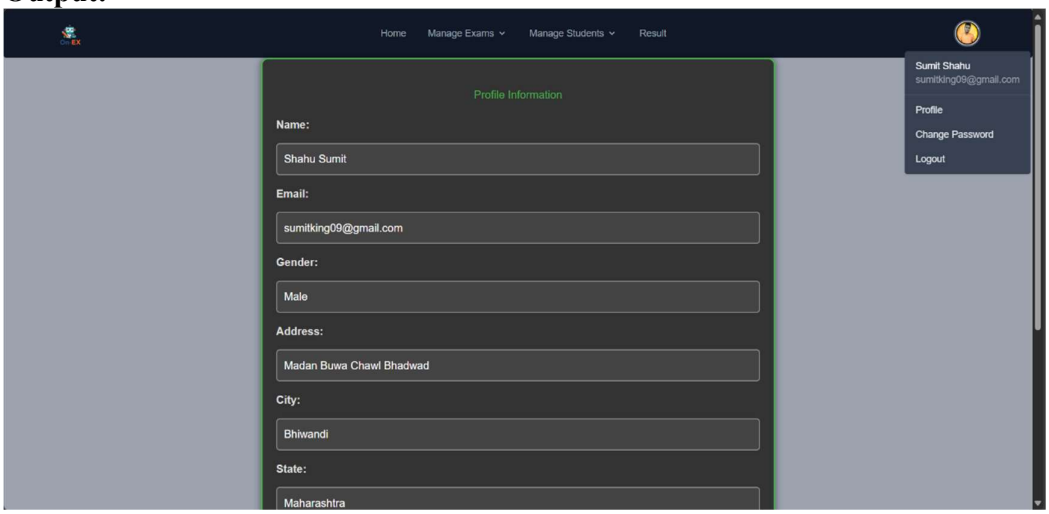
import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class AdminProfile extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      HttpSession session = request.getSession();
      String userId = (String) session.getAttribute("Admin");
      Db_Connection db = new Db_Connection();
      try (PrintWriter out = response.getWriter()) {
         try {
            String query = "SELECT email, sname, sgender, sadd, scity, state, spin,
scont, sid FROM adminlogin WHERE email = ?";
            db.Pstmt = db.Con.prepareStatement(query);
            db.Pstmt.setString(1, userId);
            db.Rst = db.Pstmt.executeQuery();
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Profile Page</title>");
            out.println("<style>");
            out.println("body { font-family: Arial, sans-serif; background-color:
#2e2e2e; color: #ccc; margin: 0; padding: 0; }");
            out.println("h1 { color: #4CAF50; text-align: center; padding: 20px 0;
}");
            out.println(".container { width: 50%; margin: 0 auto; padding: 20px;
background-color: #333; border: 3px solid #4CAF50; border-radius: 10px; box-
shadow: 0 0 10px rgba(0, 0, 0, 0.5); }");
            out.println("form { display: flex; flex-direction: column; gap: 15px; }");
            out.println("input[type='text'], input[type='password'],
input[type='date'], input[type='submit'] { padding: 10px; font-size: 1em; border:
2px solid #888; border-radius: 5px; background-color: #444; color: white; }");
            out.println("input[type='text']:focus, input[type='password']:focus,
input[type='date']:focus { border-color: #4CAF50; outline: none; }");
            out.println("input[type='submit'] { background-color: #4CAF50; color:
white; border: none; cursor: pointer; font-weight: bold; }");
            out.println("input[type='submit']:hover { background-color: #45a049;
}");
```

```
            out.println("label { font-weight: bold; color: #ddd; }");
            out.println("</style>");
            out.println("</head>");
            out.println("<body>");
            out.println("<div class='container'>");
            out.println("<h1>Profile Information</h1>");
            if (db.Rst.next()) {
                out.println("<form method='POST' action='AdminProfile'>");
                out.println("<label for='sname'>Name:</label>");
                out.println("<input type='text' name='sname' value='" +
db.Rst.getString("sname") + "' />");
                out.println("<label for='email'>Email:</label>");
                out.println("<input type='text' name='email' value='" +
db.Rst.getString("email") + "' readonly />");
                out.println("<label for='sgender'>Gender:</label>");
                out.println("<input type='text' name='sgender' value='" +
db.Rst.getString("sgender") + "' />")
                out.println("<label for='sadd'>Address:</label>");
                out.println("<input type='text' name='sadd' value='" +
db.Rst.getString("sadd") + "' />");
                out.println("<label for='scity'>City:</label>");
                out.println("<input type='text' name='scity' value='" +
db.Rst.getString("scity") + "' />");
                out.println("<label for='state'>State:</label>");
                out.println("<input type='text' name='state' value='" +
db.Rst.getString("state") + "' />");
                out.println("<label for='spin'>PIN:</label>");
                out.println("<input type='text' name='spin' value='" +
db.Rst.getString("spin") + "' />");
                out.println("<label for='scont'>Contact:</label>");
                out.println("<input type='text' name='scont' value='" +
db.Rst.getString("scont") + "' />");
                out.println("<label for='sid'>ID:</label>");
                out.println("<input type='text' name='sid' value='" +
db.Rst.getString("sid") + "' />");
                out.println("<input type='submit' value='Update Profile' />");
                out.println("</form>");
            } else {
                out.println("<p>No user found with the given ID.</p>");
            }
            out.println("</div>");
            out.println("</body>");
            out.println("</html>");
        } catch (Exception e) {
            e.printStackTrace();
            out.println("<p>Error retrieving profile information.</p>");
        } finally {
            if (db.Rst != null) try { db.Rst.close(); } catch (Exception e) {}
        }
    }
```

```
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        Db_Connection db = new Db_Connection();
        String email = request.getParameter("email");
        String sname = request.getParameter("sname");
        String sgender = request.getParameter("sgender");
        String sadd = request.getParameter("sadd");
        String scity = request.getParameter("scity");
        String state = request.getParameter("state");
        String spin = request.getParameter("spin");
        String scont = request.getParameter("scont");
        String sid = request.getParameter("sid");
        try {
            String updateQuery = "UPDATE adminlogin SET sname = ?, sgender = ?,
sadd = ?, scity = ?, state = ?, spin = ?, scont = ?, sid = ? WHERE email = ?";
            db.Pstmt = db.Con.prepareStatement(updateQuery);
            db.Pstmt.setString(1, sname);
            db.Pstmt.setString(2, sgender);
            db.Pstmt.setString(3, sadd);
            db.Pstmt.setString(4, scity);
            db.Pstmt.setString(5, state);
            db.Pstmt.setString(6, spin);
            db.Pstmt.setString(7, scont);
            db.Pstmt.setString(8, sid);
            db.Pstmt.setString(9, email);
            db.Pstmt.executeUpdate();
            response.sendRedirect("AdminHome?choice=AdminProfile"); // Redirect
back to the profile page after update
        } catch (Exception e) {
            e.printStackTrace();
        } finally {
            if (db.Pstmt != null) try { db.Pstmt.close(); } catch (Exception e) {}
        }
    }
    @Override
    public String getServletInfo() {
        return "Servlet to display and update user profile details";
    }
}
```

Output:



- **Logout.java (Logout Function)**

Code:

```
package App;

import java.io.IOException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet(name = "Logout", urlPatterns = {"/Logout"})
public class Logout extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
           throws ServletException, IOException {
       HttpSession session = request.getSession();
       session.removeAttribute("Admin");
       session.removeAttribute("Student");
       response.sendRedirect("/OnlineExam");
   }
   @Override
   protected void doGet(HttpServletRequest request, HttpServletResponse
response)
           throws ServletException, IOException {
       processRequest(request, response);
   }
   @Override
   protected void doPost(HttpServletRequest request, HttpServletResponse
response)
           throws ServletException, IOException {
       processRequest(request, response);
   }
```

```
    @Override
    public String getServletInfo() {
        return "Short description";
    }
}
```

- **Student.java (Student Home Page)**

Code:

```
package Student;

import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class Student extends HttpServlet {
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        HttpSession session = request.getSession();
        Object name = session.getAttribute("Student");
        String msg = request.getParameter("msg"),
                msgtype = request.getParameter("msgtype"),
                choice = request.getParameter("choice");
        response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {
            out.println("<!doctype html>\n"
                    + "<html>\n"
                    + "<head>\n"
                    + "  <meta charset='UTF-8'>\n"
                    + "  <meta name='viewport' content='width=device-width, initial-
scale=1.0'>\n"
                    + "  <script
src=\"https://unpkg.com/flowbite@1.5.3/dist/flowbite.js\"></script>\n"
                    + "<script src=\"https://cdn.tailwindcss.com\"></script>"
                    + "<script src=\"https://kit.fontawesome.com/fdf05f1614.js\"
crossorigin=\"anonymous\"></script>"
                    + "</head>\n"
                    + "<body class='bg-gray-400'>\n");
            out.println("<nav class='bg-purple-600 px-2 sm:px-20 py-2.5 rounded
dark:bg-gray-900'>\n"
                    + "  <div class='flex flex-wrap justify-between items-center mx-
auto'>\n"
                    + "  <a href='Student' class='flex items-center'>\n"
```

```
+ "      <img src='Images/LOGO.png' class='mr-3 h-6 sm:h-9'
alt='Flowbite Logo'>\n"
+ "  </a>\n"
+ "  <div class='flex items-center md:order-2'>\n"
+ "      <button type='button' class='flex mr-3 text-sm bg-gray-800
rounded-full md:mr-0 focus:ring-4 focus:ring-gray-300 dark:focus:ring-gray-600'
id='user-menu-button' aria-expanded='false' data-dropdown-toggle='user-
dropdown' data-dropdown-placement='bottom'>\n"
+ "        <span class='sr-only'>Open user menu</span>\n"
+ "        <img class='w-8 h-8 rounded-full'
src='Images/AdminImg.png' alt='user photo'>\n"
+ "      </button>\n"
+ "      <!-- Dropdown menu -->\n"
+ "      <div class='hidden z-50 my-4 text-base list-none bg-white
rounded divide-y divide-gray-100 shadow dark:bg-gray-700 dark:divide-gray-600'
id='user-dropdown' data-popper-reference-hidden='' data-popper-escaped='' data-
popper-placement='bottom' style='position: absolute; inset: 0px auto auto 0px;
margin: 0px; transform: translate3d(0px, 10.4px, 0px);'>\n"
+ "        <div class='py-3 px-4'>\n"
+ "          <span class='block text-sm text-gray-900 dark:text-
white'>Shahu Sumit Niranjan</span>\n"
+ "          <span class='block text-sm font-medium text-gray-500
truncate dark:text-gray-400'>"
+ name
+ "</span>\n"
+ "        </div>\n"
+ "        <ul class='py-1' aria-labelledby='user-menu-button'>\n"
+ "          <li>\n"
+ "            <a href='Student?choice=Profile' class='block py-2 px-4
text-sm text-gray-700 hover:bg-gray-100 dark:hover:bg-gray-600 dark:text-gray-
200 dark:hover:text-white'>Profile</a>\n"
+ "          </li>\n"
+ "          <li>\n"
+ "            <a href='Logout' class='block py-2 px-4 text-sm text-gray-
700 hover:bg-gray-100 dark:hover:bg-gray-600 dark:text-gray-200
dark:hover:text-white'>Logout</a>\n"
+ "          </li>\n"
+ "        </ul>\n"
+ "      </div>\n"
+ "      <button data-collapse-toggle='mobile-menu-2' type='button'
class='inline-flex items-center p-2 ml-1 text-sm text-gray-500 rounded-lg
md:hidden hover:bg-gray-100 focus:outline-none focus:ring-2 focus:ring-gray-
200 dark:text-gray-400 dark:hover:bg-gray-700 dark:focus:ring-gray-600' aria-
controls='mobile-menu-2' aria-expanded='false'>\n"
+ "        <span class='sr-only'>Open main menu</span>\n"
+ "        <svg class='w-6 h-6' aria-hidden='true' fill='currentColor'
viewBox='0 0 20 20' xmlns='http://www.w3.org/2000/svg'><path fill-
rule='evenodd' d='M3 5a1 1 0 011-1h12a1 1 0 110 2H4a1 1 0 01-1-1zM3 10a1 1 0
011-1h12a1 1 0 110 2H4a1 1 0 01-1-1zM3 15a1 1 0 011-1h12a1 1 0 110 2H4a1 1
0 01-1-1z' clip-rule='evenodd'></path></svg>\n"
```

```
                + "    </button>\n"
                + "  </div>\n"
                + "  <div class='hidden justify-between items-center w-full md:flex
md:w-auto md:order-1' id='mobile-menu-2'>\n"
                + "    <ul class='flex flex-col p-4 mt-4 bg-orange-400 rounded-lg
md:flex-row md:space-x-8 md:mt-0 md:text-sm md:font-medium dark:bg-gray-
800 md:dark:bg-gray-900 '>\n"
                + "      <li>\n"
                + "        <a href='Student?choice=StudentDashboard' class='block py-
2 pr-4 pl-3 text-gray-700 rounded hover:bg-gray-100 md:hover:bg-transparent
hover:text-blue-700 md:p-0 dark:text-gray-400 md:dark:hover:text-white
dark:hover:bg-gray-700 dark:hover:text-white md:dark:hover:bg-transparent
'>Home</a>\n"
                + "      </li>\n"
                + "      <li>\n"
                + "        <a href='Student?choice=Exam' class='block py-2 pr-4 pl-3
text-gray-700 rounded hover:bg-gray-100 md:hover:bg-transparent hover:text-
blue-700 md:p-0 dark:text-gray-400 md:dark:hover:text-white dark:hover:bg-
gray-700 dark:hover:text-white md:dark:hover:bg-transparent '>Exam</a>\n"
                + "      </li>\n"
                + "      <li>\n"
                + "        <a href='Student?choice=GiveFeedback' class='block py-2
pr-4 pl-3 text-gray-700 rounded hover:bg-gray-100 md:hover:bg-transparent
hover:text-blue-700 md:p-0 dark:text-gray-400 md:dark:hover:text-white
dark:hover:bg-gray-700 dark:hover:text-white md:dark:hover:bg-transparent
'>Feedback</a>\n"
                + "      </li>\n"
                + "    </ul>\n"
                + "  </div>\n"
                + "  </div>\n"
                + "</nav>");
            if (msg != null) {
                if (msgtype != null) {
                    if (msgtype.equals("success")) {
                        out.println("<div class='relative w-full'>"
                                + "<div id='toast-success' class='absolute top-5 right-5 flex
items-center p-4 mb-4 w-full max-w-xs text-white text-bold bg-green-400
rounded-lg shadow dark:text-gray-400 dark:bg-gray-800' role='alert'>\n"
                                + "    <div class='inline-flex flex-shrink-0 justify-center items-
center w-8 h-8 text-green-500 bg-green-100 rounded-lg dark:bg-green-800
dark:text-green-200'>\n"
                                + "        <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M16.707
5.293a1 1 0 010 1.414l-8 8a1 1 0 01-1.414 0l-4-4a1 1 0 011.414-1.414L8
12.586l7.293-7.293a1 1 0 011.414 0z' clip-rule='evenodd'></path></svg>\n"
                                + "        <span class='sr-only'>Check icon</span>\n"
                                + "    </div>\n"
                                + "    <div class='ml-3 text-sm font-normal'>"
                                + msg
```

```
                    + "</div>\n"
                    + "    <button type='button' class='ml-auto -mx-1.5 -my-1.5
bg-white text-gray-400 hover:text-gray-900 rounded-lg focus:ring-2 focus:ring-
gray-300 p-1.5 hover:bg-gray-100 inline-flex h-8 w-8 dark:text-gray-500
dark:hover:text-white dark:bg-gray-800 dark:hover:bg-gray-700' data-dismiss-
target='#toast-success' aria-label='Close'>\n"
                    + "       <span class='sr-only'>Close</span>\n"
                    + "       <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z' clip-
rule='evenodd'></path></svg>\n"
                    + "    </button>\n"
                    + "</div>"
                    + "</div>");
            } else if (msgtype.equals("error")) {
                out.println("<div class='relative w-full'><div id='toast-warning'
class='absolute top-5 right-5 flex items-center p-4 w-full max-w-xs text-white bg-
red-600 rounded-lg shadow dark:text-gray-400 dark:bg-gray-800' role='alert'>\n"
                    + "    <div class='inline-flex flex-shrink-0 justify-center items-
center w-8 h-8 text-orange-500 bg-red-100 rounded-lg dark:bg-orange-700
dark:text-orange-200'>\n"
                    + "       <svg xmlns='http://www.w3.org/2000/svg' class='w-8
h-8 text-red-600' viewBox='0 0 20 20' fill='currentColor'>\n"
                    + "      <path fill-rule='evenodd'\n"
                    + "       d='M18 10a8 8 0 11-16 0 8 8 0 0116 0zm-7 4a1 1 0
11-2 0 1 1 0 012 0zm-1-9a1 1 0 00-1 1v4a1 1 0 102 0V6a1 1 0 00-1-1z'\n"
                    + "       clip-rule='evenodd' />\n"
                    + "    </svg>\n"
                    + "<span class='sr-only'>Warning icon</span>\n"
                    + "    </div>\n"
                    + "    <div class='ml-3 text-sm font-normal'>"
                    + msg
                    + "</div>\n"
                    + "    <button type='button' class='ml-auto -mx-1.5 -my-1.5
bg-white text-gray-400 hover:text-gray-900 rounded-lg focus:ring-2 focus:ring-
gray-300 p-1.5 hover:bg-gray-100 inline-flex h-8 w-8 dark:text-gray-500
dark:hover:text-white dark:bg-gray-800 dark:hover:bg-gray-700' data-dismiss-
target='#toast-warning' aria-label='Close'>\n"
                    + "       <span class='sr-only'>Close</span>\n"
                    + "       <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z' clip-
rule='evenodd'></path></svg>\n"
                    + "    </button>\n"
```

```
                    + "</div></div>");
            } else if (msgtype.equals("delete")) {
                out.println("<div class='relative w-full'><div id='toast-danger'
class='absolute top-5 right-5 flex items-center p-4 mb-4 w-full max-w-xs text-
white bg-pink-600 rounded-lg shadow dark:text-gray-400 dark:bg-gray-800'
role='alert'>\n"
                    + "    <div class='inline-flex flex-shrink-0 justify-center items-
center w-8 h-8 text-red-500 bg-pink-100 rounded-lg dark:bg-red-800 dark:text-
red-200'>\n"
                    + "      <svg aria-hidden=\"true\" class=\"w-5 h-5\"
fill=\"currentColor\" viewBox=\"0 0 20 20\"
xmlns=\"http://www.w3.org/2000/svg\"><path fill-rule=\"evenodd\" d=\"M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z\" clip-
rule=\"evenodd\"></path></svg>\n"
                    + "       \n"
                    + "     <path fill-rule='evenodd'\n"
                    + "       d='M18 10a8 8 0 11-16 0 8 8 0 0116 0zm-7 4a1 1 0
11-2 0 1 1 0 012 0zm-1-9a1 1 0 00-1 1v4a1 1 0 102 0V6a1 1 0 00-1-1z'\n"
                    + "       clip-rule='evenodd' />\n"
                    + "   </svg>\n"
                    + "<span class='sr-only'>Error icon</span>\n"
                    + "   </div>\n"
                    + "   <div class='ml-3 text-sm font-normal'>"
                    + msg
                    + "</div>\n"
                    + "   <button type='button' class='ml-auto -mx-1.5 -my-1.5
bg-white text-gray-400 hover:text-gray-900 rounded-lg focus:ring-2 focus:ring-
gray-300 p-1.5 hover:bg-gray-100 inline-flex h-8 w-8 dark:text-gray-500
dark:hover:text-white dark:bg-gray-800 dark:hover:bg-gray-700' data-dismiss-
target='#toast-danger' aria-label='Close'>\n"
                    + "       <span class='sr-only'>Close</span>\n"
                    + "       <svg aria-hidden='true' class='w-5 h-5'
fill='currentColor' viewBox='0 0 20 20'
xmlns='http://www.w3.org/2000/svg'><path fill-rule='evenodd' d='M4.293
4.293a1 1 0 011.414 0L10 8.586l4.293-4.293a1 1 0 111.414 1.414L11.414
10l4.293 4.293a1 1 0 01-1.414 1.414L10 11.414l-4.293 4.293a1 1 0 01-1.414-
1.414L8.586 10 4.293 5.707a1 1 0 010-1.414z' clip-
rule='evenodd'></path></svg>\n"
                    + "   </button>\n"
                    + "</div></div>");
            }
        }
    }
    out.println("<div class='flex justify-center'>");
    if (name != null) {
        if (choice != null) {
            switch (choice) {
                case "Exam":
```

```
                    RequestDispatcher rd = request.getRequestDispatcher("Exam");
                    rd.include(request, response);
                    break;
                case "GiveFeedback":
                    RequestDispatcher rd2 =
request.getRequestDispatcher("GiveFeedback");
                    rd2.include(request, response);
                    break;
                case "Profile":
                    RequestDispatcher rd5 =
request.getRequestDispatcher("Profile");
                    rd5.include(request, response);
                    break;
                default:
                    RequestDispatcher rd3 =
request.getRequestDispatcher("StudentDashboard");
                    rd3.include(request, response);
            }
          } else {
            response.sendRedirect("Student?choice=StudentDashboard");\
          }
        } else {
          response.sendRedirect("Login");

        out.println("</div></body>\n"
            + "</html>");
      }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
      processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {
      processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
      return "Short description";
    }
}
```

- **StudentDashboard.java (Student Dashboard Page)**

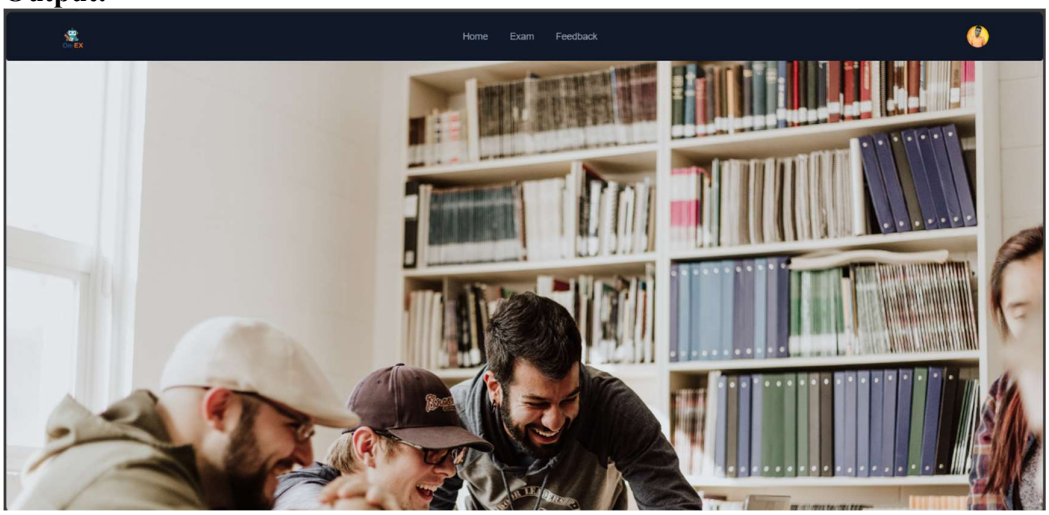
Code:

```
package Student;
```

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class StudentDashboard extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
           throws ServletException, IOException {
       response.setContentType("text/html;charset=UTF-8");
       try (PrintWriter out = response.getWriter()) {
           /* TODO output your page here. You may use following sample code. */
           out.println("<!DOCTYPE html>");
           out.println("<html>");
           out.println("<head>");
           out.println("<title>Servlet StudentDashboard</title>");
           out.println("<style>\n"+
                       " body { font-family: Arial, sans-serif; background-color: rgba(0,
0, 0, 0.329); background-image: url(Images/Student7.jpg); background-size:
cover; }\n ");
           out.println("</head>");
           out.println("<body>");
           out.println("<h1>Servlet StudentDashboard at " +
request.getContextPath() + "</h1>");
           out.println("</body>");
           out.println("</html>");
       }
   }
   @Override
   protected void doGet(HttpServletRequest request, HttpServletResponse
response)
           throws ServletException, IOException {
       processRequest(request, response);
   }
   @Override
   protected void doPost(HttpServletRequest request, HttpServletResponse
response)
           throws ServletException, IOException {
       processRequest(request, response);
   }
   @Override
   public String getServletInfo() {
       return "Short description";
   }
}
```

Output:



➢ **Exam.java (Student Exam List Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.ResultSet;
import javax.servlet.http.HttpSession;

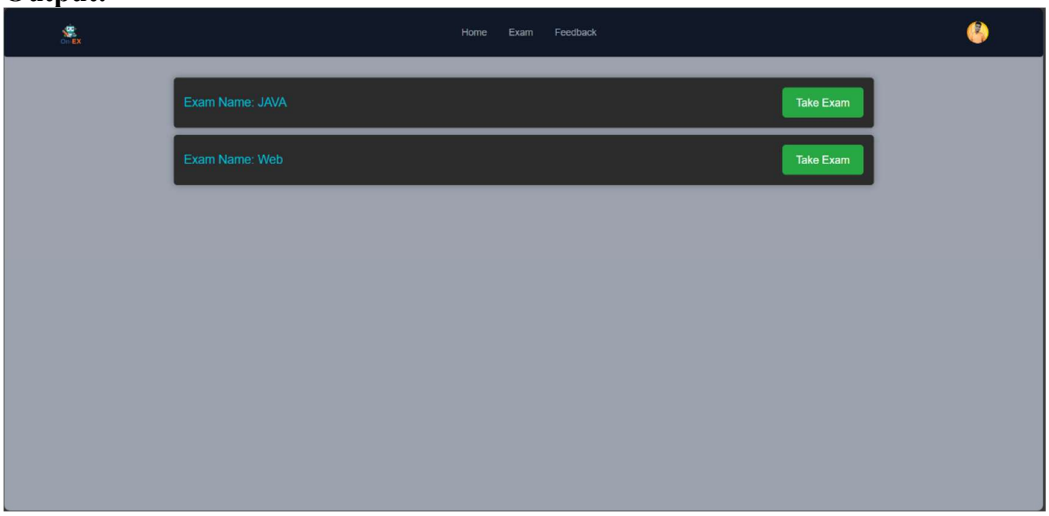
public class Exam extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         Db_Connection db = new Db_Connection();
         try {
            HttpSession session = request.getSession();
            Object name = session.getAttribute("Student");
            System.out.println("Student Email: " + name);
            String examQuery = "SELECT exam_name FROM exam";
            db.Pstmt = db.Con.prepareStatement(examQuery);
            ResultSet examRs = db.Pstmt.executeQuery();
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Exam List</title>");
            out.println("<style>");
```

```
            out.println("body { font-family: Arial, sans-serif; background-color:
#121212; color: #e0e0e0; margin: 0; padding: 0; }");
            out.println(".container { width: 70%; margin: 0 auto; padding: 20px;
}");
            out.println(".exam-list { list-style: none; padding: 0; }");
            out.println(".exam-list li { background-color: #2c2c2c; margin: 10px 0;
padding: 15px; border-radius: 5px; box-shadow: 0px 0px 10px rgba(0, 0, 0, 0.3);
display: flex; justify-content: space-between; align-items: center; }");
            out.println(".exam-list li form { margin: 0; text-align: right; }");
            out.println(".exam-list li span { font-size: 18px; color: #00bcd4; }");
            out.println(".exam-list li input[type='submit'] { padding: 10px 20px;
font-size: 16px; background-color: #28a745; color: white; border: none; border-
radius: 5px; cursor: pointer; transition: background-color 0.3s; }");
            out.println(".exam-list li input[type='submit']:hover { background-color:
#218838; }");
            out.println("</style>");
            out.println("</head>");
            out.println("<body>");
            out.println("<div class='container'>");
            out.println("<ul class='exam-list'>");
            while (examRs.next()) {
                String examName = examRs.getString("exam_name");
                out.println("<li>");
                out.println("<span>Exam Name: " + examName + "</span>");
                out.println("<form action='ExamQuestions' method='post'>");
                out.println("<input type='hidden' name='examname' value='" +
examName + "'>");
                out.println("<input type='submit' value='Take Exam'>");
                out.println("</form>");
                out.println("</li>");
            }
            out.println("</ul>");
            out.println("</div>");
            out.println("</body>");
            examRs.close();
            db.Pstmt.close();

        } catch (SQLException e) {
            out.println("<h3>Error fetching exams: " + e.getMessage() + "</h3>");
        } finally {
            try {
                if (db.Con != null) db.Con.close();
            } catch (SQLException ex) {
                out.println("<h3>Error closing database connection: " +
ex.getMessage() + "</h3>");
            }
        }
    }
```

```
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Servlet to display list of exams with dark mode styling";
    }
}
```

Output:



- **ExamQuestions.java (Student Taking Exam Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.ResultSet;
import javax.servlet.http.HttpSession;

public class ExamQuestions extends HttpServlet {
```

```
    protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {
        response.setContentType("text/html;charset=UTF-8");
        String examname = request.getParameter("examname");
        try (PrintWriter out = response.getWriter()) {
            HttpSession session = request.getSession();
            Object studentEmail = session.getAttribute("Student");
            System.out.println("Student Email: " + studentEmail);
            if (studentEmail != null && isExamTaken((String) studentEmail,
examname)) {
                out.println("<html>");
                out.println("<head><title>Exam Already Taken</title>");
                out.println("<style>\n" +
                        "    body { font-family: Arial, sans-serif; background-color: rgba(0,
0, 0, 0.329); background-image: url(Images/Student7.jpg); background-size:
cover; }\n" +
                        "    .container { max-width: 800px; margin: auto; padding: 20px;
background: rgba(255, 255, 255, 0.8); }\n" +
                        "    .section { border: 2px solid blue; border-radius: 8px; padding:
20px; margin-bottom: 20px; }\n" +
                        "    h1 { text-align: center; font-weight: bold; font-size: 2.5em;
color: #333; }\n" +
                        "    .message { padding: 20px; text-align: center; font-size: 1.5em;
color: red; border: 2px solid red; border-radius: 8px; background-color: rgba(255,
0, 0, 0.2); margin-top: 20px; }\n" +
                        "    .btn { padding: 10px 15px; border: none; border-radius: 5px;
cursor: pointer; background-color: #007BFF; color: white; text-align: center; }\n"
+                        "    .btn:hover { background-color: #0056b3; }\n" +
                        "</style>");
                out.println("</head>");
                out.println("<body>");
                out.println("<div class='container'>");
                out.println("<div class='message'>");
                out.println("<h1>You have already taken this exam!</h1>");
                out.println("<a href='Student?choice=Exam' class='btn'>Back to
Exams</a>");
                out.println("</div>");
                out.println("</div>");
                out.println("</body>");
                out.println("</html>");
                return;
            }
            out.println("<!DOCTYPE html>");
            out.println("<html>");
                out.println("<script src='https://code.jquery.com/jquery-
3.6.0.min.js'></script>");
                out.println("<script>\n" +
                        "    var tabSwitched = false;\n" +
                        "    window.addEventListener('blur', function() {\n" +
```

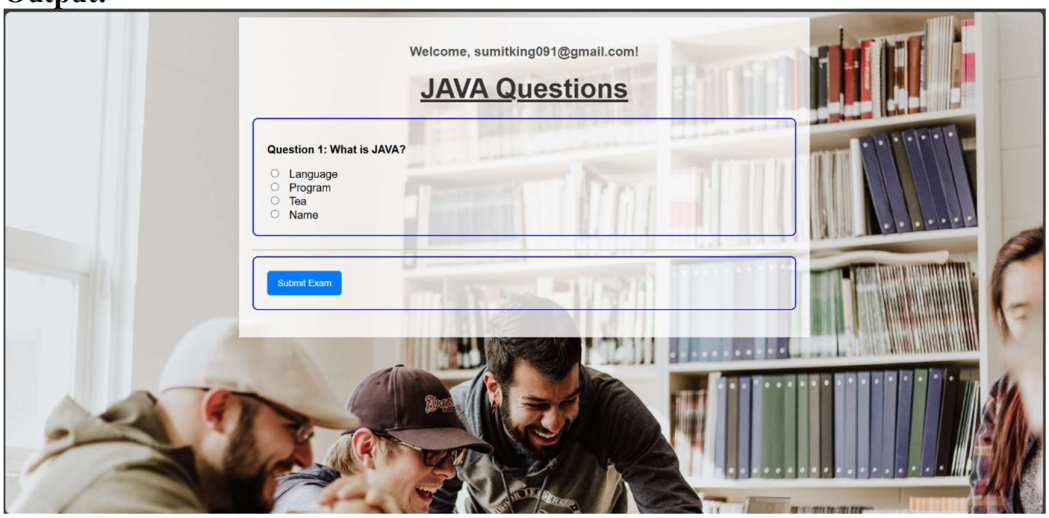
```
                "        tabSwitched = true;\n" +
                "        alert('Warning: You have switched tabs. This will be
reported!');\n" +
                "    });\n" +
                "    \n" +
                "    function saveAnswer(questionNumber, selectedOption) {\n" +
                "        $.ajax({\n" +
                "            url: 'SaveAnswer',\n" +
                "            method: 'POST',\n" +
                "            data: {\n" +
                "                examname: '" + examname + "',\n" +
                "                questionNumber: questionNumber,\n" +
                "                answer: selectedOption\n" +
                "            },\n" +
                "            success: function(response) {\n" +
                "                console.log('Answer saved successfully for question ' +
questionNumber);\n" +
                "            },\n" +
                "            error: function(xhr, status, error) {\n" +
                "                console.error('Error saving answer: ' + error);\n" +
                "            }\n" +
                "        });\n" +
                "    }\n" +
                "    \n" +
                "    $(document).ready(function() {\n" +
                "        $('input[type=radio]').change(function() {\n" +
                "            var questionNumber = $(this).attr('name').split('_')[1];\n" +
                "            var selectedOption = $(this).val();\n" +
                "            saveAnswer(questionNumber, selectedOption);\n" +
                "        });\n" +
                "    });\n" +
                "</script>");
        out.println("<head>");
        out.println("<title>" + examname + " Exam</title>");
        out.println("<style>\n" +
                "    body { font-family: Arial, sans-serif; background-color: rgba(0,
0, 0, 0.329); background-image: url(Images/Student7.jpg); background-size:
cover; }\n" +
                "    .container { max-width: 800px; margin: auto; padding: 20px;
background: rgba(255, 255, 255, 0.8); }\n" +
                "    .section { border: 2px solid blue; border-radius: 8px; padding:
20px; margin-bottom: 20px; }\n" +
                "    h1, h2 {\n" +
                "        text-align: center;\n" +
                "        font-weight: bold;\n" +
                "        text-decoration: underline;\n" +
                "        font-size: 2.5em;\n" +
                "        margin: 20px 0;\n" +
                "        color: #333;\n" +
                "    }\n" +
```

```
                "    .question-item { margin-bottom: 20px; border-bottom: 1px
solid #ccc; padding-bottom: 20px; }\n" +
                "    .btn { padding: 10px 15px; border: none; border-radius: 5px;
cursor: pointer; }\n" +
                "    .btn-submit { background-color: #007BFF; color: white; }\n" +
                "    .radio-group { margin-top: 10px; }\n" +
                "    .radio-group input { margin-right: 10px; }\n" +
                "    .student-info { margin-bottom: 20px; font-size: 1.2em; color:
#555; text-align: center; }\n" +
                "</style>");
        out.println("</head>");
        out.println("<body>");
        out.println("<div class='container'>");
        if (studentEmail != null) {
            out.println("<div class='student-info'>");
            out.println("<p><strong>Welcome, " + studentEmail +
"!</strong></p>");
            out.println("</div>");
        } else {
            out.println("<div class='student-info'>");
            out.println("<p>Student information not available. Please log in
again.</p>");
            out.println("</div>");
        }
        out.println("<h1>" + examname + " Questions</h1>");
        out.println("<form action='SubmitExam' method='post'>");
        Db_Connection db = new Db_Connection();
        try {
            String sanitizedExamTable = examname.replace(" ", "_");
            String query = "SELECT question, opt1, opt2, opt3, opt4 FROM " +
sanitizedExamTable;
            db.Pstmt = db.Con.prepareStatement(query);
            ResultSet rs = db.Pstmt.executeQuery();
            int questionCounter = 1;
            while (rs.next()) {
                String que = rs.getString("question");
                String opt1 = rs.getString("opt1");
                String opt2 = rs.getString("opt2");
                String opt3 = rs.getString("opt3");
                String opt4 = rs.getString("opt4");
                out.println("<div class='section'>");
                out.println("<p><strong>Question " + questionCounter + ": " + que +
"</strong></p>");
                out.println("<div class='radio-group'>");
                out.println("<label><input type='radio' name='answer_" + que + "'
value='" + opt1 + "' required> " + opt1 + "</label><br>");
                out.println("<label><input type='radio' name='answer_" + que + "'
value='" + opt2 + "'> " + opt2 + "</label><br>");
                out.println("<label><input type='radio' name='answer_" + que + "'
value='" + opt3 + "'> " + opt3 + "</label><br>");
```

```
                out.println("<label><input type='radio' name='answer_" + que + "'
value='" + opt4 + "'> " + opt4 + "</label>");
                out.println("</div>");
                out.println("</div>");
                out.println("<hr>");
                questionCounter++;
            }
            out.println("<div class='section'>");
            out.println("<input type='hidden' name='examname' value='" +
examname + "'>");
            out.println("<input type='submit' value='Submit Exam' class='btn btn-
submit'>");
            out.println("</div>");
            out.println("</form>");
            out.println("</div>");
            out.println("</body>");
            out.println("</html>");
            rs.close();
            db.Pstmt.close();
        } catch (SQLException e) {
            out.println("<h3>Error fetching questions: " + e.getMessage() +
"</h3>");
        } finally {
            try {
                if (db.Con != null) db.Con.close();
            } catch (SQLException ex) {
                out.println("<h3>Error closing database connection: " +
ex.getMessage() + "</h3>");
            }
        }
    }
  }
  private boolean isExamTaken(String studentEmail, String examname) {
    Db_Connection db = new Db_Connection();
    try {
        String checkQuery = "SELECT * FROM result WHERE email = ? AND
exam_name = ?";
        db.Pstmt = db.Con.prepareStatement(checkQuery);
        db.Pstmt.setString(1, studentEmail);
        db.Pstmt.setString(2, examname);
        ResultSet rs = db.Pstmt.executeQuery();
        boolean examTaken = rs.next()
        rs.close();
        db.Pstmt.close();
        return examTaken;
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (db.Con != null) db.Con.close();
```

```
            } catch (SQLException ex) {
                ex.printStackTrace();
            }
        }
        return false;
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Servlet to display exam questions for selected exam with updated
styling, and student info at the top.";
    }
}
```

Output:



- **SubmitExam.java (Student Submitting Exam Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.ResultSet;
import java.sql.SQLException;
```

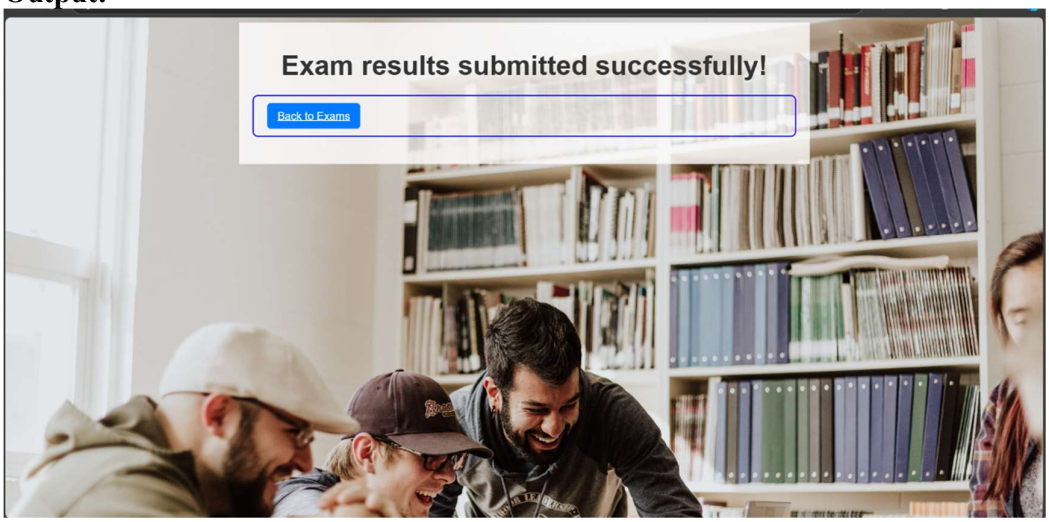
```
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

public class SubmitExam extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         Db_Connection db = new Db_Connection();
         String examName = request.getParameter("examname");
         HttpSession session = request.getSession();
         Object studentEmail = session.getAttribute("Student");
         if (studentEmail == null) {
            out.println("<html>");
            out.println("<head><title>Error</title></head>");
            out.println("<body>");
            out.println("<h3>Error: Student email is not available. Please log in
again.</h3>");
            out.println("<a href='login.html'>Login</a>");
            out.println("</body>");
            out.println("</html>");
            return;
         }
         try {
            String countQuery = "SELECT COUNT(*) AS totalQuestions FROM "
+ examName;
            db.Pstmt = db.Con.prepareStatement(countQuery);
            ResultSet rsCount = db.Pstmt.executeQuery();
            int totalQuestions = 0;
            if (rsCount.next()) {
               totalQuestions = rsCount.getInt("totalQuestions");
            }
            String examQuery = "SELECT question, ans FROM " + examName;
            db.Pstmt = db.Con.prepareStatement(examQuery);
            ResultSet rsExam = db.Pstmt.executeQuery();
            int correctAnswers = 0;
            while (rsExam.next()) {
               String questionText = rsExam.getString("question");
               String correctAnswer = rsExam.getString("ans");
               String studentAnswer = request.getParameter("answer_" +
questionText);
               if (correctAnswer != null && correctAnswer.equals(studentAnswer))
{
                  correctAnswers++;
               }
            }
```

```
            rsCount.close();
            rsExam.close();
            db.Pstmt.close();
            int score = (totalQuestions > 0) ? (correctAnswers * 100) /
totalQuestions : 0;
            String insertQuery = "INSERT INTO result (email, exam_name, score)
VALUES (?, ?, ?)";
            db.Pstmt = db.Con.prepareStatement(insertQuery);
            db.Pstmt.setString(1, (String) studentEmail);
            db.Pstmt.setString(2, examName);
            db.Pstmt.setInt(3, score);
            db.Pstmt.executeUpdate();
            db.Pstmt.close();
            db.Con.close();
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Exam Submitted</title>");
            out.println("<style>\n"
                    + "    body { font-family: Arial, sans-serif; background-color:
rgba(0, 0, 0, 0.329); background-image: url(Images/Student7.jpg); background-
size: cover; }\n"
                    + "    .container { max-width: 800px; margin: auto; padding: 20px;
background: rgba(255, 255, 255, 0.8); }\n"
                    + "    .section { border: 2px solid blue; border-radius: 8px; padding:
20px; margin-bottom: 20px; }\n"
                    + "    h1, h2 {\n"
                    + "        text-align: center; font-weight: bold; font-size: 2.5em;
margin: 20px 0; color: #333; }\n"
                    + "    .btn { padding: 10px 15px; border: none; border-radius: 5px;
cursor: pointer; }\n"
                    + "    .btn-submit { background-color: #007BFF; color: white; }\n"
                    + "    .student-info { margin-bottom: 20px; font-size: 1.2em; color:
#555; text-align: center; }\n"
                    + "</style>");
            out.println("</head>");
            out.println("<body>");
            out.println("<div class='container'>");
            out.println("<h1>Exam results submitted successfully!</h1>");
            out.println("<div class='section'>");
            out.println("<a href='Student?choice=Exam' class='btn btn-
submit'>Back to Exams</a>");
            out.println("</div>");
            out.println("</div>");
            out.println("</body>");
            out.println("</html>");
        } catch (SQLException e) {
            out.println("<h3>Error inserting exam result: " + e.getMessage() +
"</h3>");
        }
```

```
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    public String getServletInfo() {
        return "Servlet that stores exam results (exam name, student email, and
score) in the result table, fetching correct answers from the exam table";
    }
}
```

Output:



- **SaveAnswer.java (Student's Response saving for handling failure during Exam Page)**

Code:

```
package Student;

import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
import java.sql.SQLException;
import javax.servlet.http.HttpSession;

public class SaveAnswer extends HttpServlet {
   @Override
   protected void doPost(HttpServletRequest request, HttpServletResponse
response)
         throws ServletException, IOException {
      response.setContentType("text/plain");
      String examname = request.getParameter("examname");
      String questionNumber = request.getParameter("questionNumber");
      String answer = request.getParameter("answer");
      HttpSession session = request.getSession();
      String studentEmail = (String) session.getAttribute("Student");
      Db_Connection db = new Db_Connection();
      try (PrintWriter out = response.getWriter()) {
         String query = "INSERT INTO student_answers (email, exam_name,
question_number, answer) "
                  + "VALUES (?, ?, ?, ?) ON DUPLICATE KEY UPDATE answer
= ?";
         db.Pstmt = db.Con.prepareStatement(query);
         db.Pstmt.setString(1, studentEmail);
         db.Pstmt.setString(2, examname);
         db.Pstmt.setInt(3, Integer.parseInt(questionNumber));
         db.Pstmt.setString(4, answer);
         db.Pstmt.setString(5, answer);
         db.Pstmt.executeUpdate();
         out.println("Answer saved successfully");
      } catch (SQLException e) {
         e.printStackTrace();
      } finally {
         try {
            if (db.Con != null) db.Con.close();
         } catch (SQLException ex) {
            ex.printStackTrace();
         }
      }
   }
   @Override
   public String getServletInfo() {
      return "Servlet to save answers using AJAX";
   }
}
```

- **GiveFeedback.java (Student Giving Feedback Page)**

Code:

```
package Feedback;

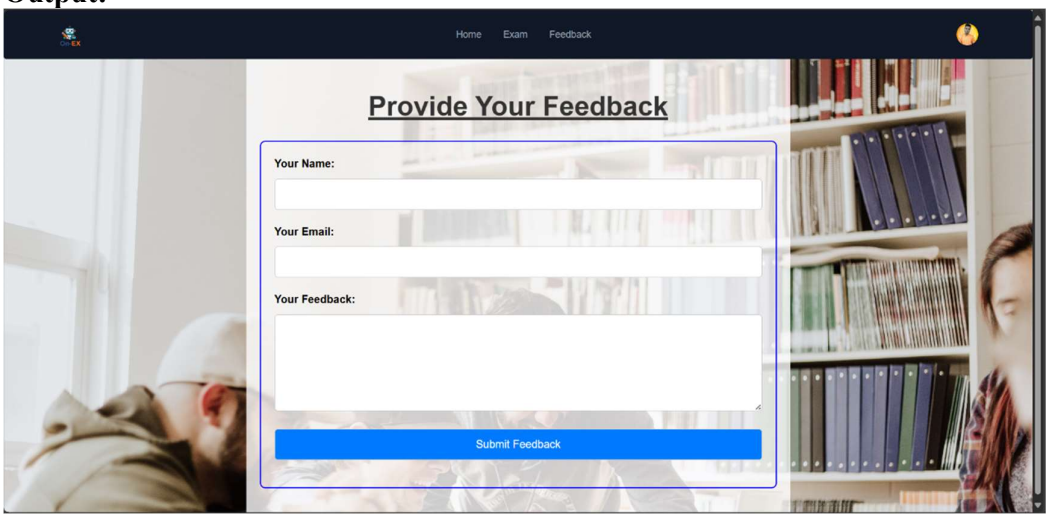
import DB.Db_Connection;
import java.io.IOException;
```

```
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class GiveFeedback extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         out.println("<!DOCTYPE html>");
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Give Feedback</title>");
         out.println("<style>\n" +
                  "    body { font-family: Arial, sans-serif; background-color: rgba(0,
0, 0, 0.329); background-image: url(Images/Student7.jpg); background-size:
cover; }\n" +
                  "    .container { max-width: 800px; margin: auto; padding: 20px;
background: rgba(255, 255, 255, 0.8); }\n" +
                  "    .section { border: 2px solid blue; border-radius: 8px; padding:
20px; margin-bottom: 20px; }\n" +
                  "    h1, h2 {\n" +
                  "        text-align: center;\n" +
                  "        font-weight: bold;\n" +
                  "        text-decoration: underline;\n" +
                  "        font-size: 2.5em;\n" +
                  "        margin: 20px 0; \n" +
                  "        color: #333; }/\n" +
                  "    .form-group { margin-bottom: 20px; }\n" +
                  "    .form-group label { display: block; font-weight: bold; margin-
bottom: 10px; }\n" +
                  "    .form-group input, .form-group textarea { width: 100%;
padding: 10px; border: 1px solid #ccc; border-radius: 5px; }\n" +
                  "    .btn { padding: 10px 15px; border: none; border-radius: 5px;
cursor: pointer; background-color: #007BFF; color: white; text-align: center; }\n"
+                    "    .btn:hover { background-color: #0056b3; }\n" +
                  "</style>");
         out.println("</head>");
         out.println("<body>");
         out.println("<div class='container'>");
         out.println("<h1>Provide Your Feedback</h1>");
         out.println("<form action='SubmitFeedback' method='post'>");
         out.println("<div class='section'>");
         out.println("<div class='form-group'>");
         out.println("<label for='name'>Your Name:</label>");
         out.println("<input type='text' id='name' name='name' required>");
         out.println("</div>");
```

```
            out.println("<div class='form-group'>");
            out.println("<label for='email'>Your Email:</label>");
            out.println("<input type='email' id='email' name='email' required>");
            out.println("</div>");
            out.println("<div class='form-group'>");
            out.println("<label for='feedback'>Your Feedback:</label>");
            out.println("<textarea id='feedback' name='feedback' rows='5'
required></textarea>");
            out.println("</div>");
            out.println("<div class='form-group'>");
            out.println("<input type='submit' value='Submit Feedback' class='btn'>");
            out.println("</div>");
            out.println("</div>");
            out.println("</form>");
            out.println("</div>");
            out.println("</body>");
            out.println("</html>");
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

Output:



- **SubmitFeedback.java (Student Submitting Feedback Page)**

Code:

```
package Feedback;

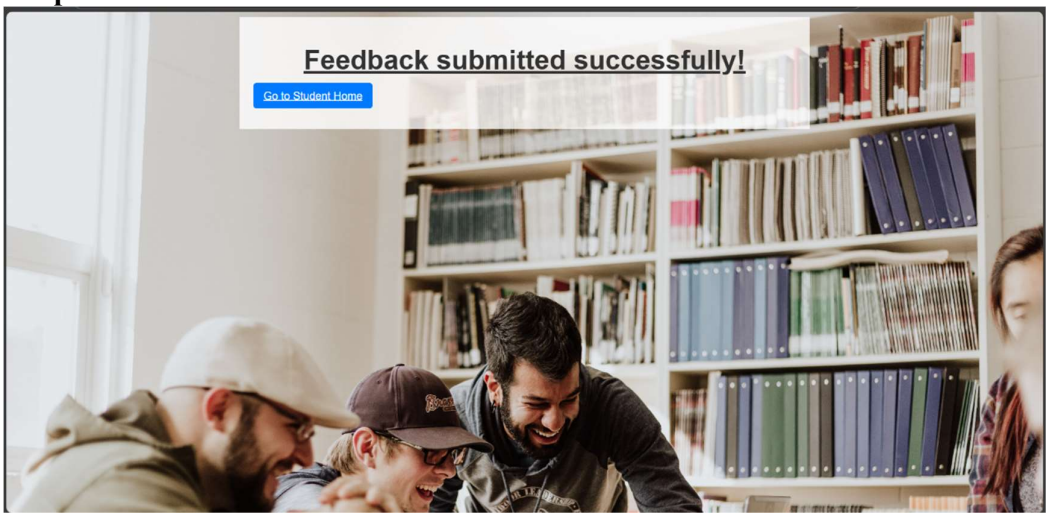
import DB.Db_Connection;
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class SubmitFeedback extends HttpServlet {
   protected void processRequest(HttpServletRequest request,
HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      String name = request.getParameter("name");
      String email = request.getParameter("email");
      String feedbackText = request.getParameter("feedback");
      try (PrintWriter out = response.getWriter()) {
         try {
            Db_Connection db = new Db_Connection();
            String sql = "INSERT INTO feedback (name, email, feedback)
VALUES (?, ?, ?)";
            db.Pstmt = db.Con.prepareStatement(sql);
            db.Pstmt.setString(1, name);
            db.Pstmt.setString(2, email);
            db.Pstmt.setString(3, feedbackText);
            int rowsAffected = db.Pstmt.executeUpdate();
            if (rowsAffected > 0) {
               out.println("<head>\n" +
               "    <title>Submit Feedback</title>\n" +
               "    <style>\n" +
               "        body {\n" +
               "            font-family: Arial, sans-serif;\n" +
               "            background-color: rgba(0, 0, 0, 0.329);\n" +
               "            background-image: url(Images/Student7.jpg);\n" +
               "            background-size: cover;\n" +
               "        }\n" +
               "\n" +
               "        .container {\n" +
               "            max-width: 800px;\n" +
               "            margin: auto;\n" +
               "            padding: 20px;\n" +
               "            background: rgba(255, 255, 255, 0.8);\n" +
               "        }\n" +
               "\n" +
               "        .section {\n" +
               "            border: 2px solid blue;\n" +
```

```
"            border-radius: 8px;\n" +
"            padding: 20px;\n" +
"            margin-bottom: 20px;\n" +
"        }\n" +
"\n" +
"        h1, h2 {\n" +
"            text-align: center;\n" +
"            font-weight: bold;\n" +
"            text-decoration: underline;\n" +
"            font-size: 2.5em;\n" +
"            margin: 20px 0;\n" +
"            color: #333;\n" +
"        }\n" +
"\n" +
"        .form-group {\n" +
"            margin-bottom: 20px;\n" +
"        }\n" +
"\n" +
"        .form-group label {\n" +
"            display: block;\n" +
"            font-weight: bold;\n" +
"            margin-bottom: 10px;\n" +
"        }\n" +
"\n" +
"        .form-group input, .form-group textarea {\n" +
"            width: 100%;\n" +
"            padding: 10px;\n" +
"            border: 1px solid #ccc;\n" +
"            border-radius: 5px;\n" +
"        }\n" +
"\n" +
"        .btn {\n" +
"            padding: 10px 15px;\n" +
"            border: none;\n" +
"            border-radius: 5px;\n" +
"            cursor: pointer;\n" +
"            background-color: #007BFF;\n" +
"            color: white;\n" +
"            text-align: center;\n" +
"        }\n" +
"\n" +
"        .btn:hover {\n" +
"            background-color: #0056b3;\n" +
"        }\n" +
"\n" +
"        .btn:focus {\n" +
"            outline: none;\n" +
"        }\n" +
"    </style>\n" +
"</head>");
```

```
                out.println("<div class='container'>");
                out.println("<h1>Feedback submitted successfully!</h1>");
            } else {
                out.println("<html><body>");
                out.println("<div class='container'>");
                out.println("<h1>Failed to submit feedback!</h1>");
            }
            out.println("<div class='form-group'>");
            out.println("<a href='Student' class='btn'>Go to Student Home</a>");
            out.println("</div>");
            out.println("</div>");
            out.println("</body></html>");
            db.Pstmt.close();
            db.Con.close();
        } catch (Exception e) {
            e.printStackTrace();
            response.getWriter().println("<h1>Error occurred while submitting
feedback. Please try again later.</h1>");
        }
    }
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }\
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

Output:



TESTING

1. Introduction to Testing

Testing is a crucial phase in the software development lifecycle that ensures the web application functions as intended. For this project, we applied both manual and automated testing techniques to validate the application's functionality, performance, and user experience. The primary focus was to ensure that the integration of servlets, JSP pages, and the MySQL database provided a seamless user experience, and that business logic executed correctly.

2. Testing Environment

- Development Platform: NetBeans IDE
- Technology Stack: Java Servlets, JSP, and MySQL
- Server: Apache Tomcat
- Database: MySQL

3. Types of Testing Conducted

a. Unit Testing

- Objective: To test individual components such as servlets, business logic, and data access objects.
- Tools Used: JUnit was used for unit testing servlets and database operations.
- Approach: Each servlet method was tested by simulating HTTP requests and validating the responses. JUnit tests covered core functionalities such as session management, form submissions, and database CRUD operations.

b. Integration Testing

- Objective: To ensure proper interaction between the servlets, JSP pages, and the MySQL database.
- Approach: We tested the flow from JSP pages (front-end) to servlets (back-end) and then to the database (MySQL). This included validation of the data passing between layers and proper synchronization between form submissions and database operations. Tools like Postman were also used for testing HTTP requests.

c. Functional Testing

- Objective: To verify that each feature of the web application works as per the requirements.
- Approach: Manual testing was performed on each functionality, such as:
    - User registration and login.
    - Profile management.
    - Exam creation and question addition.
    - Fetching exam data and questions from the database.
    - Handling user sessions and exam submissions.

Test cases were designed based on user stories and functional requirements, and each scenario was executed to ensure correct system behavior.

d. Database Testing

- Objective: To verify the integrity of data stored and retrieved from the MySQL database.
- Approach: Manual queries were executed to check the correctness of data insertion, updates, and deletions in tables such as exam, questions, and stu_info. The relationships between tables were tested to ensure data consistency.

e. User Interface Testing

- Objective: To ensure that the user interface provides a seamless experience across browsers.
- Approach: The application's UI was tested manually in different browsers (Chrome, Firefox, and Edge) to check for consistency in layout, forms, navigation, and responsiveness. We used browser developer tools to inspect the design and ensure it was in line with expected standards.

f. Performance Testing

- Objective: To evaluate the performance of the application under normal and peak load conditions.
- Approach: We used Apache JMeter to simulate concurrent users accessing the application. This helped measure the load-handling capacity, page response times, and database query performance under stress. The results were analyzed to identify potential bottlenecks.

ADVANTAGES

The Java-based web application developed using NetBeans, Servlets, JSP, and MySQL provides several key advantages that enhance the functionality, reliability, and usability of the exam portal:

1. Tab-Switching Trackin:
The system includes a tab-switching detection feature that allows administrators to monitor and flag any instances where students switch tabs during an exam. This adds a layer of security to prevent cheating and ensures that students remain focused on the exam page.

2. Session Data Persistence:
The application is designed to save exam progress in the session, allowing students to resume their exams seamlessly in case of device failure or browser crash. This prevents data loss and ensures that students do not lose their progress during unexpected interruptions.

3. Restricted Access to Answers and Results:
Students are not able to access their answers or results until they are released by the administrator, ensuring the integrity of the exam process. This prevents students from knowing the correct answers prematurely and sharing them with others.

4. Device Flexibility:
The application allows students to use any device—such as laptops, tablets, or smartphones—to take the exam. This flexibility improves accessibility and convenience, ensuring that students can participate in exams from a variety of platforms and locations.

5. Simplified User and Exam Management:
The system provides an easy-to-use interface for updating and deleting users and exams. Administrators can easily add, modify, or remove users and exam data, reducing the complexity of managing large datasets and ensuring that only current and authorized information is maintained.

6. Real-Time Data Handling:
With the use of MySQL as the database backend, the application provides real-time data handling for exams, user information, and results. This ensures that all changes are immediately reflected in the system without delay, making data management efficient.

7. Scalability and Maintenance:
The modular nature of the application makes it easier to scale and maintain. As the system is built using Servlets and JSP, updates to the codebase, including adding new features or modifying existing ones, can be done with minimal disruption to the ongoing processes.

8. Efficient Data Storage and Retrieval:
MySQL, being a relational database management system, ensures efficient storage and retrieval of exam data, user details, and results. The structured query language (SQL) used allows for quick search, insertion, updating, and deletion of records, improving the overall performance of the application.

LIMITATIONS

While the Java-based web application developed using NetBeans, Servlets, JSP, and MySQL provides a functional platform for conducting exams, several limitations have been identified during its development:

1. Question Type Restriction:
Currently, the system is limited to handling only one type of question format—Multiple Choice Questions (MCQs). This limitation restricts the flexibility for educators to include other types of questions such as descriptive answers, true/false, or matching-type questions, which may be important for certain types of assessments.

2. Lack of Image Insertion:
At this moment, the system does not support the insertion of images in questions or answers. This limits the capability of the application to host questions that require visual content, such as diagrams or graphs, which may be essential for subjects like mathematics, science, or engineering.

3. Inability to Track Copying Across Devices:
The application is unable to track or prevent students from copying exam content using other devices. If a student accesses the exam on one device and copies or shares the content through another device, the system currently lacks the functionality to detect or prevent this kind of cheating.

4. No Geofencing or Location Restriction:
The system does not have a feature to enforce location-based restrictions, such as geofencing or IP address-based approval, for taking exams. This means that students can access the exam from any location, making it difficult to limit exams to a controlled or supervised environment.

5. Basic User Authentication:
The current authentication system is basic and lacks additional layers of security, such as two-factor authentication (2FA) or biometric verification, making it vulnerable to unauthorized access if user credentials are compromised.

6. Limited Browser Compatibility:
The application has not been extensively tested across multiple browsers or devices, which could lead to inconsistent user experiences. For instance, the exam page might function differently on older browsers or mobile devices, affecting accessibility for some users.

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FUTURE SCOPE

The Java-based web application developed using NetBeans, Servlets, JSP, and MySQL has laid a solid foundation for conducting online examinations. However, several enhancements and new features can be integrated to improve its functionality, security, and scalability in the future. Based on the analysis of the current advantages and limitations, the following areas offer potential for future development:

1. Support for Multiple Question Types:
Currently, the system supports only Multiple Choice Questions (MCQs). A future enhancement could include the ability to handle various question formats such as descriptive answers, fill-in-the-blanks, true/false, and matching-type questions. This would make the platform more versatile for diverse exam types and subjects.

2. Image and Media Integration:
Incorporating support for image and media file uploads in questions and answers can significantly enhance the system's utility. This would allow instructors to create exams with visual components, such as diagrams, graphs, or multimedia content, which is especially useful in fields like science, engineering, and art.

3. Advanced Cheating Prevention Mechanisms:
While tab-switching tracking is currently available, more robust cheating prevention measures could be introduced. These could include webcam monitoring, screen recording, or keystroke tracking, alongside machine learning algorithms to detect suspicious behavior patterns during the exam.

4. Location-Based Access Control:
Implementing geofencing or IP-based restrictions to control where students can take the exam will enhance security and reduce the likelihood of dishonest behavior. This feature would ensure that exams are taken in authorized locations only, such as within a school or designated test center.

5. Mobile-Friendly Design and Testing:
Although the system allows students to take exams from any device, the user interface could be optimized for mobile platforms to ensure a consistent experience across devices. Implementing responsive design techniques would improve usability on smartphones and tablets, making the system more accessible to a broader audience.

6. Enhanced User Authentication:
Currently, the system uses basic user authentication. Upgrading to a more secure multi-factor authentication (MFA) system, incorporating methods such as biometric verification (fingerprint or facial recognition) or one-time password (OTP) verification, would provide stronger security against unauthorized access.

7. Exam Time Management Enhancements:
Adding a feature to track inactivity during the exam, such as automatic timeouts after a period of inactivity, would further enhance security. This can prevent students from leaving their screens idle for long periods or switching to other tabs without triggering consequences.

8. Real-Time Exam Monitoring Dashboard:
A live monitoring dashboard for administrators could be introduced, allowing them to track students' activities in real time, such as time spent on each question, number of tab switches, and login attempts. This would enable more proactive exam supervision and help identify potential cheating in real time.

9. Improved Reporting and Analytics:
Adding advanced reporting features and analytics would allow instructors to gain insights into student performance trends, question difficulty levels, and time taken per question. This data can help instructors refine their exams for future sessions and provide more tailored feedback to students.

10. Automatic Grading and Feedback System:
Expanding the system with an automatic grading module that provides immediate feedback to students after the exam could improve student experience. Once secure result release mechanisms are in place, this could reduce the time required for manual grading and enhance the efficiency of the assessment process.

11. Integration with Learning Management Systems (LMS):
Future versions of the application could be integrated with popular learning management systems (LMS) like Moodle or Blackboard. This would allow educators to manage exams, course content, and student records in a centralized platform, enhancing the overall educational experience.

12. Scalability for Large-Scale Use:
To accommodate a larger number of students and exams, the system could be migrated to a cloud infrastructure. This would improve scalability, performance, and reliability, particularly when handling high traffic during peak exam periods.

CONCLUSION

The development of this Java-based web application using Servlets, JSP, and MySQL provided a comprehensive understanding of web application architecture and database integration. The use of NetBeans as the development environment facilitated efficient coding, debugging, and deployment, while adhering to industry-standard practices for MVC (Model-View-Controller) design.

Through this project, we successfully implemented various functionalities, including user authentication, profile management, and exam portal management. The integration of MySQL allowed for robust data handling, ensuring that the application is capable of managing and retrieving data efficiently.

The use of Servlets enabled us to manage request and response cycles seamlessly, while JSP (JavaServer Pages) allowed for dynamic content generation, improving the user experience. The database schema, designed in MySQL, efficiently supported data storage and retrieval, and proved to be scalable for future enhancements or data-intensive operations.

Overall, the project demonstrated the power and flexibility of Java technologies in developing secure, efficient, and user-friendly web applications. The experience gained from this project in combining backend programming with database management will be invaluable for future developments.

This web application provides a solid foundation for expansion, allowing for future features such as advanced user analytics, better UI designs, and real-time interaction capabilities. With further refinement and optimization, the application can scale to meet the demands of larger user bases and more complex functionalities.

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